Retail Analytics: Predicting Customer Purchase Behavior Business

Introduction:

Welcome to RetailVC's Retail Analytics Solutions, a leading provider of data-driven solutions for retail businesses. RetailVC specializes in using advanced analytics techniques to help retailers optimize their operations, enhance customer experiences, and drive business growth. Its mission is to empower retail organizations with actionable insights that enable them to make informed decisions, improve performance, and remain competitive in today's dynamic marketplace.

Problem Statement:

The goal of this capstone project is to develop a comprehensive understanding of customer behavior and preferences based on retail transaction data. The aim is to analyze customers' demographic information, purchasing patterns, and interactions with marketing campaigns to identify key insights. These insights will inform strategic decision-making and drive business growth. Specifically, the focus will be on predicting customers' responses to marketing campaigns and optimizing promotional strategies to maximize their effectiveness and return on investment (ROI).

Note: Use the **marketing_campaign.csv** dataset for performing the below tasks

Excel Tasks:

1. **Data Exploration**:

   Task 1: Statistical Summary

   Create a statistical summary for numerical features

   RESULTS

   The first task in the data inspection was to visually examine some of the data to determine the quality and categorization. The first observation of the 30 columns of data is that some columns are missing data (i.e., “Income”) and others are miscategorized (i.e., “Dt_Customer” column categorized as “General”). It was determined by this data analyst to leave the “income” field that was missing data BLANK because the survey participant may be unemployed. It was noted that all fields are NUMERIC fields with the exception of the following, which are CATEGORICAL fields:

   1. Education
   2. Marital_Status

   Here are the results of the statistical summary of the numeric field of the data set to be analyzed

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines
count	2240	2240	2240	2240	2216	2240	2240	2240	2240	2240
mean	5592.159821	1968.805804	N/A	N/A	52247.25135	0.444196429	0.50625	7/1/2013	49.109375	303.9357143
std	3246.662198	11.98406946	N/A	N/A	25173.07666	0.538398098	0.544538231	9/25/1900	28.96245281	336.5973926
min	0	1893	N/A	N/A	1730	0	0	1/8/2012	0	0
25%	2828.25	1959	N/A	N/A	35303	0	0	1/5/2013	24	23.75
50%	5458.5	1970	N/A	N/A	51381.5	0	0	7/1/2013	49	173.5
75%	8427.75	1977	N/A	N/A	68522	1	1	12/10/2013	74	504.25
max	11191	1996	N/A	N/A	666666	2	2	12/6/2014	99	1493

	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts	MntGoldProds	NumDealsPurchases	NumWebPurchases	NumCatalogPurchase	NumStorePurchases	NumWebVisitsMonth
count	2240	2240	2240	2240	2240	2240	2240	2240	2240	2240
mean	26.30223214	166.95	37.52544643	27.06294643	44.021875	2.325	4.084821429	2.662053571	5.790178571	5.316517857
std	39.77343376	225.7153725	54.6289794	41.28049849	52.16743891	1.932237501	2.778714147	2.923100656	3.250958146	2.42664501
min	0	0	0	0	0	0	0	0	0	0
25%	1	16	3	1	9	1	2	0	3	3
50%	8	67	12	8	24	2	4	2	5	6
75%	33	232	50	33	56	3	6	4	8	7
max	199	1725	259	263	362	15	27	28	13	20

	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Complain	Z_CostContact	Z_Revenue	Response	Age
count	2240	2240	2240	2240	2240	2240	2240	2240	2240	2240
mean	0.072767857	0.074553571	0.072767857	0.064285714	0.013392857	0.009375	3	11	0.149107143	55.19419643
std	0.25981307	0.262728285	0.25981307	0.245315974	0.114975606	0.096391168	0	0	0.356273586	11.98406946
min	0	0	0	0	0	0	3	11	0	28
25%	0	0	0	0	0	0	3	11	0	47
50%	0	0	0	0	0	0	3	11	0	54
75%	0	0	0	0	0	0	3	11	0	65
max	1	1	1	1	1	1	3	11	1	131

Figure 1: Statistical Summary of Numeric Values in the Dataset

Task 2: Line Chart

Create a line chart for the number of enrollments by year

RESULTS

Created a Pivot table to count the number of enrollments based on birth year by ID.

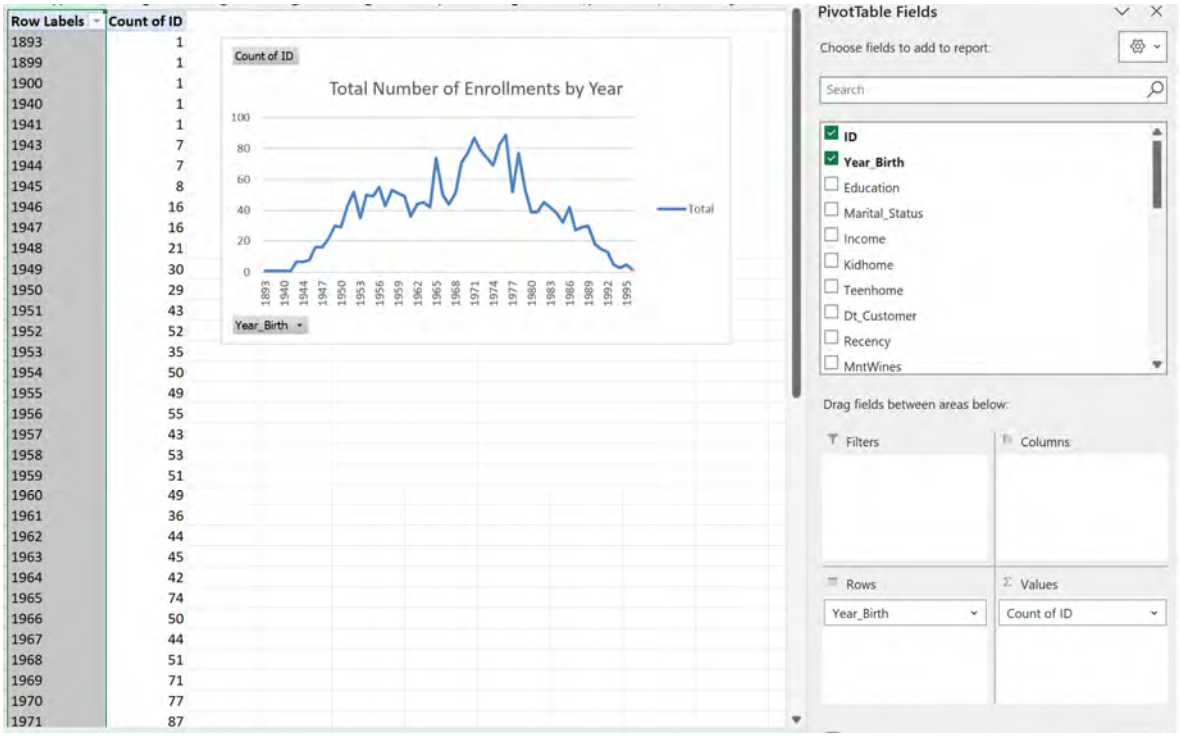


Figure 2: Setup Steps for Determining Number of Enrollments per Year

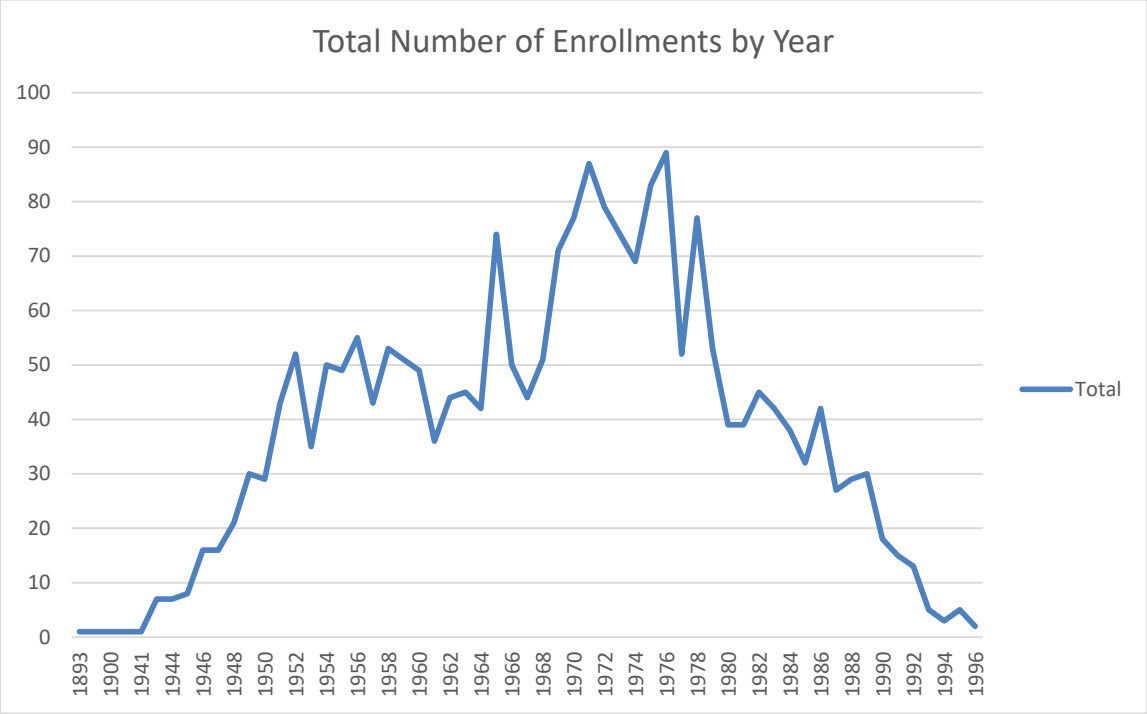


Figure 3: Output Chart for Number of Enrollments per Year

Task 3: Cross-Tabulated Count

Give a cross-tabulated count for response values against education

RESULTS

Created a Pivot table to determine the count of the education level of all respondents by ID.

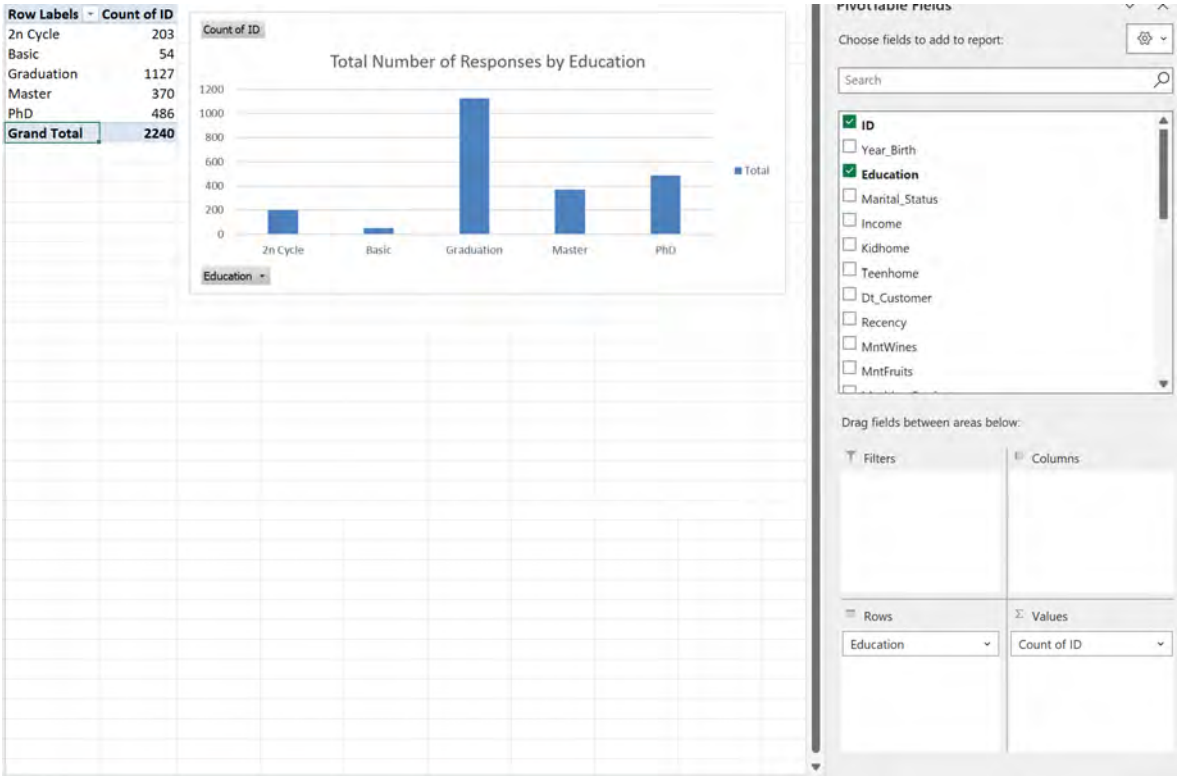


Figure 4: Setup Steps for Determining Count of Response Against Education

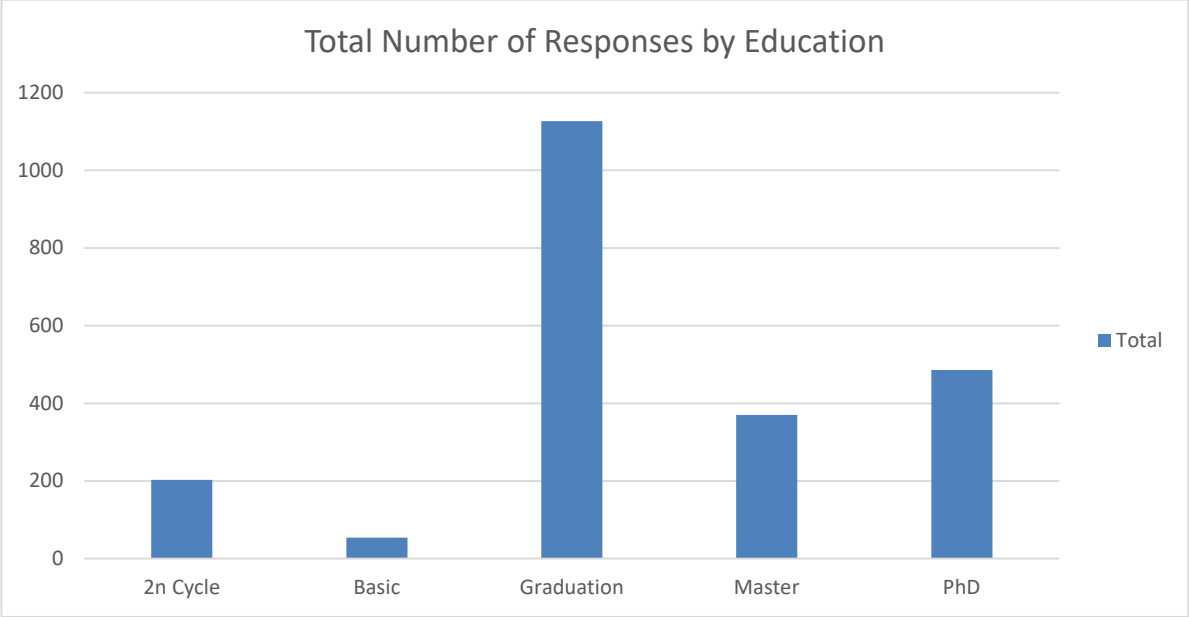


Figure 5: Output Chart of Count of Response Against Education

Task 4: Boxplot

Make a boxplot on income and write your observations

RESULTS

Extract from the dataset the column associated with "Income" to analyze the salary ranges of the respondents (NOTE: MS Excel cannot generate a blockplot from data generated from a pivot table).

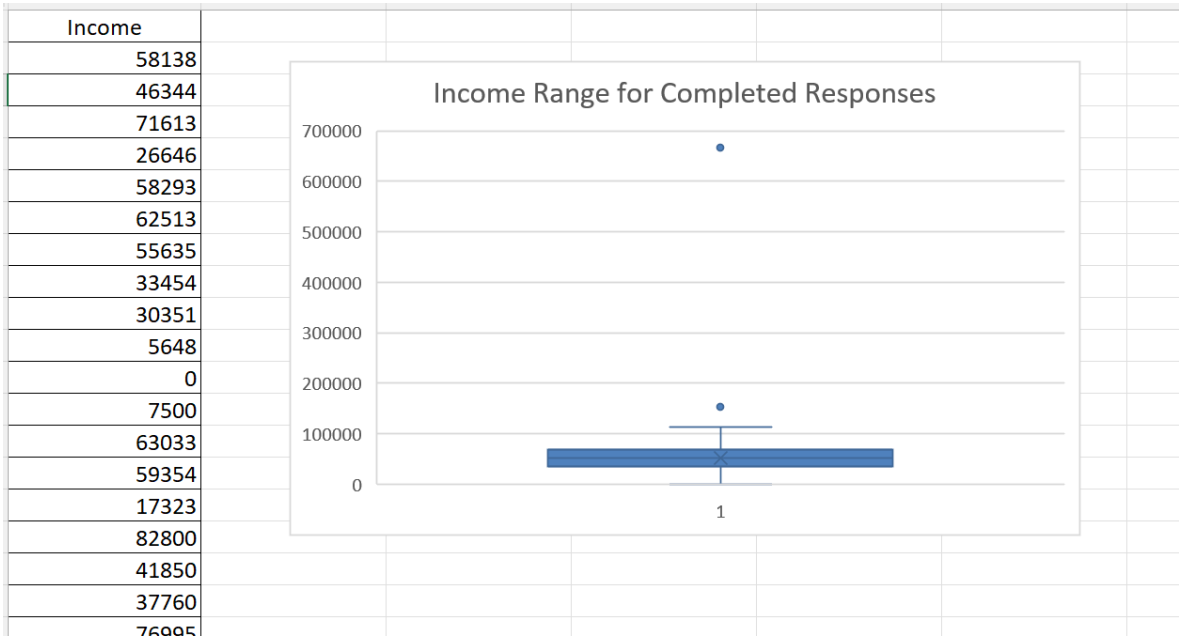


Figure 6: Setup Steps for Determining Income of All Respondents

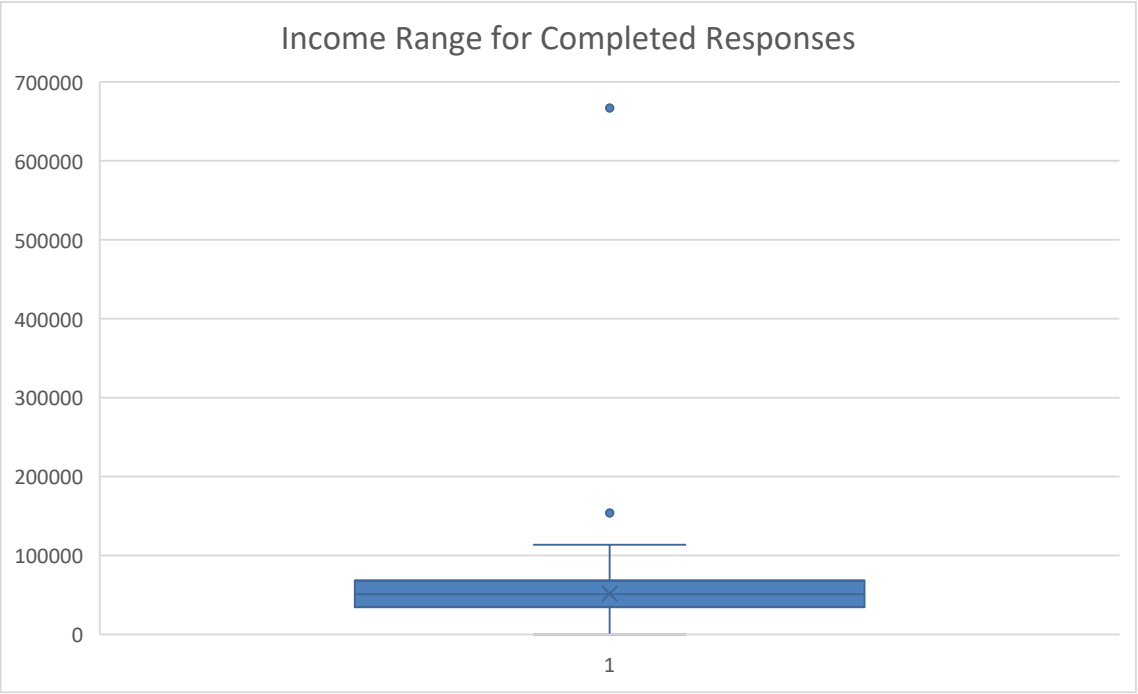


Figure 7: Output Boxplot Chart of Income of All Respondents

Based on my numerical analysis, the average income is $52,247. There are eight distinguished outliers above $118,350, with the highest two being $162,397 and $666,66. Median income is approximately $51,381. On the opposite end of the spectrum, there are 24 respondents who registered $0 annual income.

Task 5: Calculate the Age

Calculate the age of customers and make a histogram of that

RESULTS

Created a Pivot table to determine age ranges of all respondents by ID. Grouped them by an age range of ten (10) starting with age twenty-five (25)

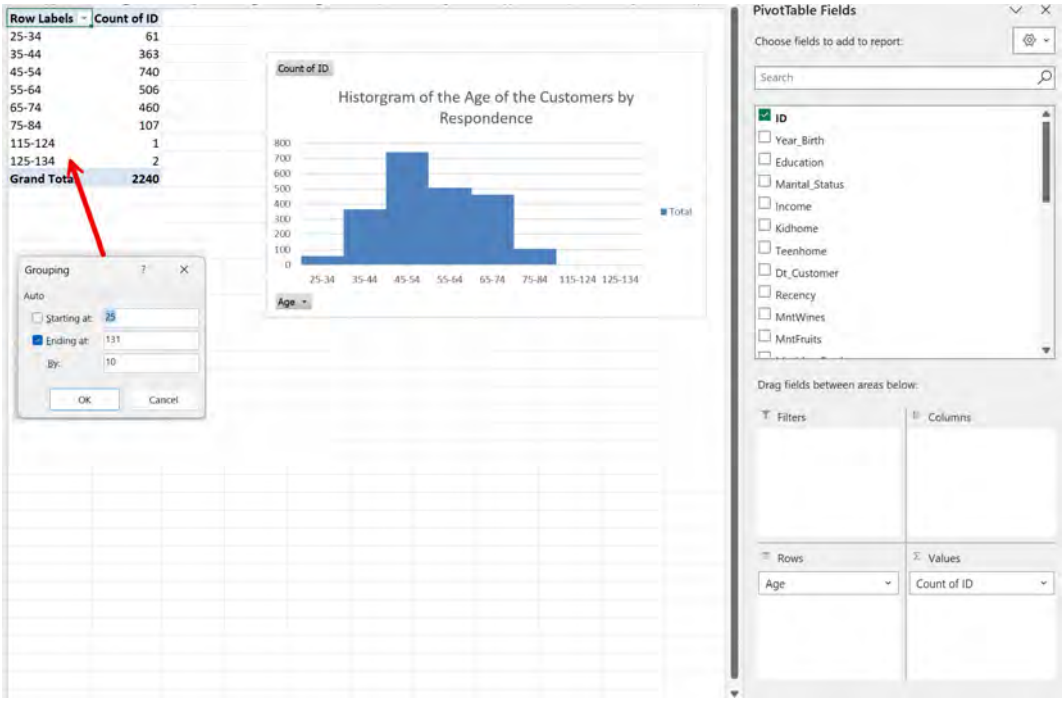


Figure 8: Setup Steps for Determining the Age Ranges of All Respondents

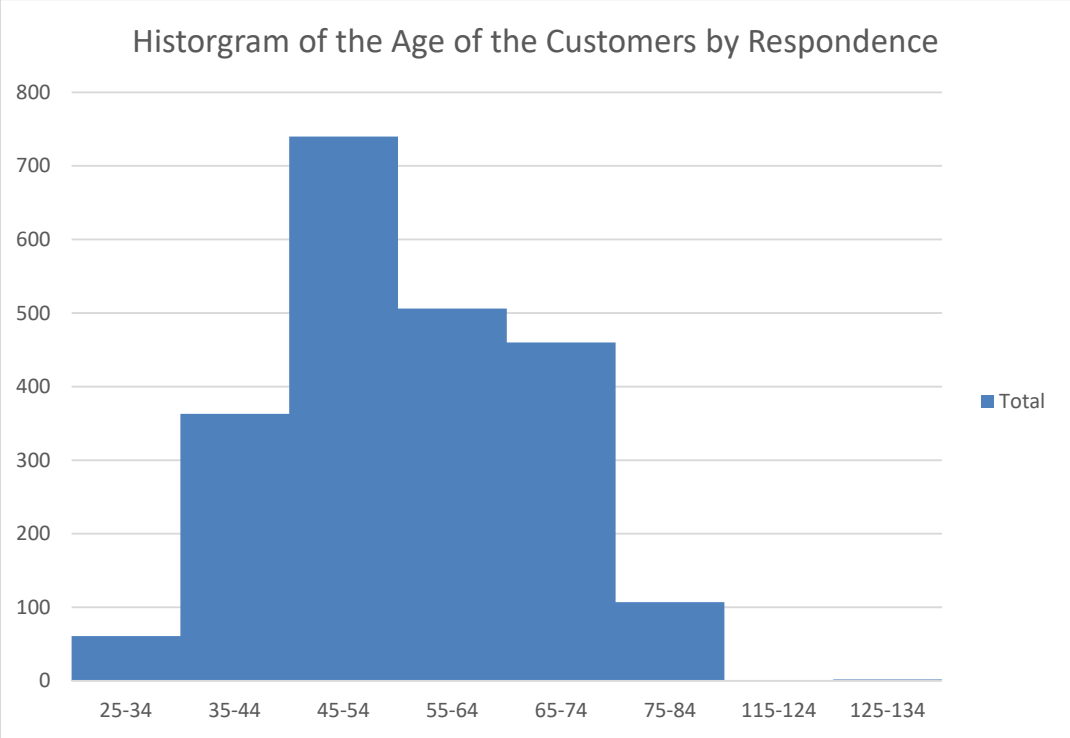


Figure 9: Output Histogram of the Age Ranges of All Respondents

Task 6: Visualize the Response

Visualize the response against “Marital_Status”

RESULTS

Created a Pivot table to determine the marital status of all respondents by ID.

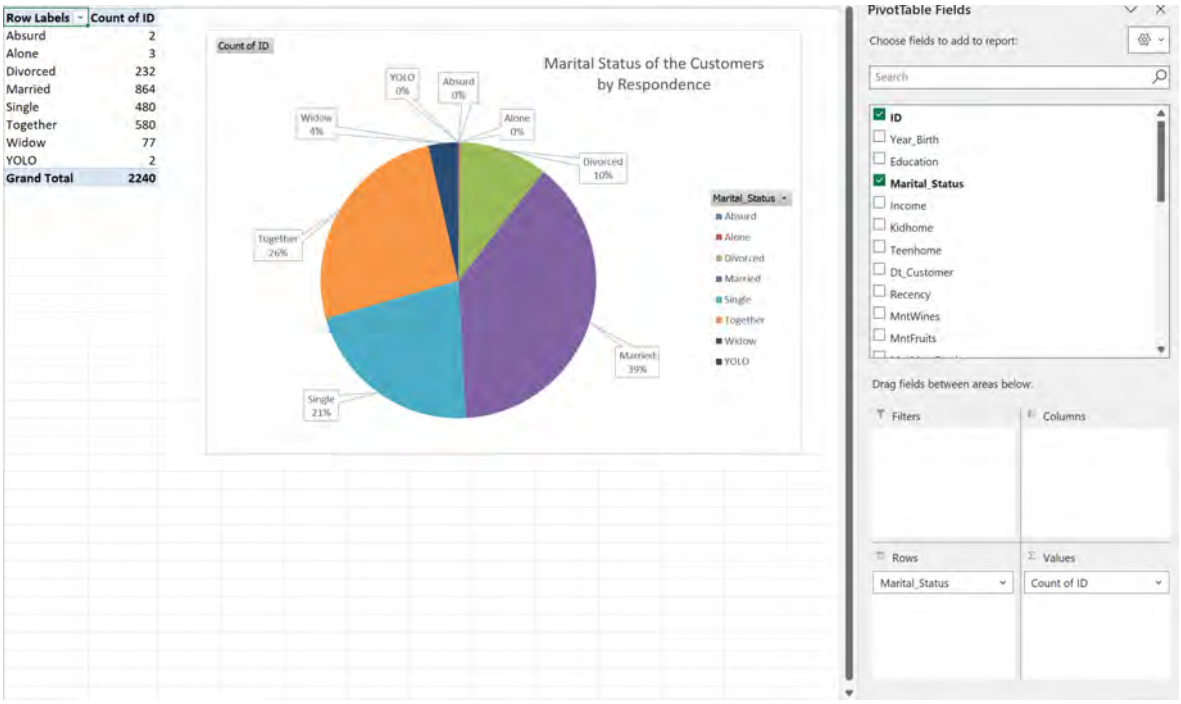


Figure 10: Setup Steps for Determining the Marital Status of All Respondents

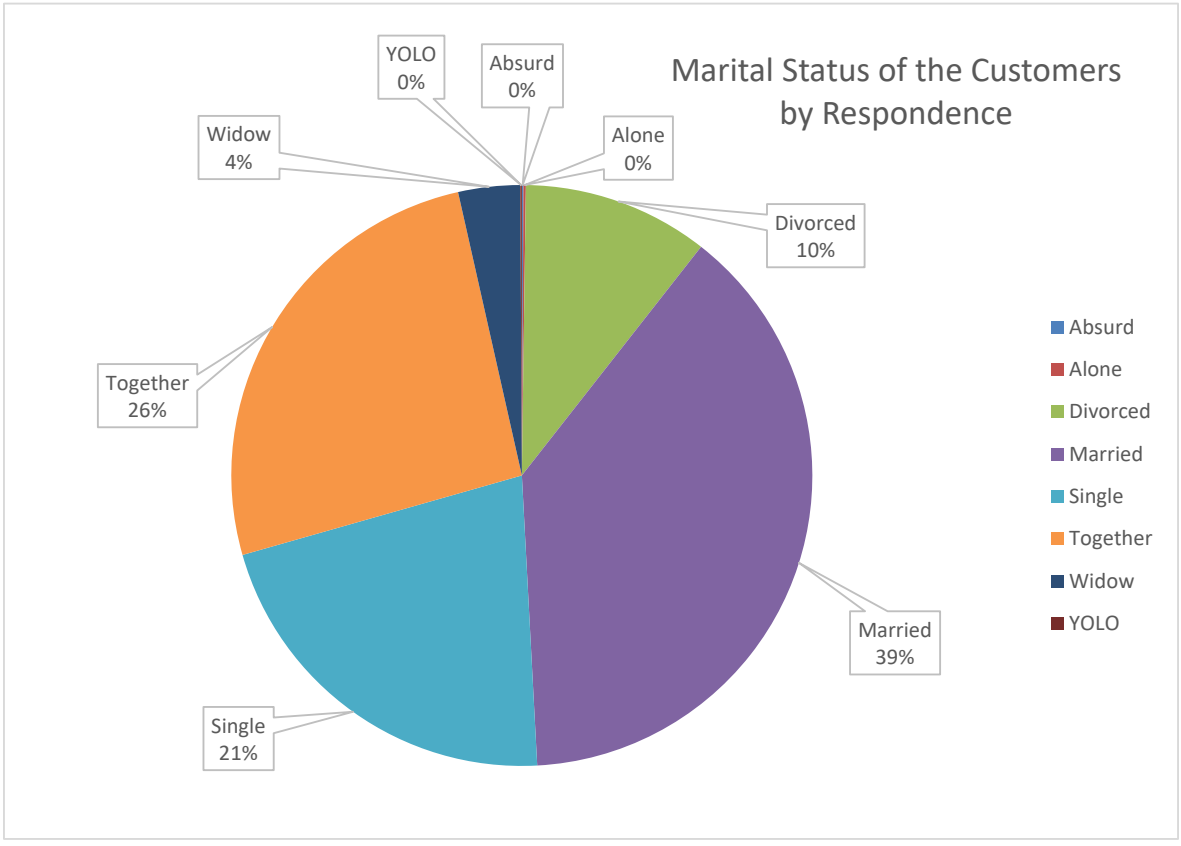


Figure 11: Output Pie Chart of the Marital Status of All Respondents

PRELIMINARY CONCLUSION FROM EXCEL DATA REVIEW

Key Findings:

- Demographics: The average customer is 55 years old. A significant portion of customers are married or single, with a PhD and a Graduation being the most common educational levels.
- Income: Average income is $52,247. Outliers above $118,350 were identified. Median income is approximately $51,381.
- Marketing Response:
    - Highest positive response rates are observed among customers with PhDs (20.8%) and Graduations (13.5%).
    - Married customers have higher total counts but lower response rates than singles.
- Enrollment Trends: Enrollments peaked in certain years, with a steady flow observed across the dataset's time span.
- Age Distribution: The majority of customers fall within the 40- to 70-year-old age range, indicating a mature customer base.
- Campaign Strategy: Optimizing campaigns based on marital status and education may improve response rates.

Recommendations:

- Tailor marketing messages by education level and marital status.
- Focus promotional strategies on demographics with higher observed response rates.
- Consider income segmentation for targeting premium products.
- Revisit customer acquisition efforts to balance age distribution and diversify engagement.

SQL Tasks:

2. **Data Loading:**
    - Create a schema named "retail_data"
    - Set "retail_data" as the default schema
    - Create tables to store the retail transaction data
    - Set ct_customer as the datetime field while loading the data and apply the appropriate date format **(NOTE: Assignment correction by instructor. It is "dt_customer", not "ct_customer")**

Task 1: Create SQL Database

RESULTS

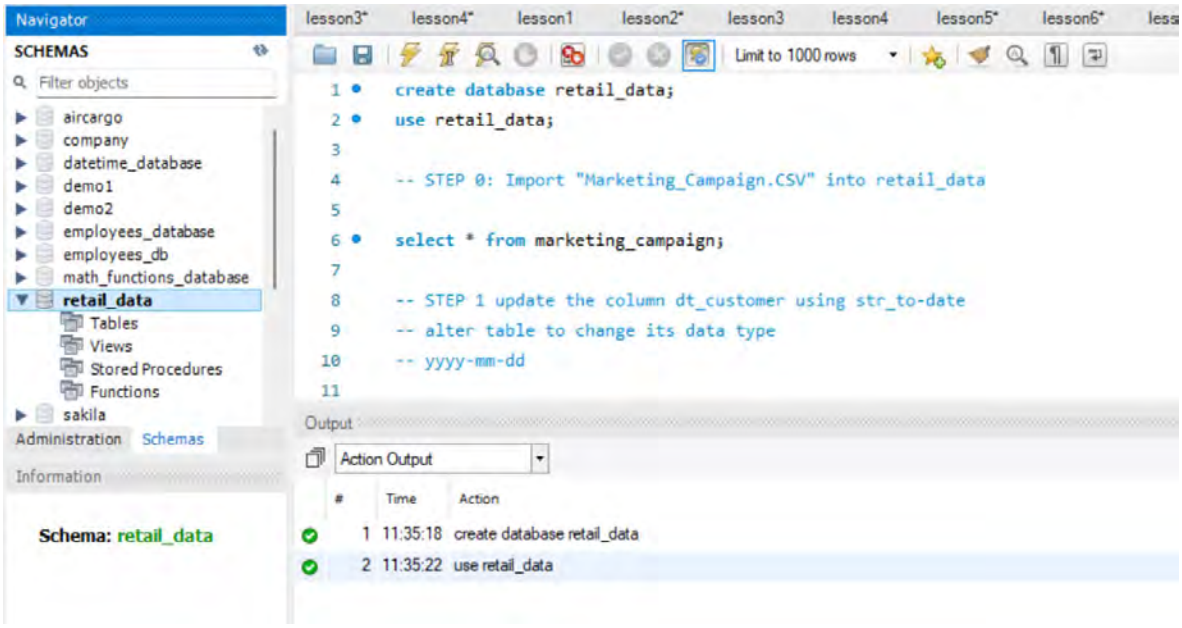


Figure 12a: Setup Steps for creating "Marketing_Campaign" SQL Database

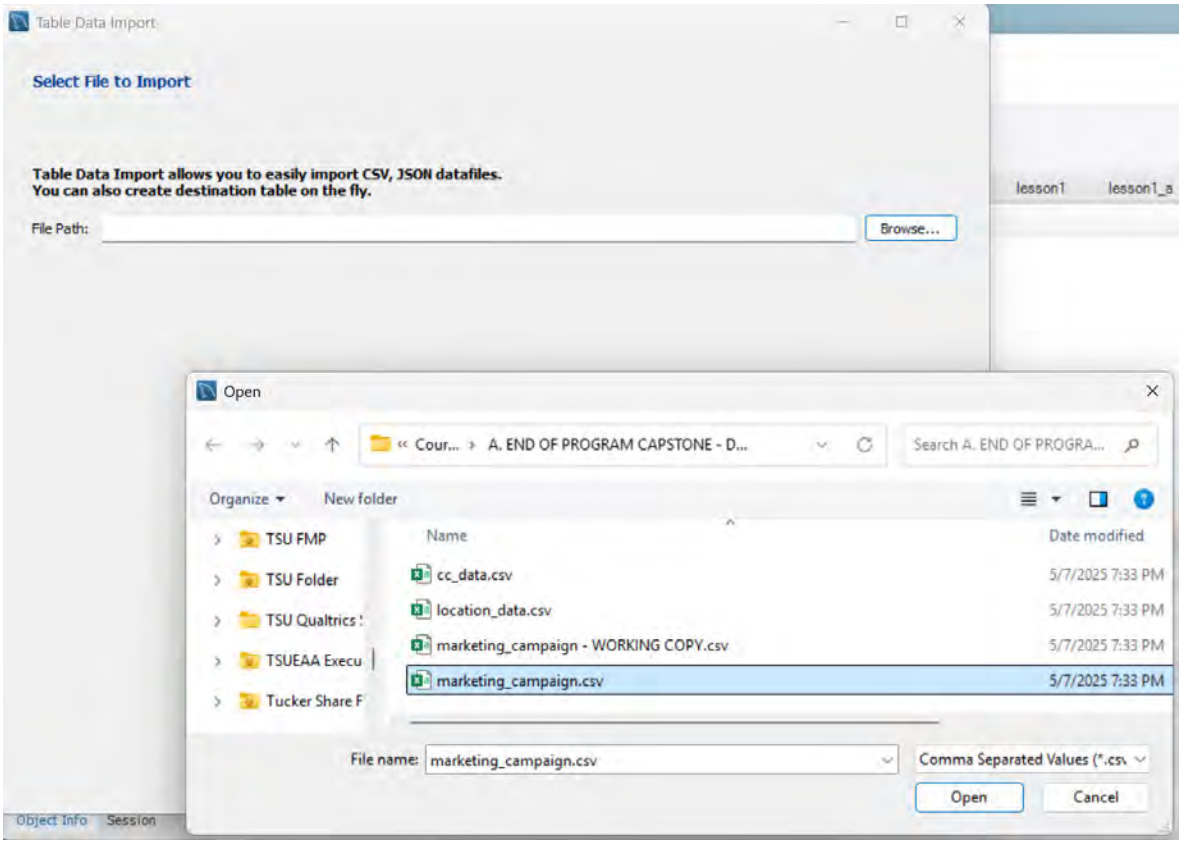


Figure 12b: Setup Steps for creating "Marketing_Campaign" SQL Database

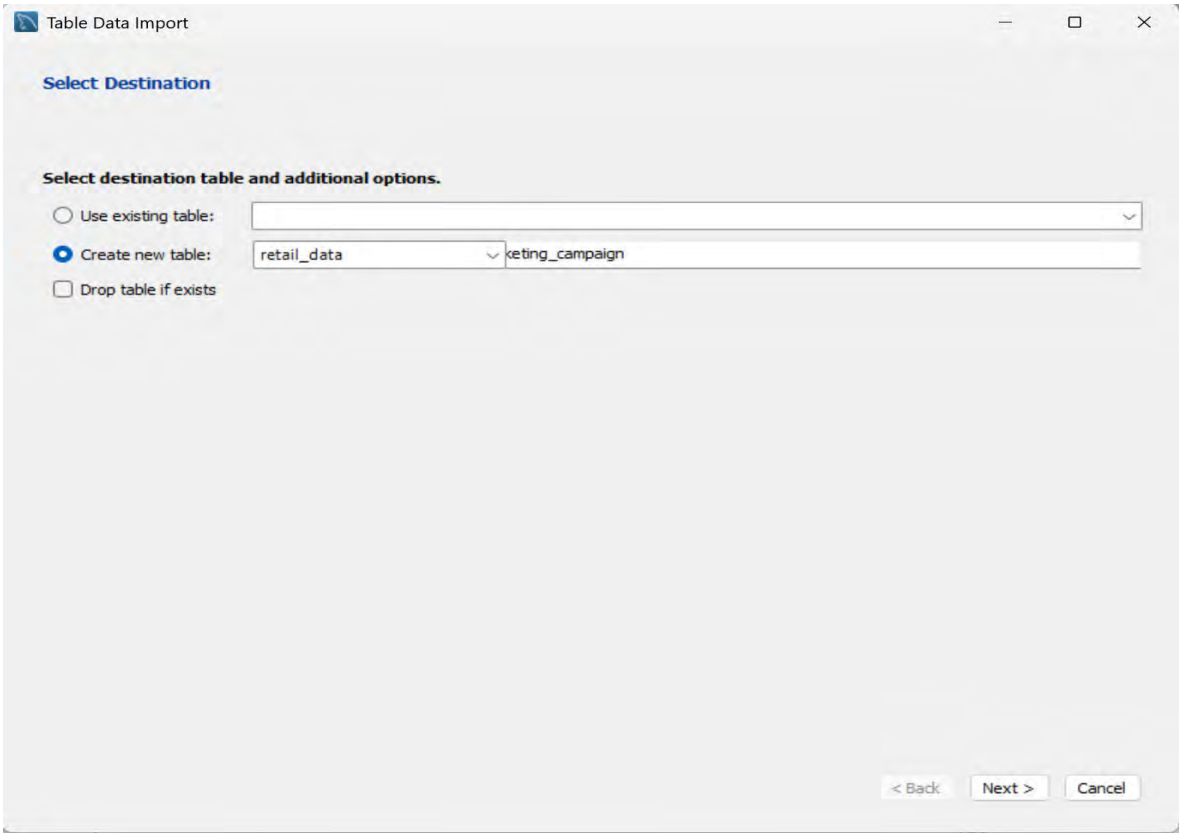


Figure 12c: Setup Steps for creating "Marketing_Campaign" SQL Database

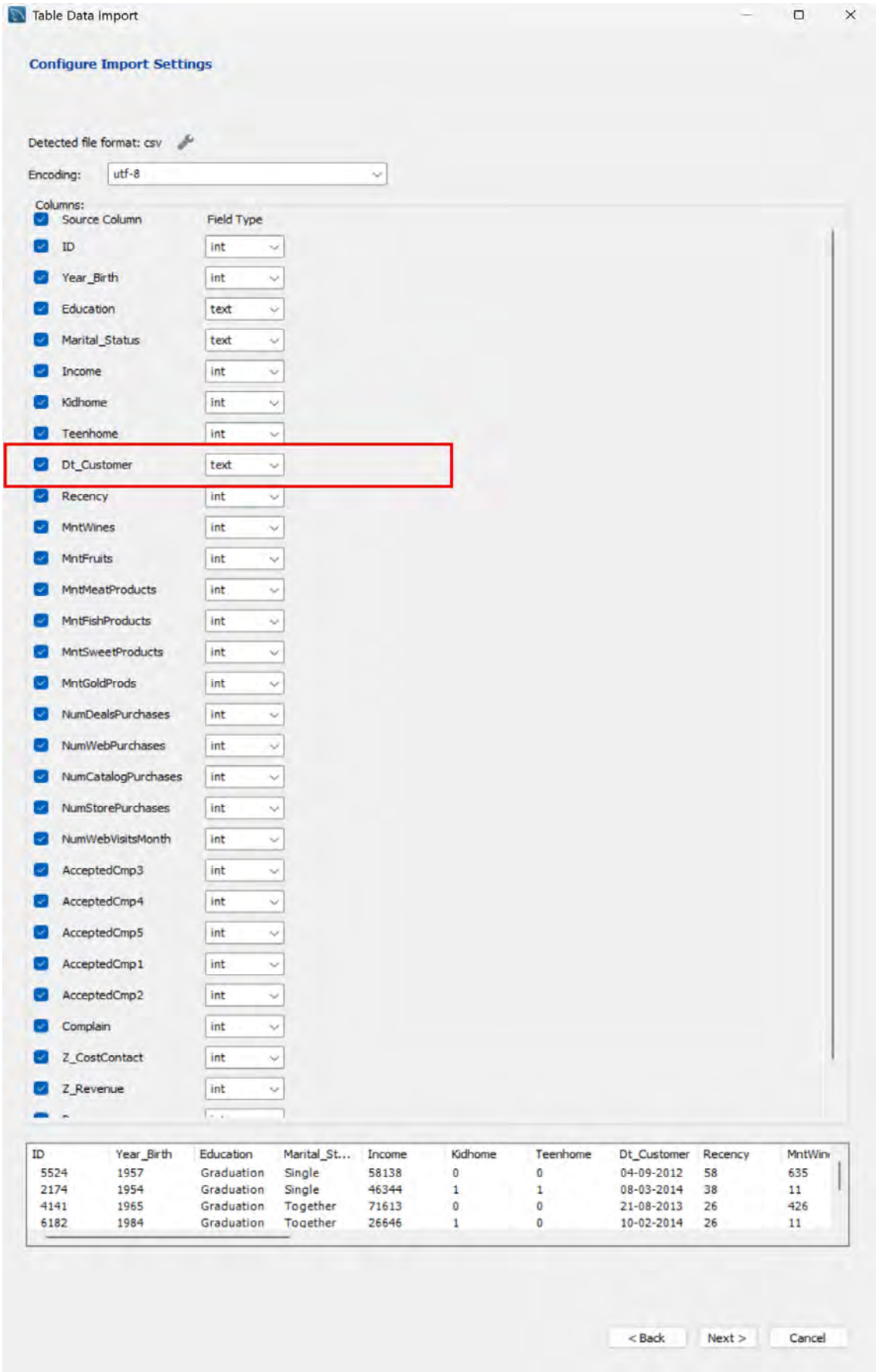


Figure 12d: Setup Steps for creating "Marketing_Campaign" SQL Database

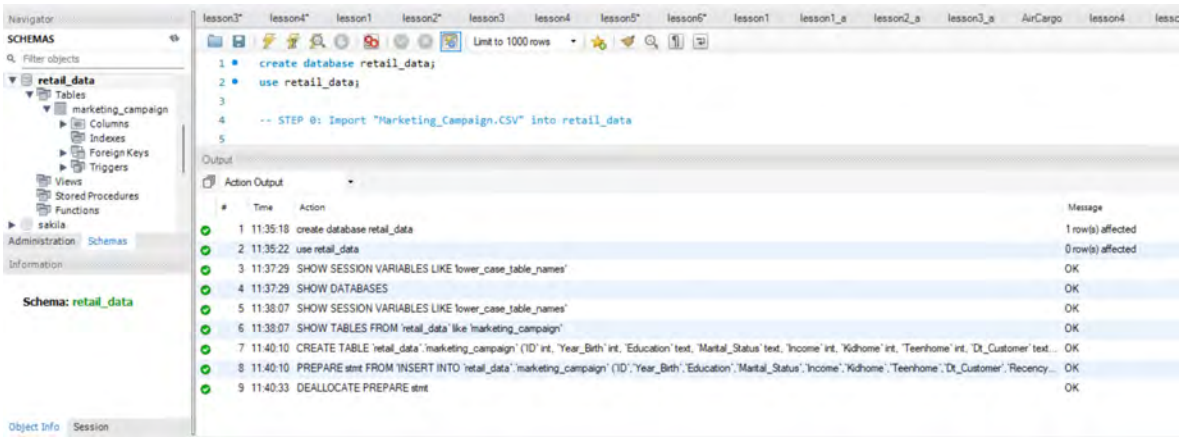


Figure 12e: Setup Steps for creating "Marketing_Campaign" SQL Database

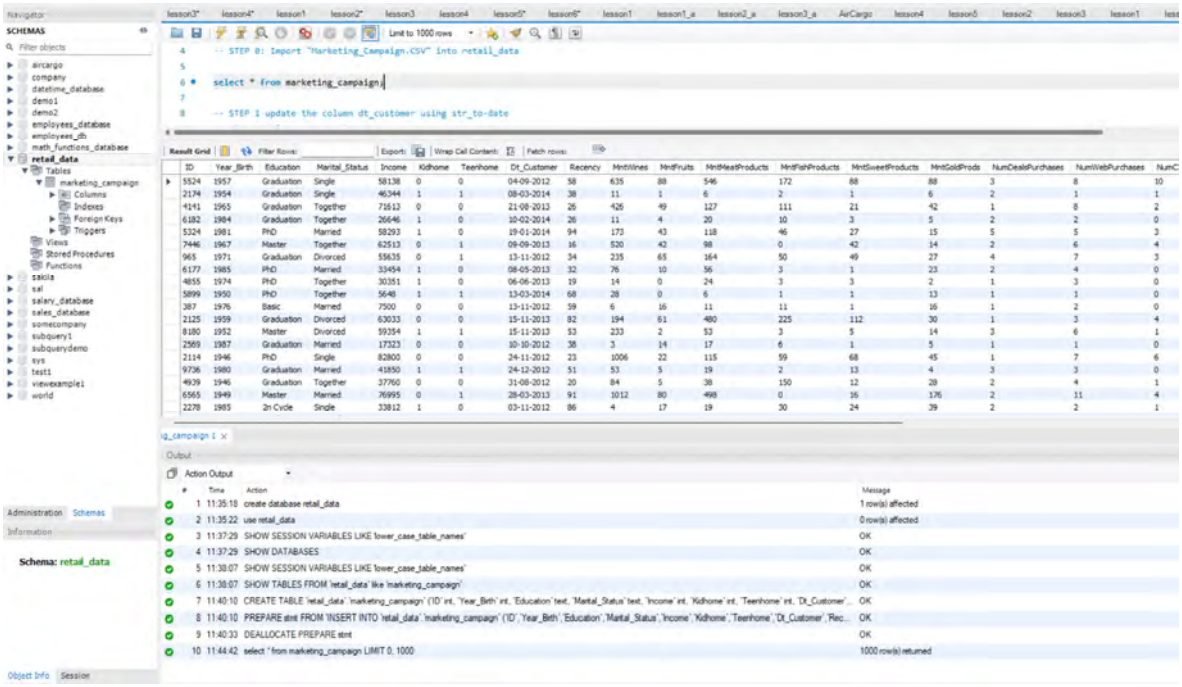


Figure 13: Results of creating "Marketing_Campaign" SQL Database

Before changing the type for "Dt_customer", had to execute the following steps in MySQL Workbench:

1. Go to Edit → Preferences
2. Choose SQL Editor
3. Uncheck "Safe Updates"
4. Click OK and Reconnect to your server
5. Select 'Reconnect to DBMS" icon on action bar

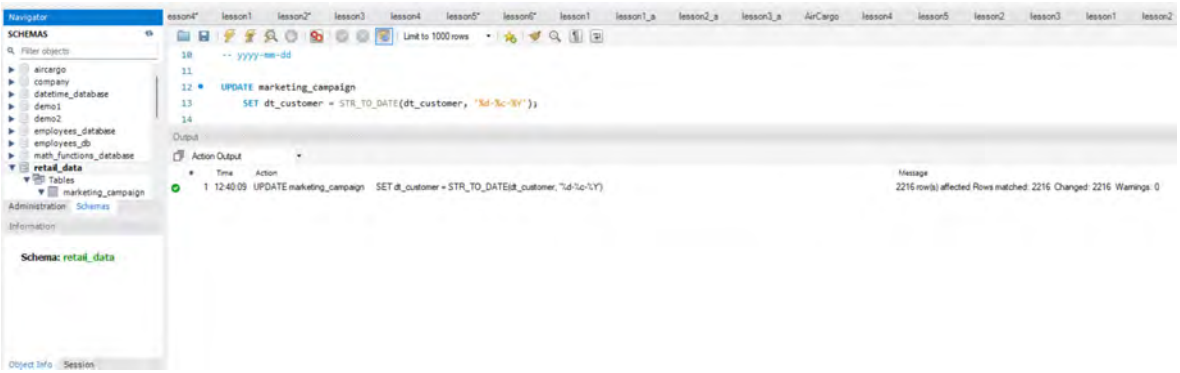


Figure 14: Results of Changing Type for "DT_customer" in SQL Database

Data Preprocessing:

Task 2: Calculate Number of Encounters

- Calculate the total number of customer encounters in the marketing campaign dataset

RESULTS

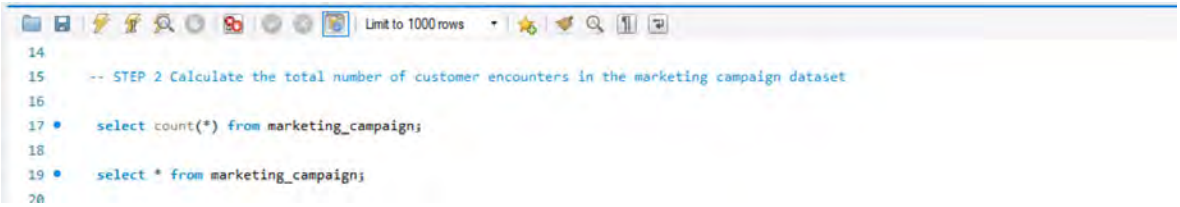


Figure 15: Setup Steps for Calculating the Total Number of Customer Encounters

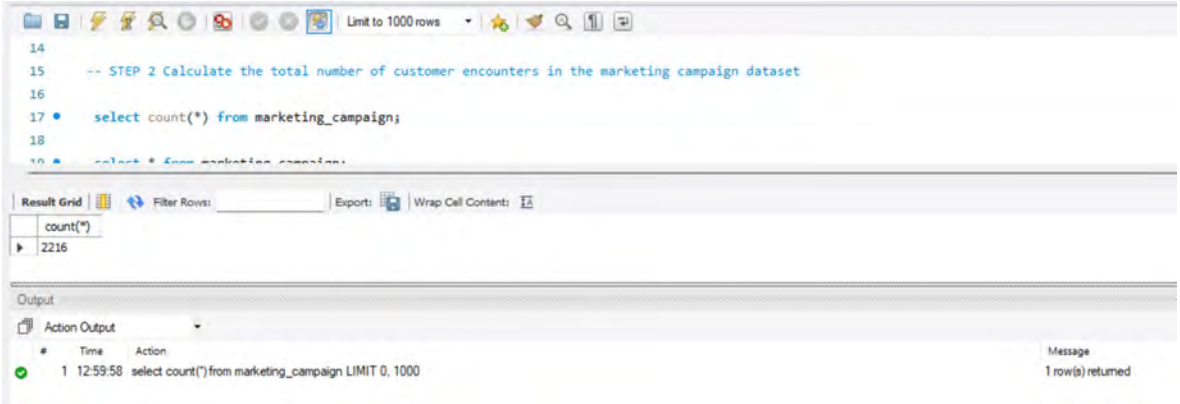


Figure 16: Results of Calculating the Total Number of Customer Encounters

Task 3: Determine the Top 10 Most Purchased

- Identify the top 10 most purchased products in the dataset, such as Wines, Meat Products, etc.

RESULTS

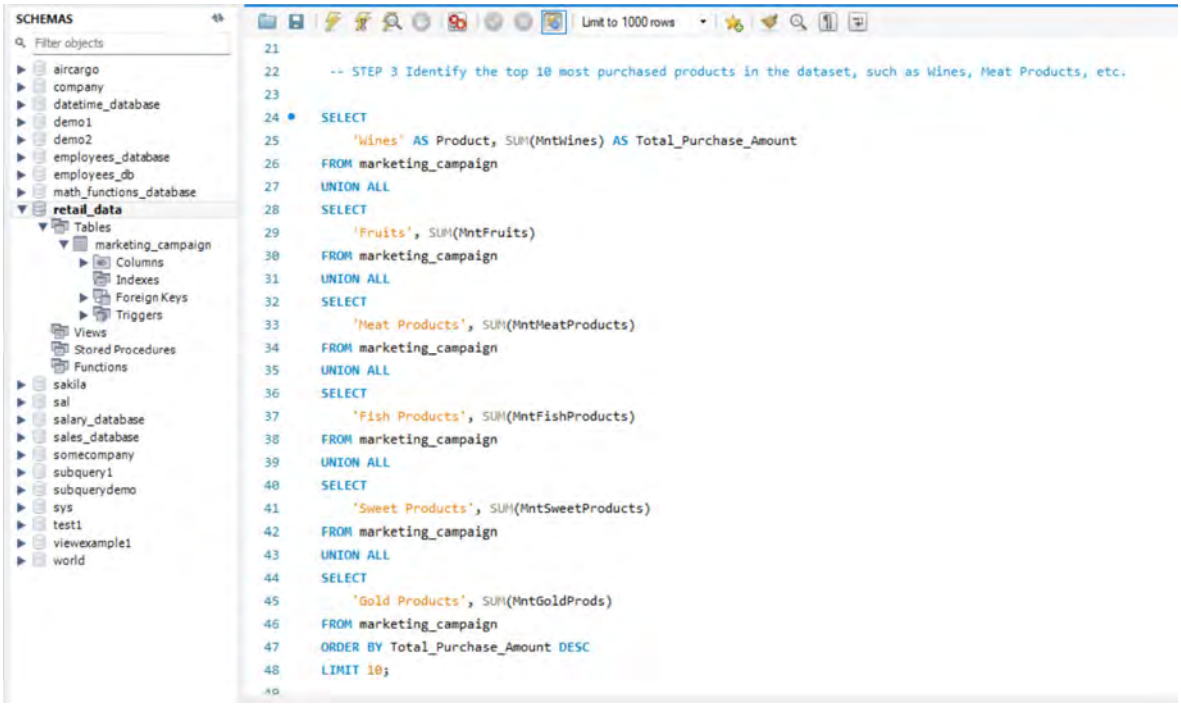


Figure 17: Setup Steps for Identifying the Top 10 Most Purchased Products

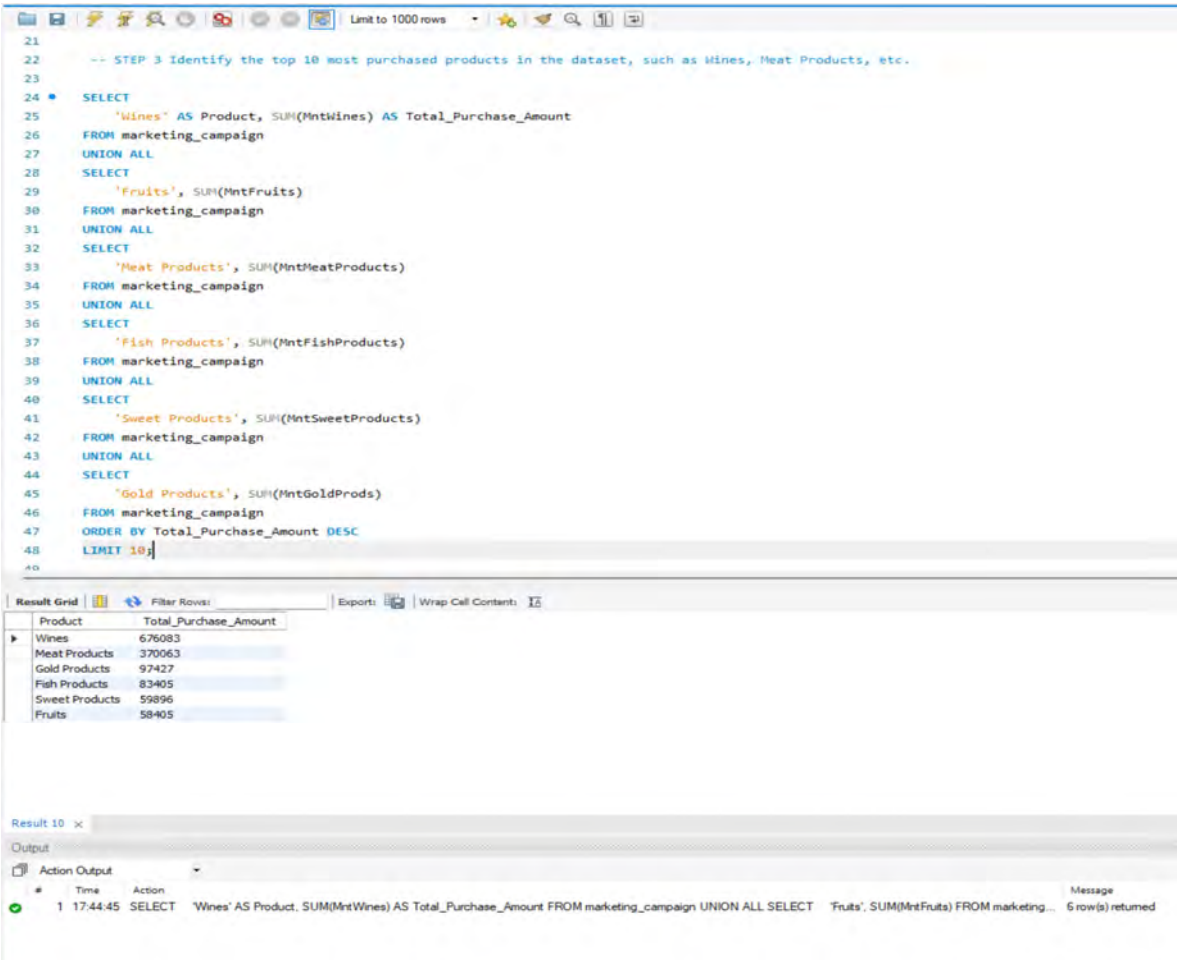


Figure 18: Results of Identifying the Top 10 Most Purchased Products

Task 4: Calculate Number of Responses

- Find the count of response values

RESULTS

```
49
50    -- STEP 4 Find the count of response values
51
52 •  SELECT
53        Response,
54        COUNT(*) AS Count_Of_Customers
55    FROM
56        marketing_campaign
57    GROUP BY
58        Response;
59
```

Figure 19: Setup Steps for Counting the Number of Responses

```
49
50    -- STEP 4 Find the count of response values
51
52 •  SELECT
53        Response,
54        COUNT(*) AS Count_Of_Customers
55    FROM
56        marketing_campaign
57    GROUP BY
58        Response;
59
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Response	Count_Of_Customers
1	333
0	1883

Result 11

Output

Action Output

#	Time	Action	Message
1	17:47:25	SELECT Response, COUNT(*) AS Count_Of_Customers FROM marketing_campaign GROUP BY Response LIMIT 0, 1000	2 row(s) returned

Figure 20: Results of Counting the Number of Responses

Task 5: Distribution of Customers

- Determine the distribution of customers based on their education level and marital status

RESULTS

```
59
60    -- STEP 5 Determine the distribution of customers based on their education level and marital status
61
62 •  select Education, Marital_status, count(*) from marketing_campaign
63    group by  Education, Marital_status;
64
```

Figure 21: Setup Steps for Determining the Distribution of Customers Based on Education and Marital Status

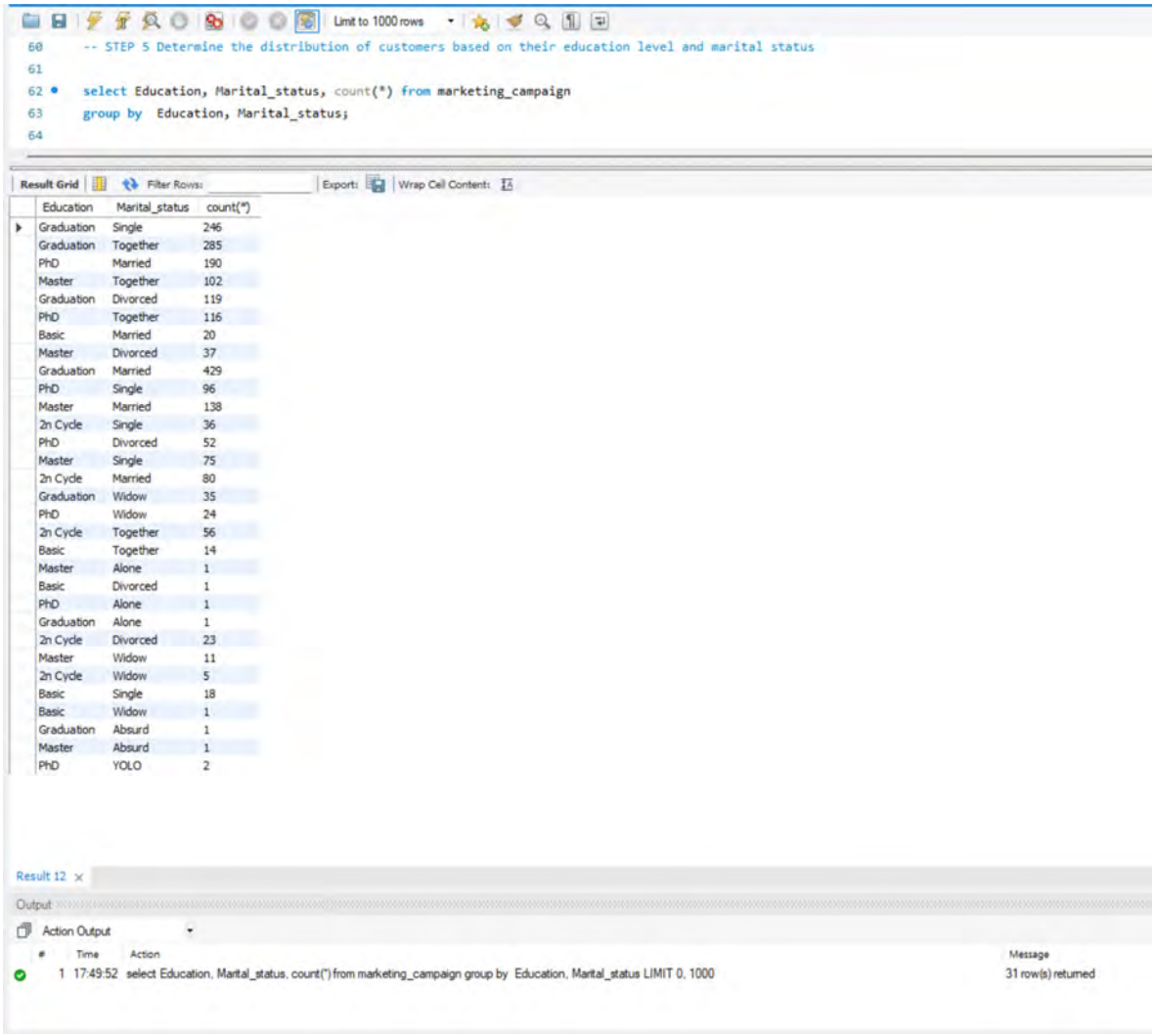


Figure 22: Results of Determining the Distribution of Customers Based on Education and Marital Status

Task 6: Average Income

- Identify the average income of customers who participated in the marketing campaign

RESULTS

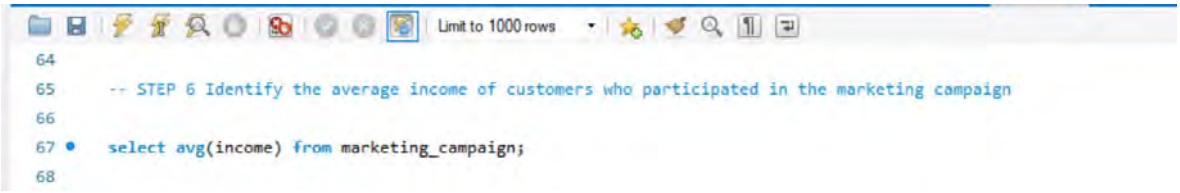


Figure 23: Setup Steps for Determining Average Income of Customers who Participated in Marketing Campaign

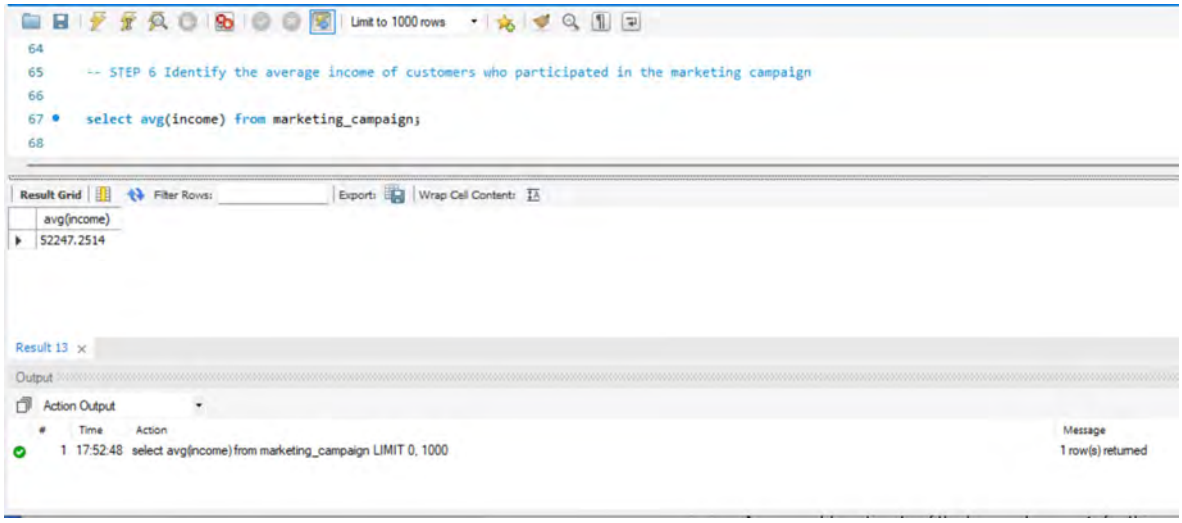


Figure 24: Results of Determining the Average Income of Customers who Participated in Marketing Campaign

Task 7: Number of Promotions Accepted

- Calculate the total number of promotions accepted by customers in each campaign

RESULTS



Figure 25: Setup Steps for Determining the Number of Promotions Accepted by Customers in Each Campaign

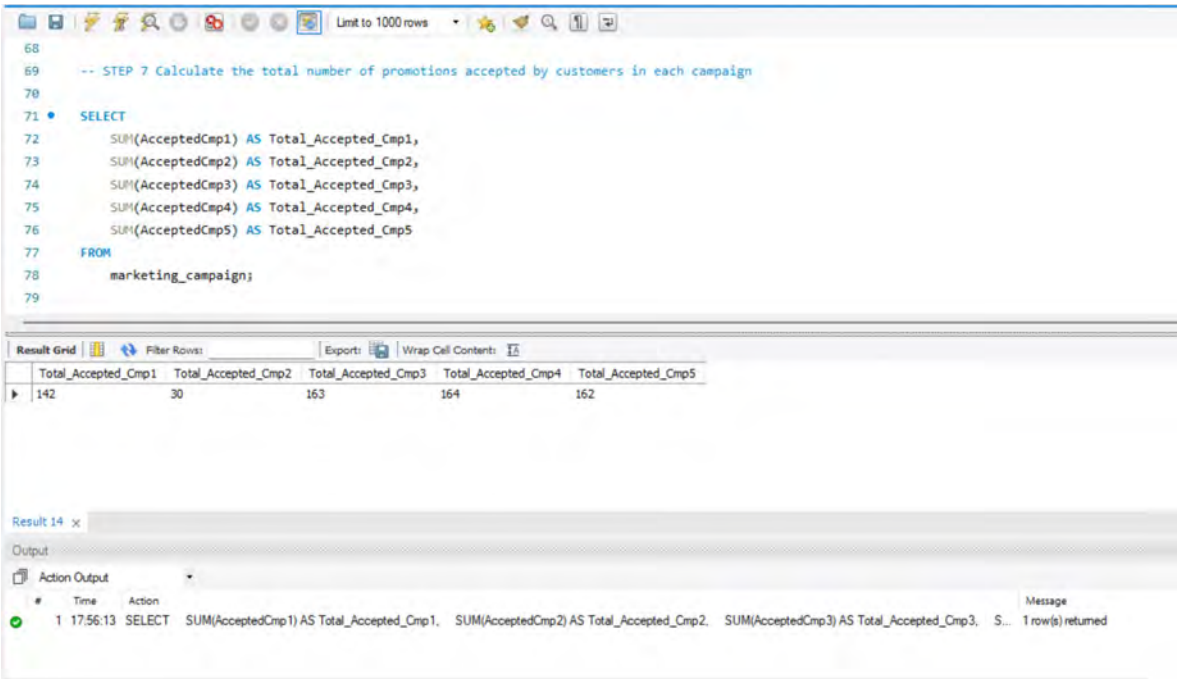


Figure 26: Results of Determining the Number of Promotions Accepted by Customers in Each Campaign

Task 8: Responses to Last Campaign

- Identify the distribution of customers' responses to the last campaign

RESULTS



Figure 27: Setup Steps for Determining the Distribution of Customers' Response to the Last Campaign

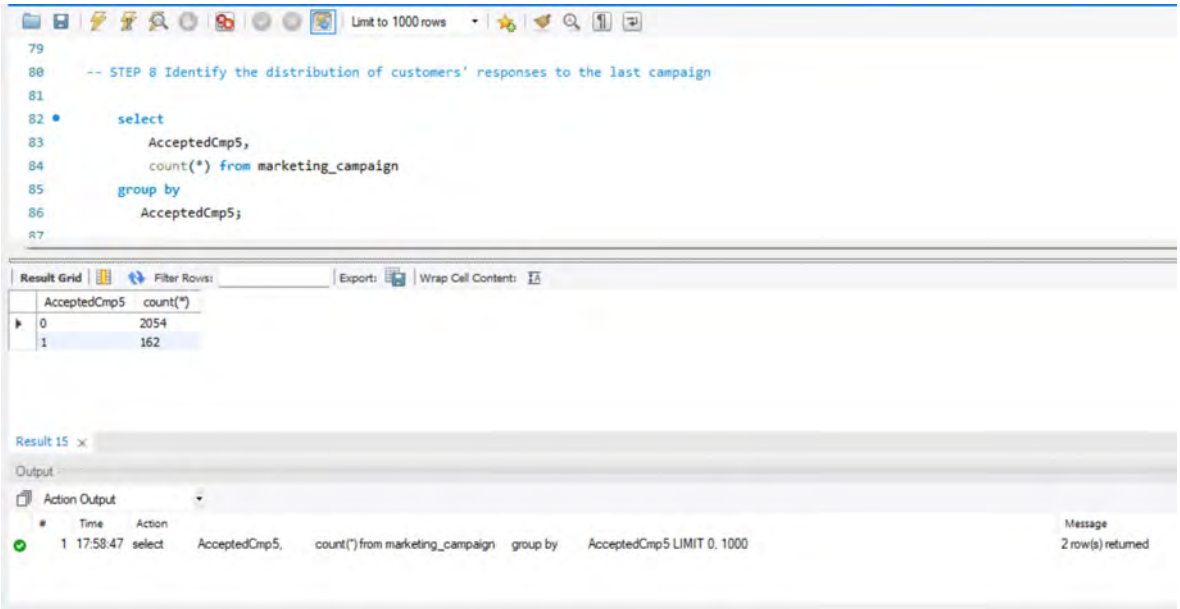


Figure 28: Results of Determining the Distribution of Customer Responses to the Last Campaign

Task 9: Youth in Households

- Calculate the average number of children and teenagers in customers' households

RESULTS

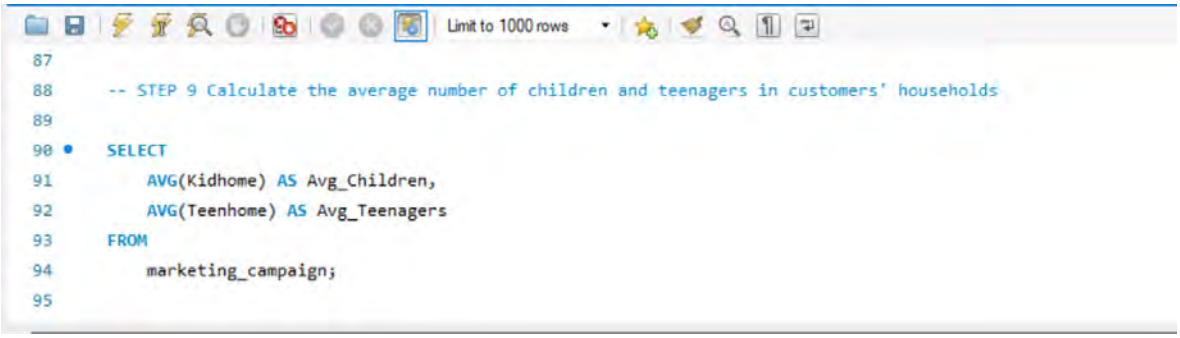


Figure 29: Setup Steps for Determining the Average Number of Children and Teenagers in Customer Households

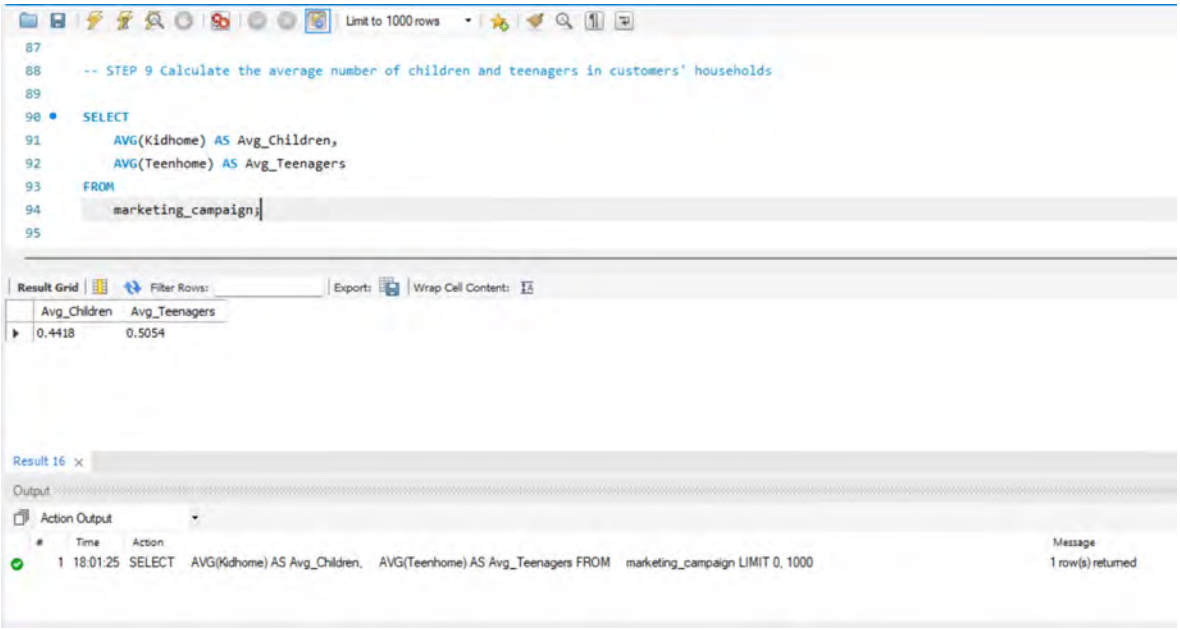


Figure 30: Results of Determining the Average Number of Children and Teenagers in Customer Households

Task 10: Calculate Age Minus Birth Year

- Create an Age column by subtracting year_birth from the current year

RESULTS

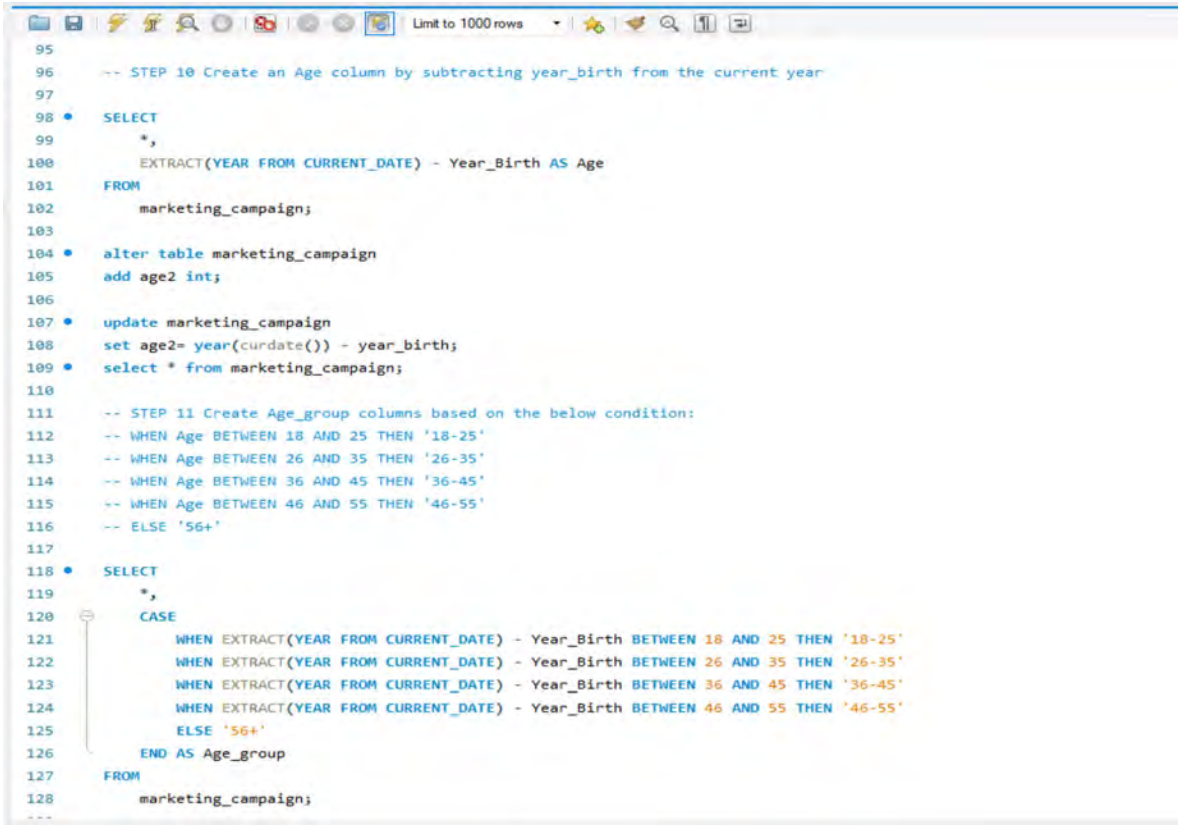


Figure 31: Setup Steps for Determining the Age by Subtracting "year_birth" from Current Year

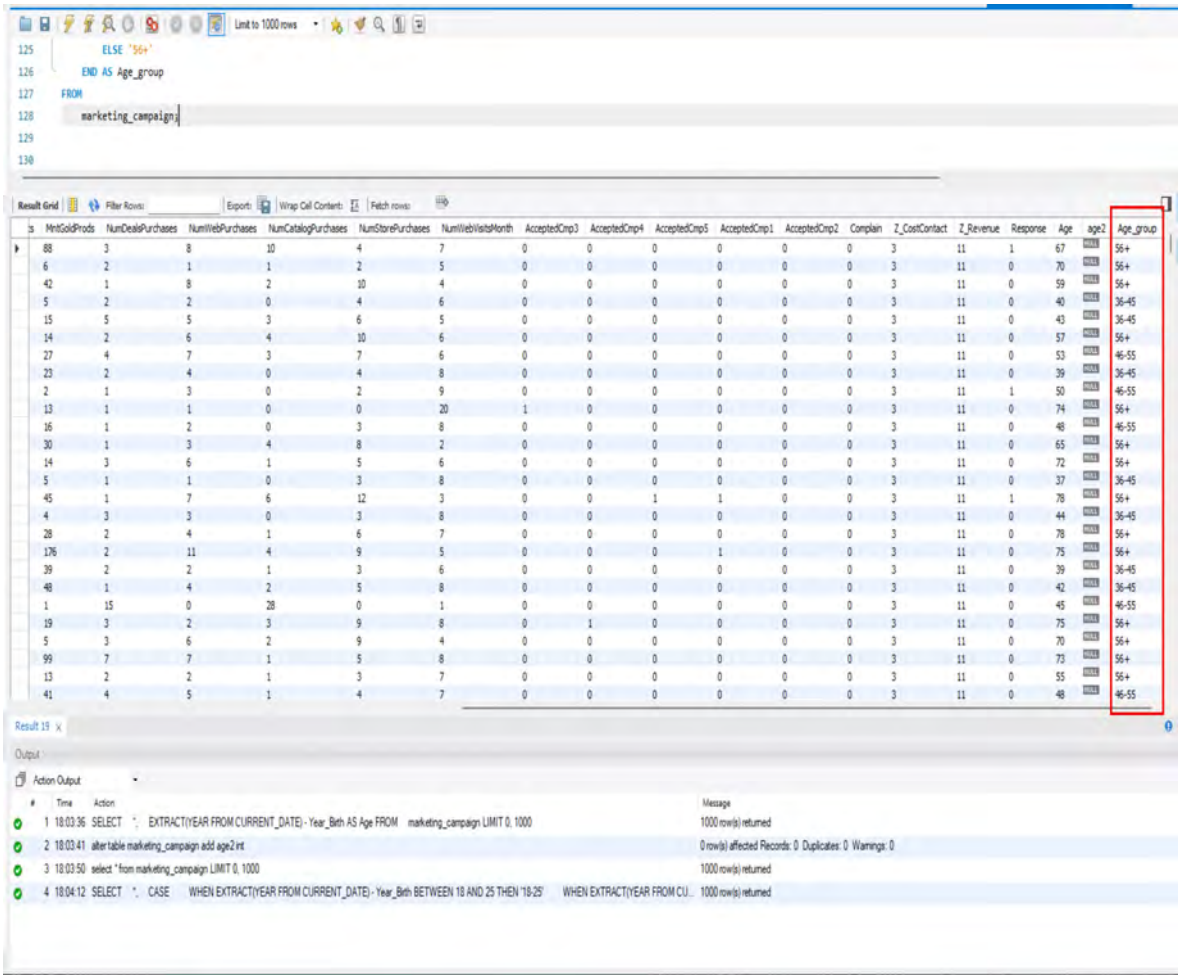
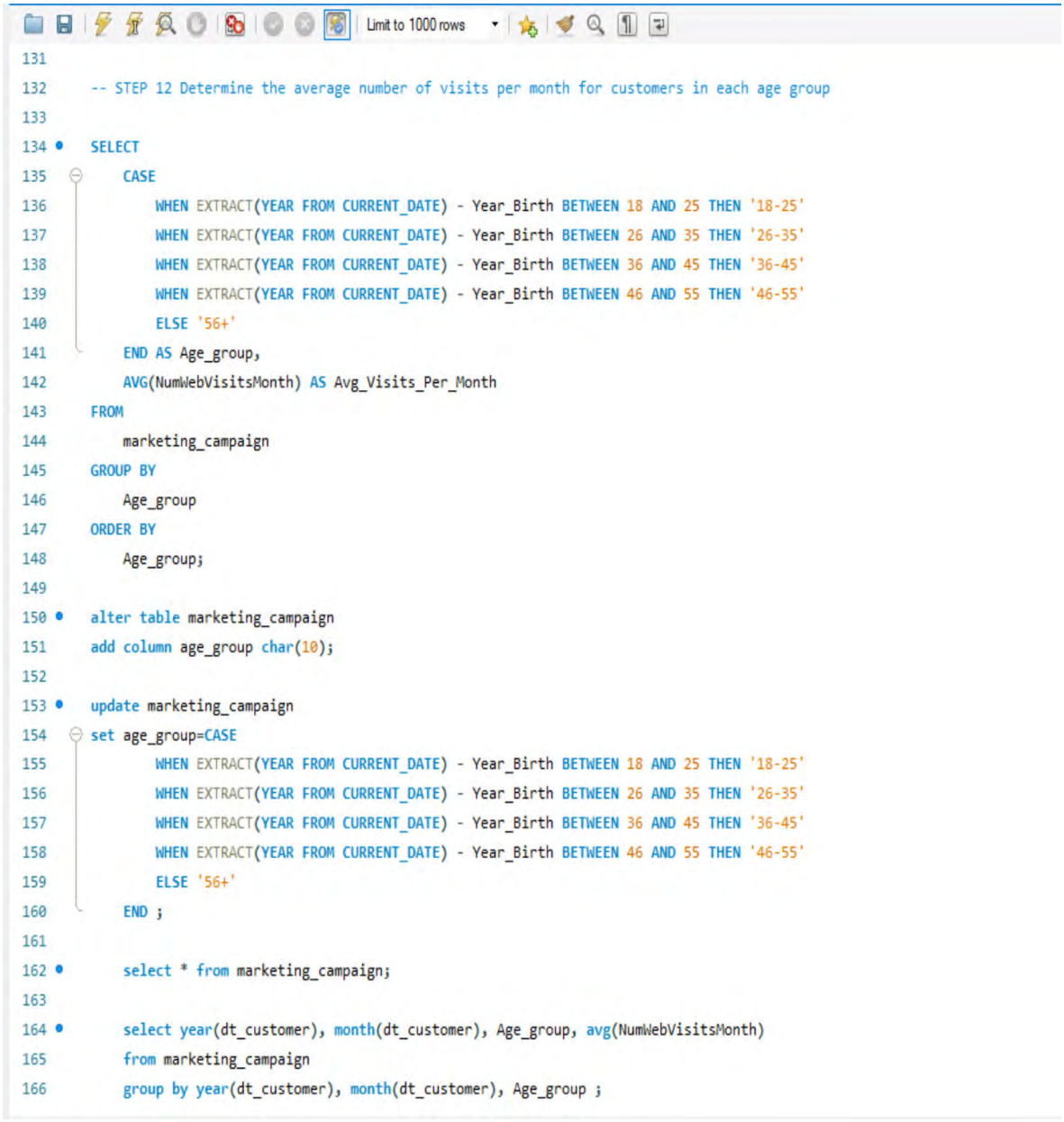


Figure 32: Results of Determining the Age by Subtracting "year_birth" from Current Year

Task 11: Age by Age Groups

- Create Age_group columns based on the below condition: WHEN Age BETWEEN 18 AND 25 THEN '18-25'
  WHEN Age BETWEEN 26 AND 35 THEN '26-35'
  WHEN Age BETWEEN 36 AND 45 THEN '36-45'
  WHEN Age BETWEEN 46 AND 55 THEN '46-55'
  ELSE '56+'

RESULTS



```
131
132     -- STEP 12 Determine the average number of visits per month for customers in each age group
133
134 •   SELECT
135         CASE
136             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 18 AND 25 THEN '18-25'
137             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 26 AND 35 THEN '26-35'
138             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 36 AND 45 THEN '36-45'
139             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 46 AND 55 THEN '46-55'
140             ELSE '56+'
141         END AS Age_group,
142         AVG(NumWebVisitsMonth) AS Avg_Visits_Per_Month
143     FROM
144         marketing_campaign
145     GROUP BY
146         Age_group
147     ORDER BY
148         Age_group;
149
150 •   alter table marketing_campaign
151     add column age_group char(10);
152
153 •   update marketing_campaign
154     set age_group=CASE
155             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 18 AND 25 THEN '18-25'
156             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 26 AND 35 THEN '26-35'
157             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 36 AND 45 THEN '36-45'
158             WHEN EXTRACT(YEAR FROM CURRENT_DATE) - Year_Birth BETWEEN 46 AND 55 THEN '46-55'
159             ELSE '56+'
160         END ;
161
162 •       select * from marketing_campaign;
163
164 •       select year(dt_customer), month(dt_customer), Age_group, avg(NumWebVisitsMonth)
165         from marketing_campaign
166         group by year(dt_customer), month(dt_customer), Age_group ;
```

Figure 33: Setup Steps for Creating “Age_group” based on Age Ranges

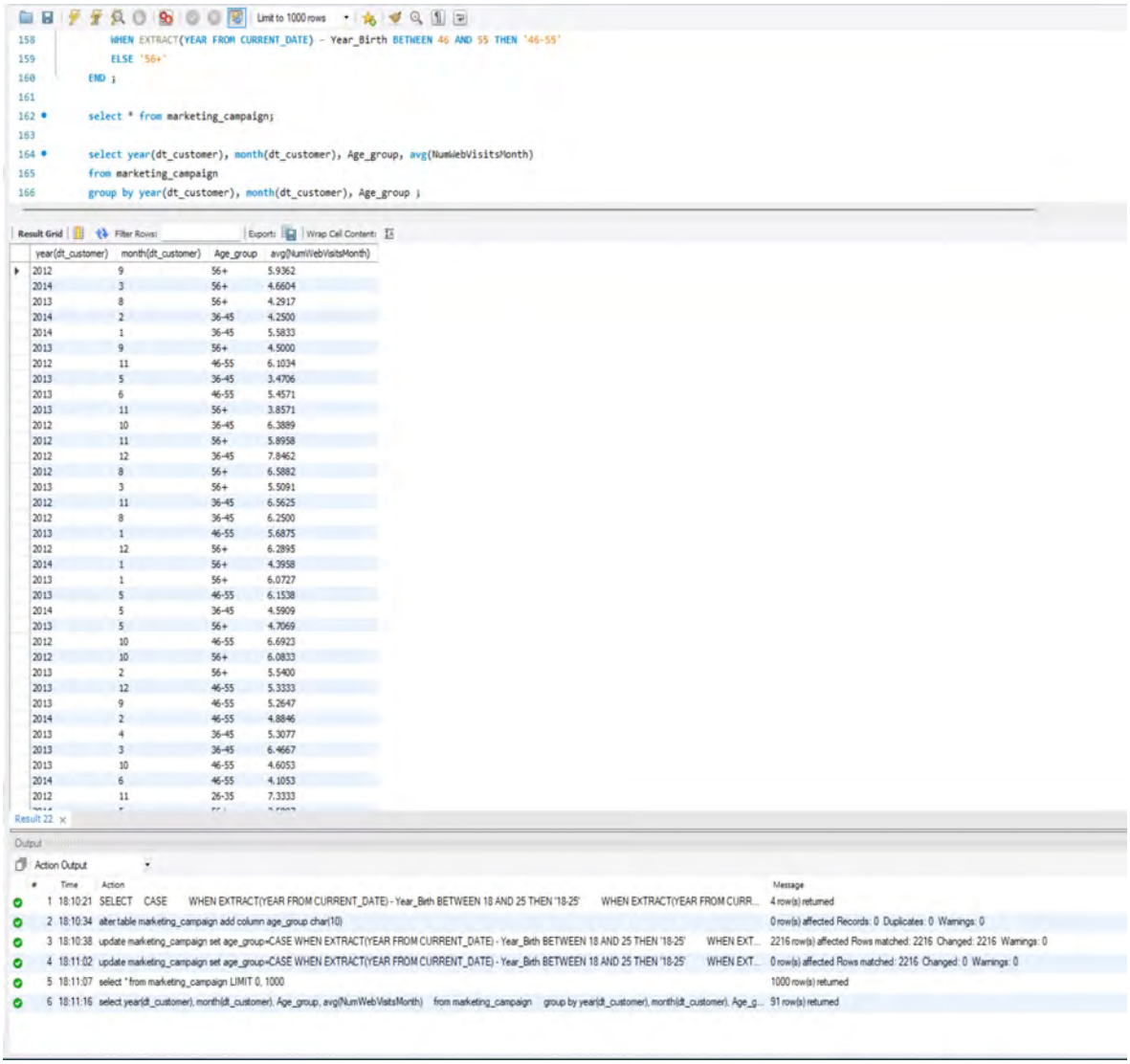


Figure 34: Results of Creating "Age_group" based on Age Ranges

Task 12: Average Visits per Month

- Determine the average number of visits per month for customers in each age group

RESULTS

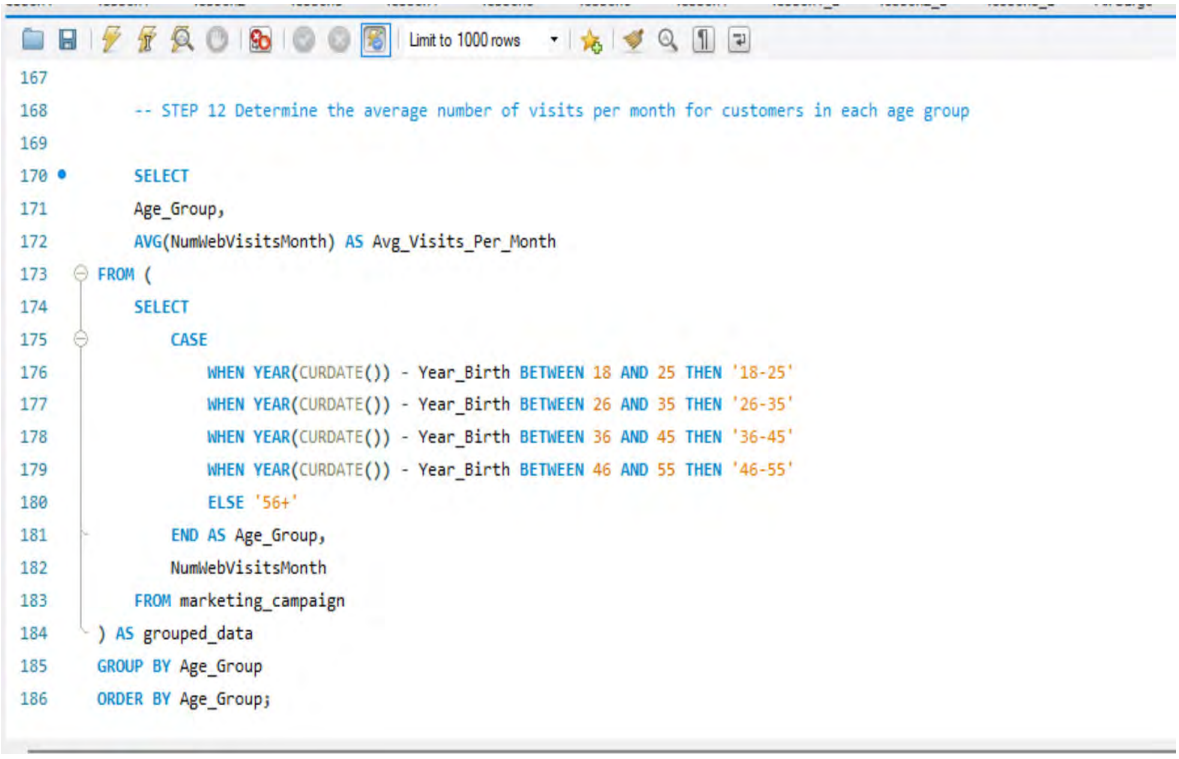


Figure 35: Setup Steps for Determining Average Visits per Month by Customers in Each Age Group

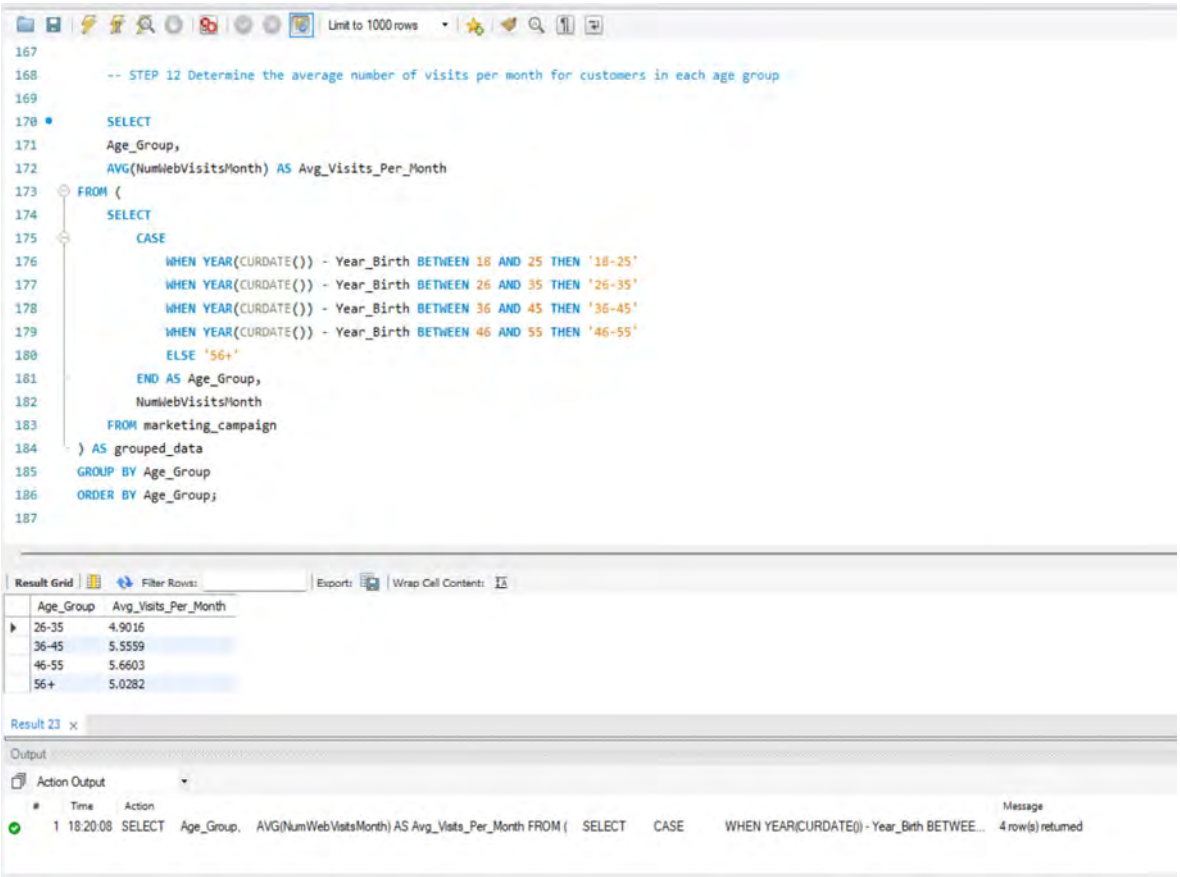


Figure 36: Results of Determining Average Visits per Month by Customers in Each Age Group

Python Tasks:

3. Exploratory Data Analysis

Task 1: General Data Overview:

a. Check the first few rows of the dataset to understand its structure

b. Check the data types of each column

c. Check for any missing values in the dataset

RESULTS

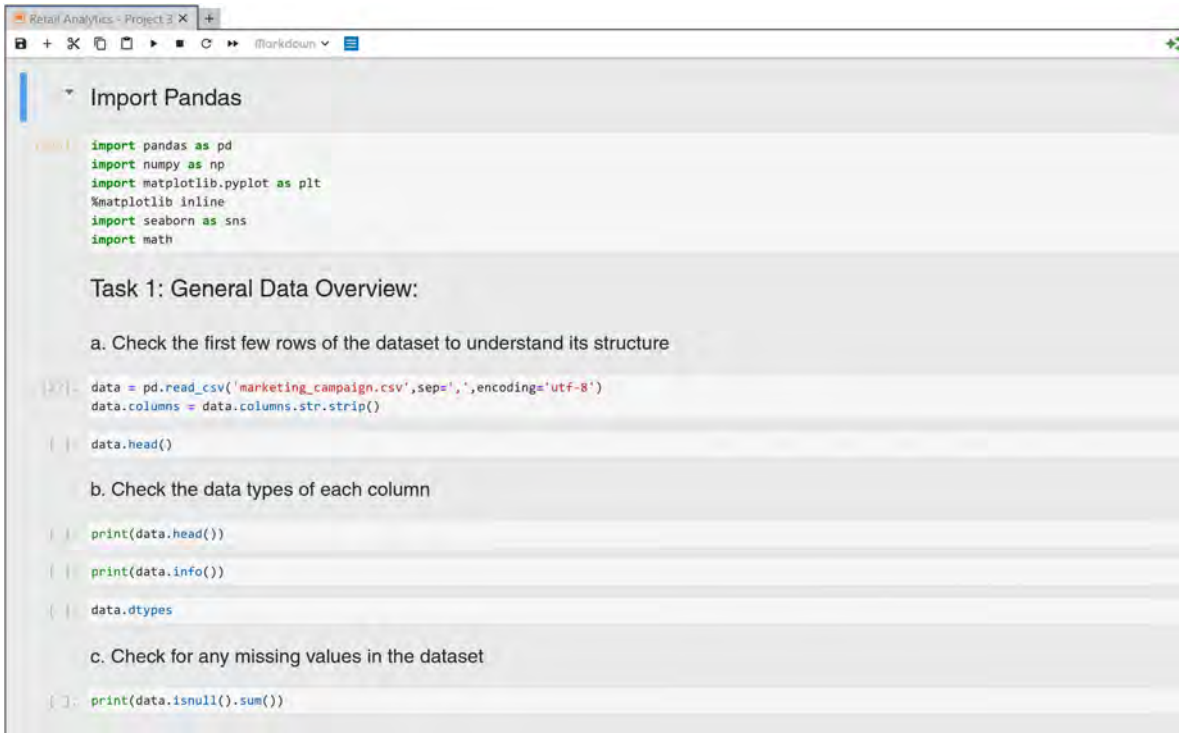


Figure 37: Setup Steps for Performing Exploratory Data Analysis

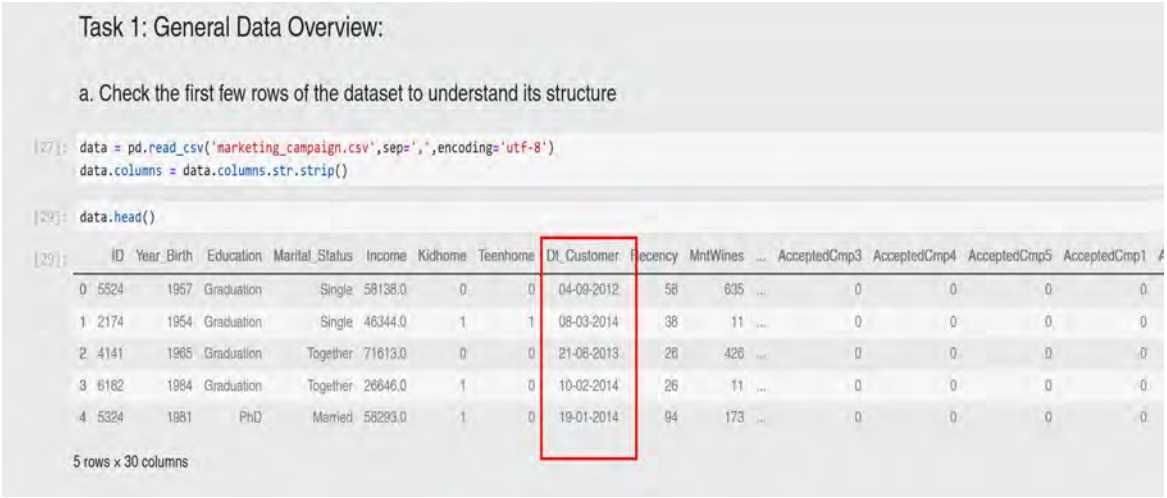
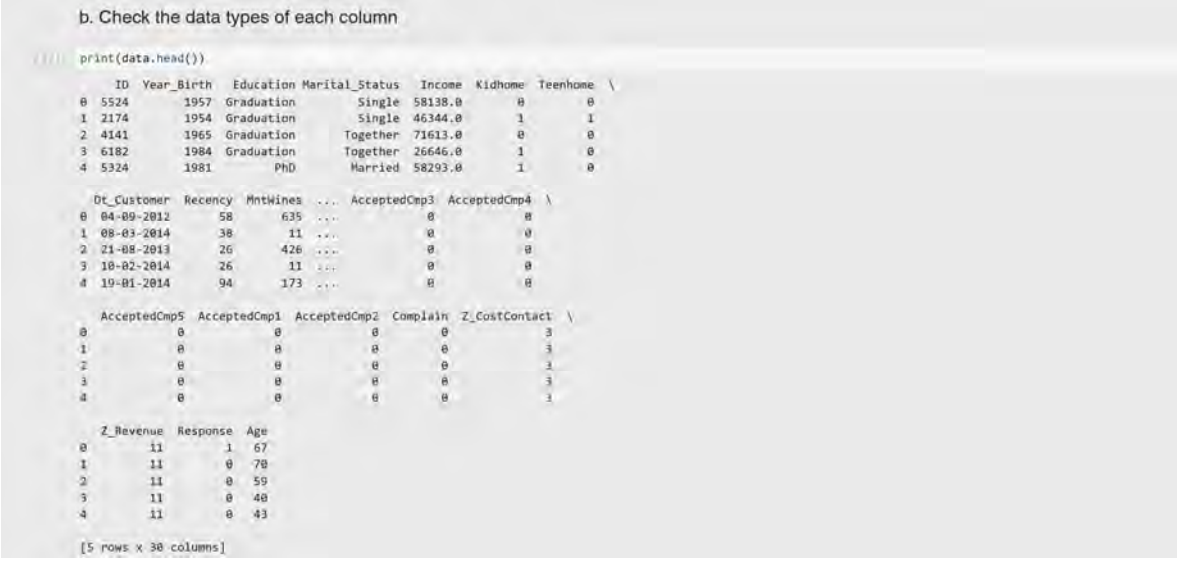


Figure 38a: Results of Performing Exploratory Data Analysis



b. Check the data types of each column

```
print(data.head())
```

```
     ID  Year_Birth   Education Marital_Status   Income  Kidhome  Teenhome  \
0  5524        1957  Graduation         Single  58138.0        0         0
1  2174        1954  Graduation         Single  46344.0        1         1
2  4141        1965  Graduation       Together  71613.0        0         0
3  6182        1984  Graduation       Together  26646.0        1         0
4  5324        1981         PhD        Married  58293.0        1         0

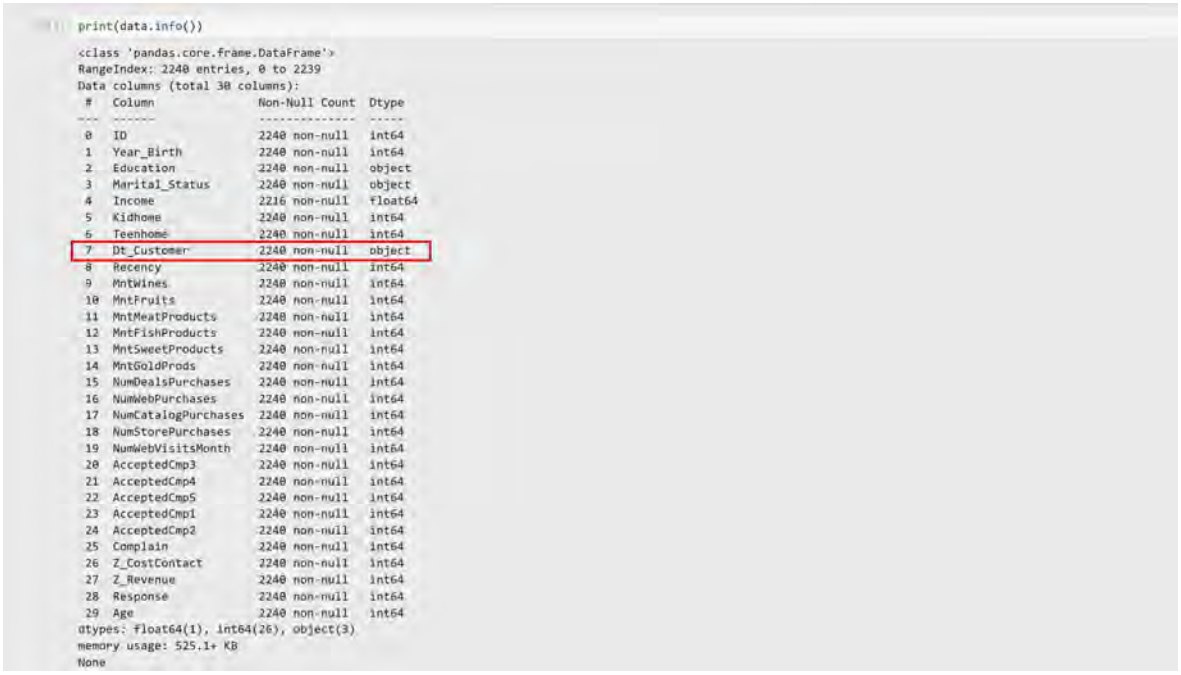
  Dt_Customer  Recency  MntWines  ...  AcceptedCmp3  AcceptedCmp4  \
0  04-09-2012       58       635  ...             0             0
1  08-03-2014       38        11  ...             0             0
2  21-08-2013       26       426  ...             0             0
3  10-02-2014       26        11  ...             0             0
4  19-01-2014       94       173  ...             0             0

   AcceptedCmp5  AcceptedCmp1  AcceptedCmp2  Complain  Z_CostContact  \
0             0             0             0         0              3
1             0             0             0         0              3
2             0             0             0         0              3
3             0             0             0         0              3
4             0             0             0         0              3

   Z_Revenue  Response  Age
0         11         1   67
1         11         0   70
2         11         0   59
3         11         0   40
4         11         0   43

[5 rows x 30 columns]
```

Figure 38b: Results of Performing Exploratory Data Analysis



```
print(data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 30 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   ID                   2240 non-null   int64
 1   Year_Birth           2240 non-null   int64
 2   Education            2240 non-null   object
 3   Marital_Status       2240 non-null   object
 4   Income               2216 non-null   float64
 5   Kidhome              2240 non-null   int64
 6   Teenhome             2240 non-null   int64
 7   Dt_Customer          2240 non-null   object
 8   Recency              2240 non-null   int64
 9   MntWines             2240 non-null   int64
 10  MntFruits            2240 non-null   int64
 11  MntMeatProducts      2240 non-null   int64
 12  MntFishProducts      2240 non-null   int64
 13  MntSweetProducts     2240 non-null   int64
 14  MntGoldProds         2240 non-null   int64
 15  NumDealsPurchases    2240 non-null   int64
 16  NumWebPurchases      2240 non-null   int64
 17  NumCatalogPurchases  2240 non-null   int64
 18  NumStorePurchases    2240 non-null   int64
 19  NumWebVisitsMonth    2240 non-null   int64
 20  AcceptedCmp3         2240 non-null   int64
 21  AcceptedCmp4         2240 non-null   int64
 22  AcceptedCmp5         2240 non-null   int64
 23  AcceptedCmp1         2240 non-null   int64
 24  AcceptedCmp2         2240 non-null   int64
 25  Complain             2240 non-null   int64
 26  Z_CostContact        2240 non-null   int64
 27  Z_Revenue            2240 non-null   int64
 28  Response             2240 non-null   int64
 29  Age                  2240 non-null   int64
dtypes: float64(1), int64(26), object(3)
memory usage: 525.1+ KB
None
```

Figure 38c: Results of Performing Exploratory Data Analysis



```
data.dtypes

ID                     int64
Year_Birth             int64
Education             object
Marital_Status        object
Income               float64
Kidhome                int64
Teenhome               int64
Dt_Customer           object
Recency                int64
MntWines               int64
MntFruits              int64
MntMeatProducts        int64
MntFishProducts        int64
MntSweetProducts       int64
MntGoldProds           int64
NumDealsPurchases      int64
NumWebPurchases        int64
NumCatalogPurchases    int64
NumStorePurchases      int64
NumWebVisitsMonth      int64
AcceptedCmp3           int64
AcceptedCmp4           int64
AcceptedCmp5           int64
AcceptedCmp1           int64
AcceptedCmp2           int64
Complain               int64
Z_CostContact          int64
Z_Revenue              int64
Response               int64
Age                    int64
dtype: object
```

Figure 38d: Results of Performing Exploratory Data Analysis



```
c. Check for any missing values in the dataset

print(data.isnull().sum())

ID                     0
Year_Birth             0
Education              0
Marital_Status         0
Income                24
Kidhome                0
Teenhome               0
Dt_Customer            0
Recency                0
MntWines               0
MntFruits              0
MntMeatProducts        0
MntFishProducts        0
MntSweetProducts       0
MntGoldProds           0
NumDealsPurchases      0
NumWebPurchases        0
NumCatalogPurchases    0
NumStorePurchases      0
NumWebVisitsMonth      0
AcceptedCmp3           0
AcceptedCmp4           0
AcceptedCmp5           0
AcceptedCmp1           0
AcceptedCmp2           0
Complain               0
Z_CostContact          0
Z_Revenue              0
Response               0
Age                    0
dtype: int64
```

Figure 38e: Results of Performing Exploratory Data Analysis

Task 2: Descriptive Statistics:

a. Compute summary statistics for numerical columns (mean, median, min, max, and standard deviation)
b. Explore the distribution of numerical variables using histograms or box plots

RESULTS

Task 2: Descriptive Statistics:

a. Compute summary statistics for numerical columns (mean, median, min,

```
# Basic summary statistics: mean, std, min, 25%, 50% (median), 75%, max
summary_stats = data.describe()

# Display the result
summary_stats.describe()
```

b. Explore the distribution of numerical variables using histograms or box plots

```
data.columns
```

```
numerical_col=['Year_Birth', 'Income', 'Kidhome',
       'Teenhome', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases', 'Age']
```

```
#  Plot a Histogram

plt.figure(figsize=(12,18))
for i , col in enumerate(numerical_col):
    plt.subplot(5,4,i+1)
    sns.displot(data[col], bins=30)
    plt.title(f'Histogram - {col}')
    plt.xlabel(col)
    plt.ylabel('Count')
plt.tight_layout
plt.show()
```

Figure 39: Setup Steps for Performing Descriptive Analysis

a. Compute summary statistics for numerical columns (mean, median, min,

```
# Basic summary statistics: mean, std, min, 25%, 50% (median), 75%, max
summary_stats = data.describe()

# Display the result
summary_stats.describe()
```

	ID	Year_Birth	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	...	AcceptedCmp3	AcceptedCm
count	8.000000	8.000000	8.000000	8.000000	8.000000	8.000000	8.000000	8.000000	8.000000	8.000000	...	8.000000	8.000
mean	4873.040252	1751.973734	112904.853502	280.497824	280.506349	320.508978	634.379138	318.384458	584.083172	332.019303	...	280.166573	280.167
std	3603.929757	710.454501	225013.172775	791.756794	791.755289	776.190009	803.364616	779.164335	678.422980	775.502761	...	791.892363	791.892
min	0.000000	11.984069	1730.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	...	0.000000	0.000
25%	2681.187500	1942.500000	19433.807496	0.000000	0.000000	27.721840	138.062500	6.250000	54.250000	9.750000	...	0.000000	0.000
50%	4352.581090	1969.402902	43342.250000	0.491297	0.525394	49.054688	320.266553	29.651116	196.332686	43.782723	...	0.036384	0.037
75%	6301.057366	1981.750000	56315.938515	1.250000	1.250000	80.250000	751.437500	79.580075	805.250000	105.721735	...	0.444860	0.447
max	11191.000000	2240.000000	666666.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	...	2240.000000	2240.000

8 rows × 27 columns

b. Explore the distribution of numerical variables using histograms or box plots

```
data.columns
```

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response',
       'Age'],
      dtype='object')
```

```
numerical_col=['Year_Birth', 'Income', 'Kidhome',
       'Teenhome', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases', 'Age']
```

Figure 40a: Results of Performing Descriptive Analysis

```
[52]: # Plot a Histogram

plt.figure(figsize=(12,18))
for i , col in enumerate(numerical_col):
    plt.subplot(5,4,i+1)
    sns.displot(data[col], bins=30)
    plt.title(f'Histogram - {col}')
    plt.xlabel(col)
    plt.ylabel('Count')
plt.tight_layout
plt.show()
```

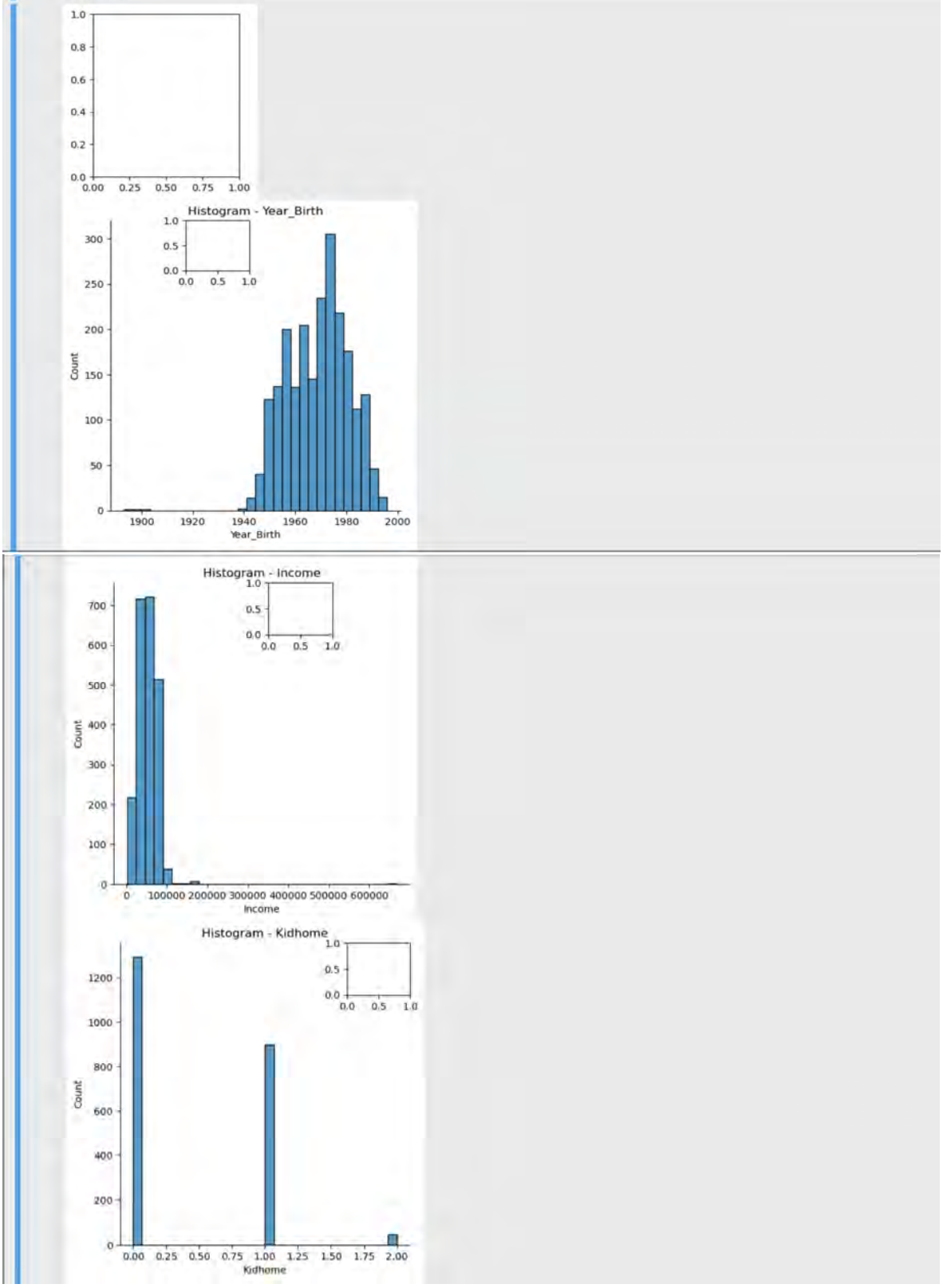


Figure 40b: Results of Performing Descriptive Analysis

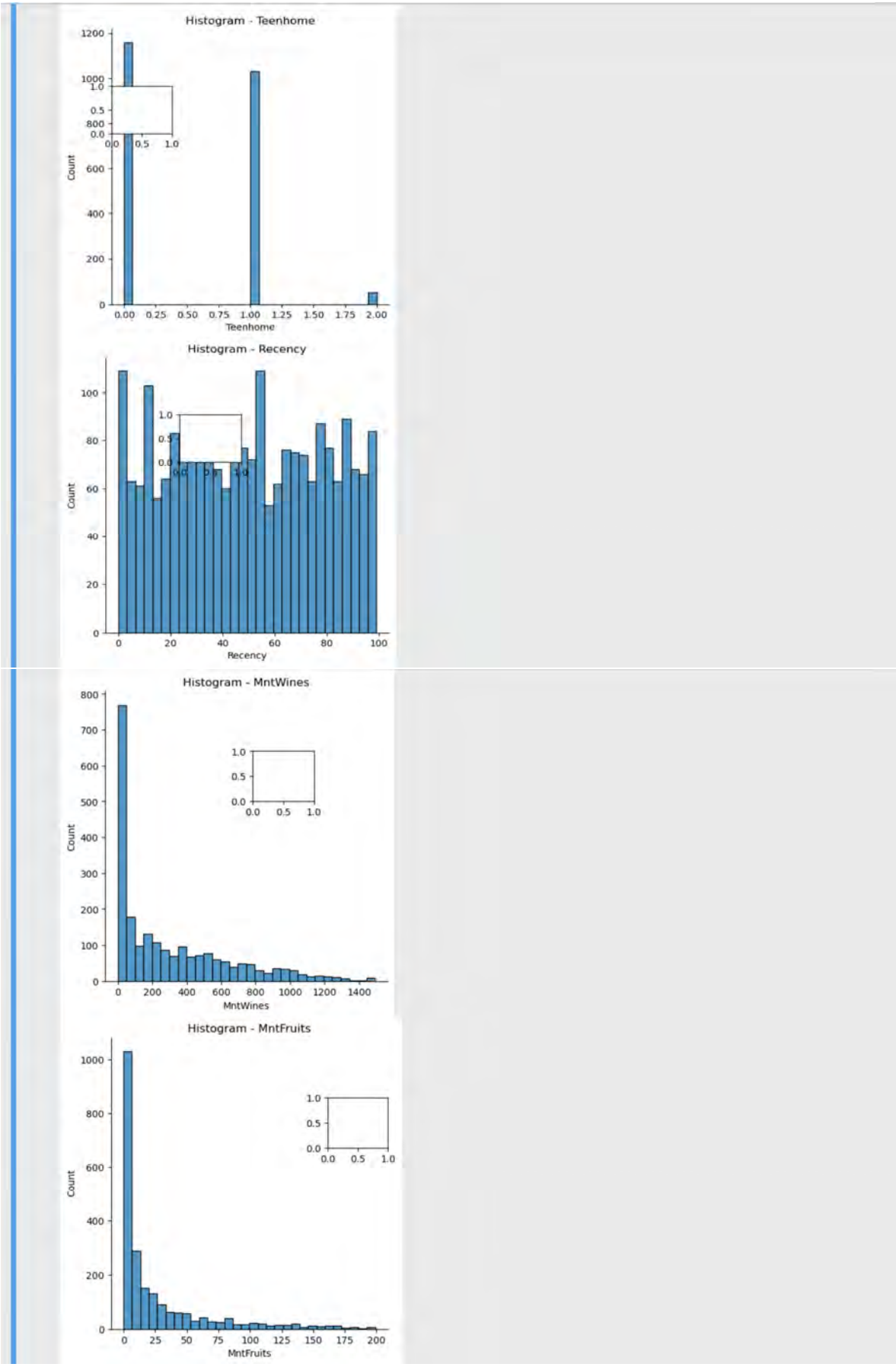


Figure 40c: Results of Performing Descriptive Analysis

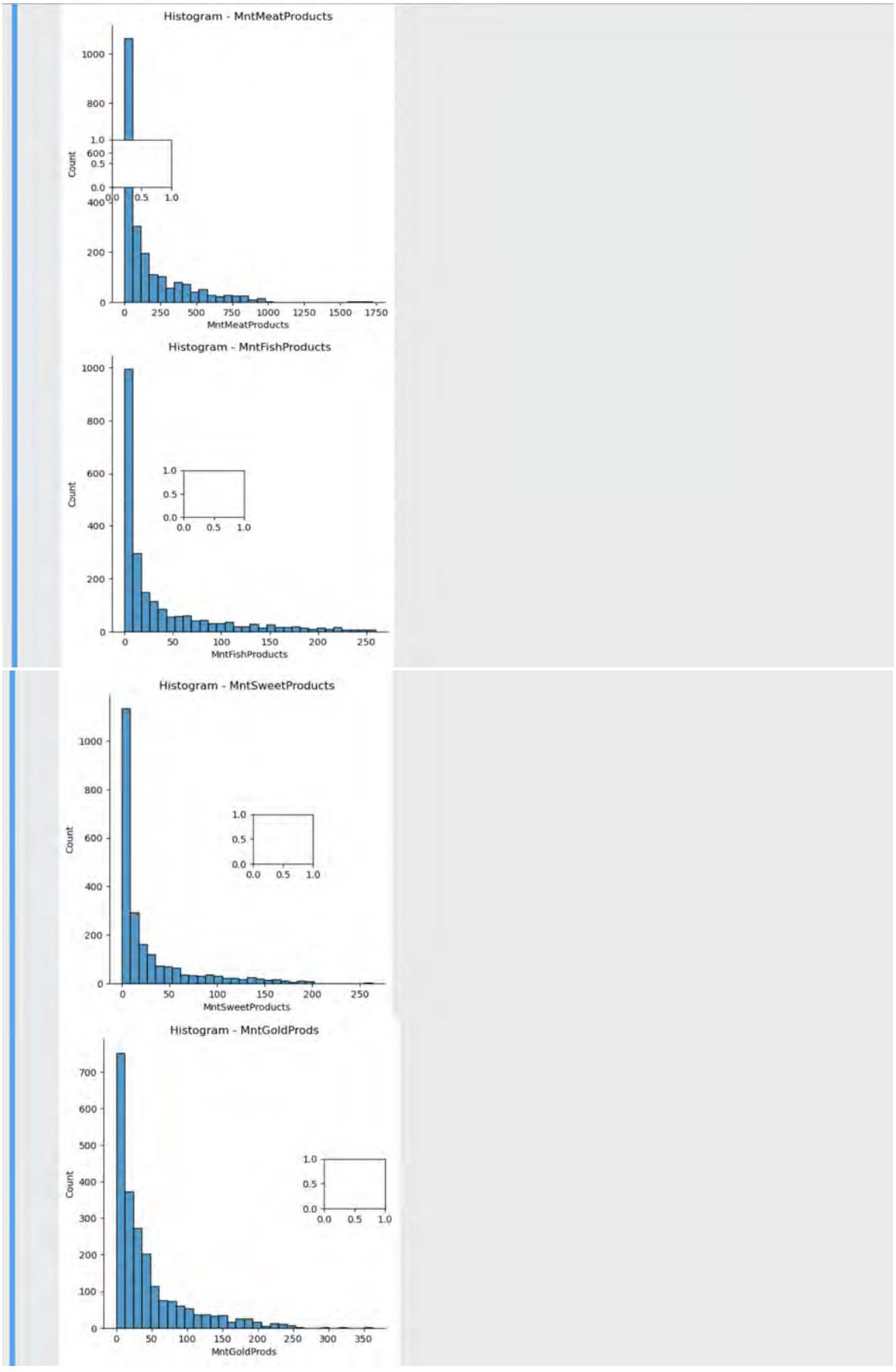


Figure 40d: Results of Performing Descriptive Analysis

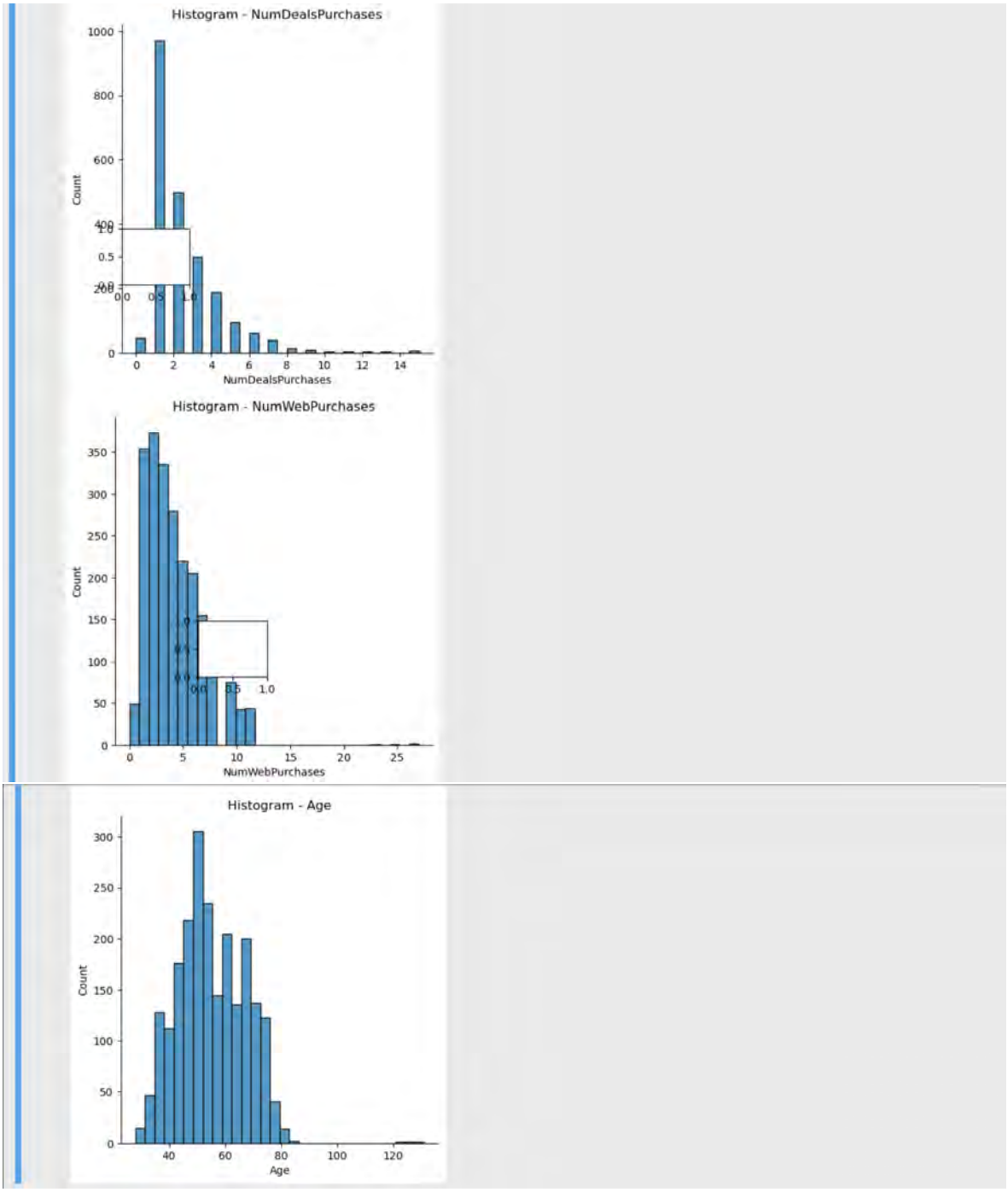


Figure 40e: Results of Performing Descriptive Analysis

Task 3: Univariate Analysis:

a. Explore the distribution of each numerical variable using histograms or kernel density plots
b. Explore the distribution of each categorical variable using bar plots or pie charts
c. Identify outliers in numerical variables using box plots or scatter plots

RESULTS

Task 3: Univariate Analysis:

a. Explore the distribution of each numerical variable using histograms or kernal density plots

```
data.columns
```

```
categorical_col=['Education', 'Marital_Status', 'Response',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain',]
```

```
# Plot a Histogram

plt.figure(figsize=(12,18))
for i , col in enumerate(categorical_col):
    plt.subplot(5,4,i+1)
    data[col].value_counts().plot(kind='bar')
    plt.title(f'Barplot - {col}')
    plt.xlabel(col)
    plt.ylabel('Count')
plt.tight_layout
plt.show()
```

b. Explore the distribution of each categorical variable using bar plots or pie chart

```
numerical_col=['Year_Birth', 'Income', 'Kidhome',
       'Teenhome', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases', 'Age']
```

```
# Plot box plots for each numerical column
plt.figure(figsize=(16, 20))
for i, col in enumerate(numerical_col, 1):
    plt.subplot((len(numerical_col) + 2) // 3, 3, i)
    sns.boxplot(x=data[col])
    plt.title(f'Boxplot of {col}')
    plt.tight_layout()

plt.show()
```

Figure 41a: Setup Steps for Performing Univariate Analysis

c. Identify outliers in numerical variables using box plots or scatter plots

```
plt.figure(figsize=(6, 5))
sns.scatterplot(data=data, x='Income', y='MntWines')
plt.title('Income vs Wine Purchases')
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(16, 20))
plot_num = 1

for col_x in ['Income', 'Age']:  # Independent variables
    for col_y in numerical_col:
        if col_y.startswith('Mnt') and col_y != col_x:
            plt.subplot(4, 2, plot_num)
            sns.scatterplot(data=data, x=col_x, y=col_y)
            plt.title(f'{col_x} vs {col_y}')
            plot_num += 1
            if plot_num > 8:  # limit to first 8 plots
                break
    if plot_num > 8:
        break

plt.tight_layout()
plt.show()
```

Figure 41b: Setup Steps for Performing Univariate Analysis

a. Explore the distribution of each numerical variable using histograms or kernal density plots

```
data.columns
```

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response',
       'Age'],
      dtype='object')
```

```
categorical_col=['Education', 'Marital_Status', 'Response',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain',]
```

Figure 42a: Results of Performing Univariate Analysis with Histogram Plots

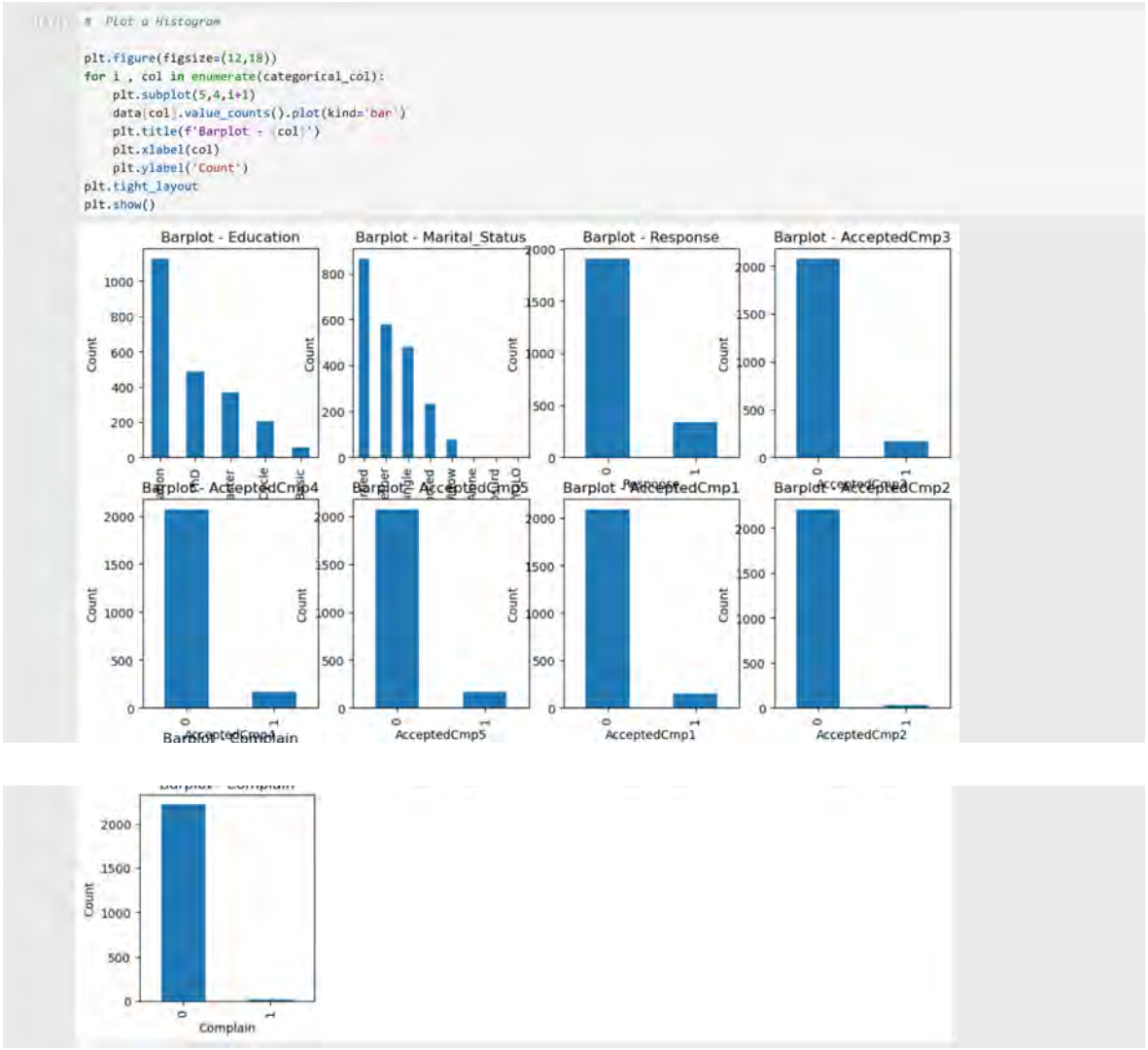


Figure 42b: Results of Performing Univariate Analysis with Histogram Plots

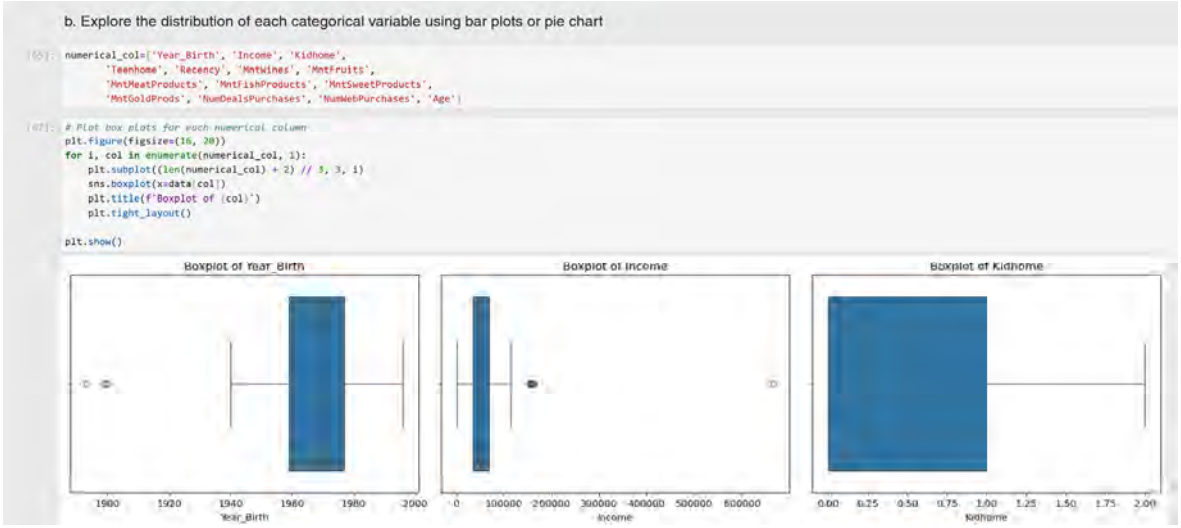


Figure 43a: Results of Performing Univariate Analysis with Box Plots

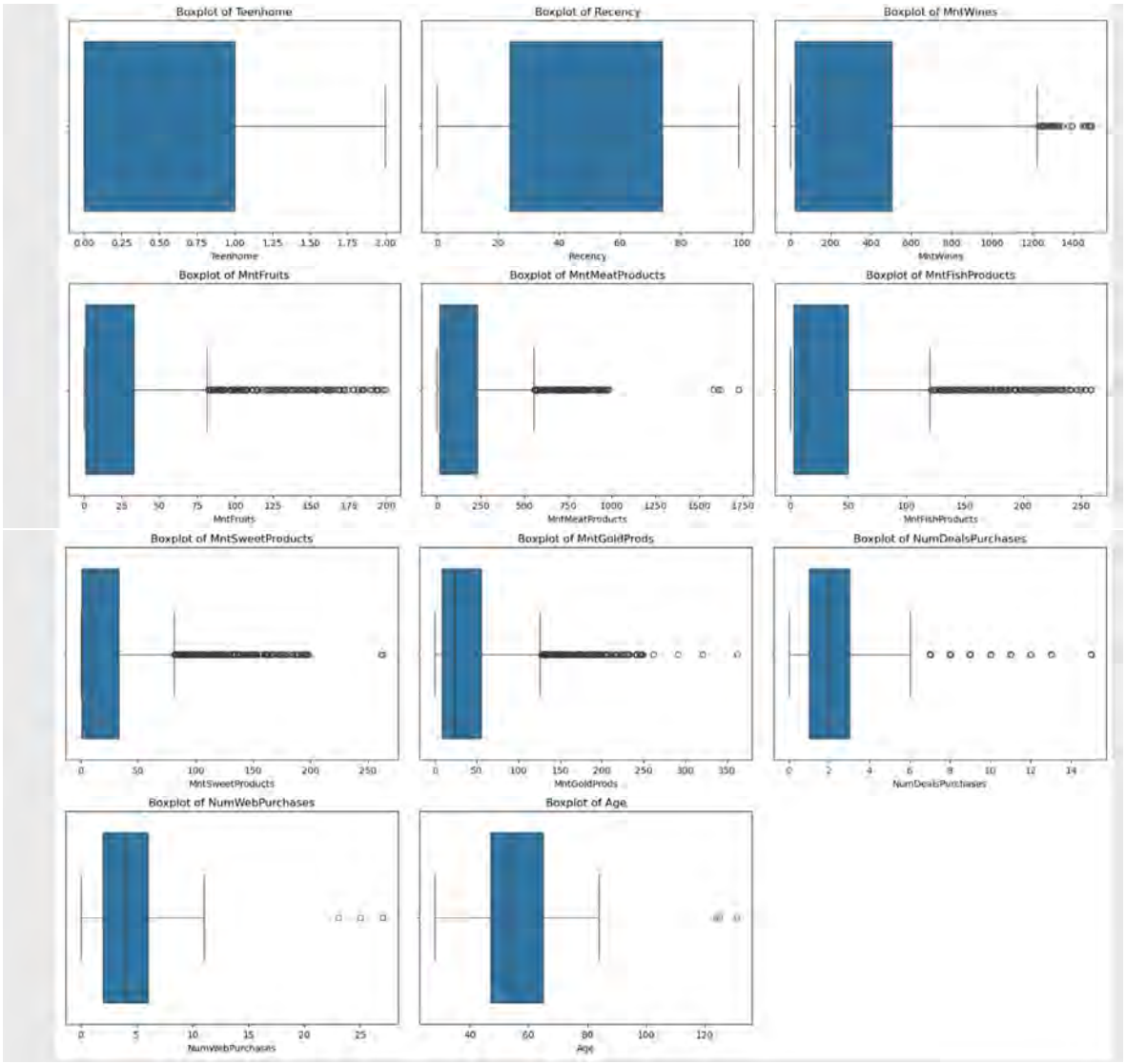


Figure 43b: Results of Performing Univariate Analysis with Box Plots

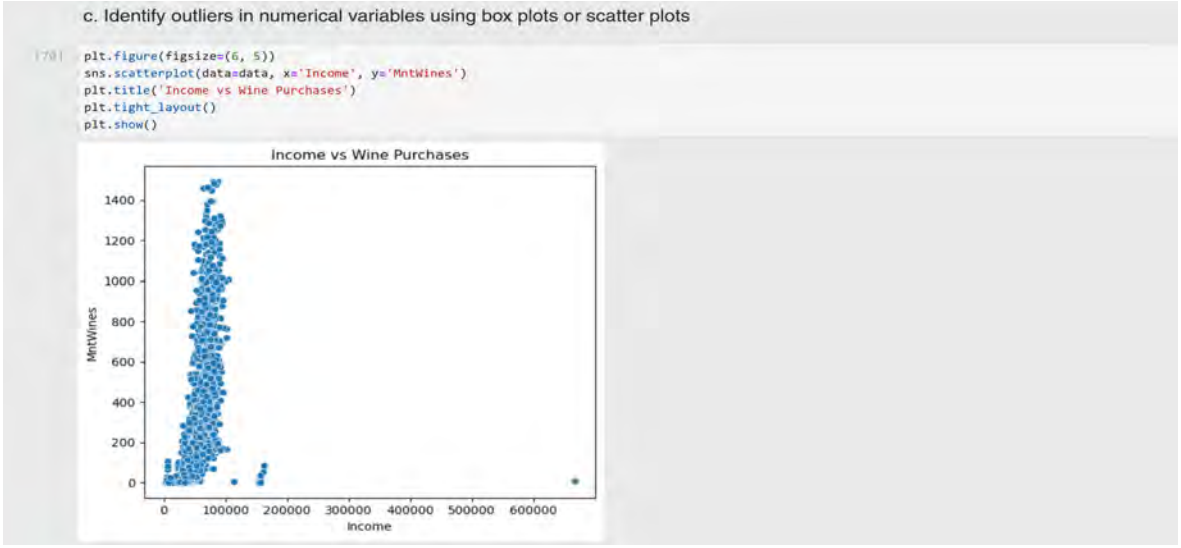


Figure 44a: Results of Performing Univariate Analysis with Scatter Plots

```
plt.figure(figsize=(16, 20))
plot_num = 1

for col_x in ['Income', 'Age']:  # Independent variables
    for col_y in numerical_col:
        if col_y.startswith('Mnt') and col_y != col_x:
            plt.subplot(4, 2, plot_num)
            sns.scatterplot(data=data, x=col_x, y=col_y)
            plt.title(f'{col_x} vs {col_y}')
            plot_num += 1
            if plot_num > 8:  # Limit to first 8 plots
                break
    if plot_num > 8:
        break

plt.tight_layout()
plt.show()
```

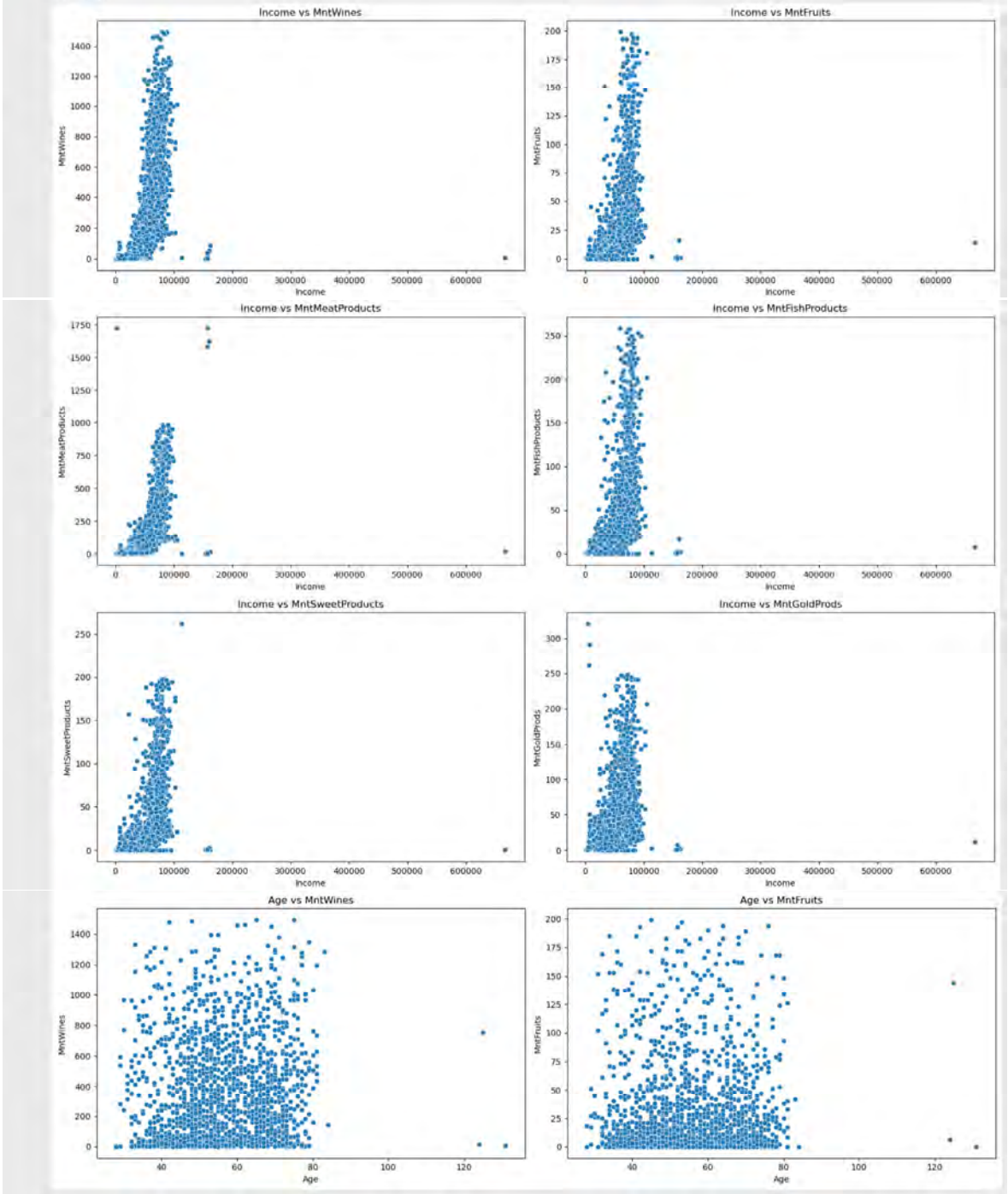


Figure 44b: Results of Performing Univariate Analysis with Scatter Plots

Task 4: Bivariate Analysis:

a. Explore the relationship between numerical variables and the target variable (**Response**) using scatter plots or correlation matrices

b. Explore the relationship between categorical variables and the target variable using bar plots or chi-square tests
c. Explore the relationship between numerical and categorical variables using box plots or violin plots

RESULTS

Task 4: Bivariate Analysis:

a. Explore the relationship between numerical variables and the target variable (Response) using scatter plots or correlation matr

```
# Select numeric columns except Response
numerical_col = data.select_dtypes(include=['int64', 'float64']).columns.drop("Response")

# Determine subplot layout
num_plots = len(numerical_col)
cols = 4
rows = math.ceil(num_plots / cols)

plt.figure(figsize=(16, 4 * rows))

for i, col in enumerate(numerical_col):
    plt.subplot(rows, cols, i + 1)
    sns.scatterplot(x=data[col], y=data['Response'])  # Corrected Y-axis
    plt.title(f'{col} vs Response')
    plt.xlabel(col)
    plt.ylabel('Response')  # Changed from 'Count'

plt.tight_layout()
plt.show()
```

```
categorical_vars = ['Education', 'Marital_Status']
numerical_vars = ['Income', 'Age', 'NumWebVisitsMonth']

for cat in categorical_vars:
    for num in numerical_vars:
        plt.figure(figsize=(8, 5))
        sns.boxplot(data=data, x=cat, y=num)
        plt.title(f'{num} by {cat}')
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.show()
```

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.stripplot(data=data, x='Response', y='Income', ax=axes[0], jitter=True)
axes[0].set_title('Income vs Response')

sns.stripplot(data=data, x='Response', y='Age', ax=axes[1], jitter=True)
axes[1].set_title('Age vs Response')

plt.tight_layout()
plt.show()
```

Figure 45a: Setup Steps for Performing Bivariate Analysis

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

plt.figure(figsize=(8, 5))
sns.scatterplot(x='Response', y='Income', data=data)
plt.title('Income vs Response')
plt.show()

plt.figure(figsize=(8, 5))
sns.scatterplot(x='Response', y='Age', data=data)
plt.title('Age vs Response')
plt.show()

plt.tight_layout()
plt.show()
```

b. Explore the relationship between categorical variables and the target variable using bar plots or chi-square tests

```
categorical_cols = data.select_dtypes(include=['object']).columns

# Plot Response distribution by each categorical variable
for idx, col in enumerate(categorical_cols,1):
    plt.subplot(2,2,idx)
    sns.countplot(x=col, hue='Response', data=data)
    plt.title(f'{col} by Response')
    plt.xticks(rotation=45)
    plt.ylabel('Count')
    plt.tight_layout()
plt.show()
```

c. Explore the relationship between numerical and categorical variables using box plots or violin plots

```
categorical_vars = ['Education', 'Marital_Status']
numerical_vars = ['Income', 'Age', 'NumWebVisitsMonth']

for cat in categorical_vars:
    for num in numerical_vars:
        plt.figure(figsize=(8, 5))
        sns.boxplot(data=data, x=cat, y=num)
        plt.title(f'{num} by {cat}')
        plt.xticks(rotation=45)
        plt.tight_layout()
plt.show()
```

Figure 45b: Setup Steps for Performing Bivariate Analysis

a. Explore the relationship between numerical variables and the target variable (Response) using scatter plots or correlation matrices

```
# Select numeric columns except Response
numerical_col = data.select_dtypes(include=['int64', 'float64']).columns.drop("Response")

# Determine subplot layout
num_plots = len(numerical_col)
cols = 4
rows = math.ceil(num_plots / cols)

plt.figure(figsize=(16, 4 * rows))

for i, col in enumerate(numerical_col):
    plt.subplot(rows, cols, i + 1)
    sns.scatterplot(x=data[col], y=data['Response'])  # Corrected Y-axis
    plt.title(f'{col} vs Response')
    plt.xlabel(col)
    plt.ylabel('Response')  # Changed from 'Count'

plt.tight_layout()
plt.show()
```

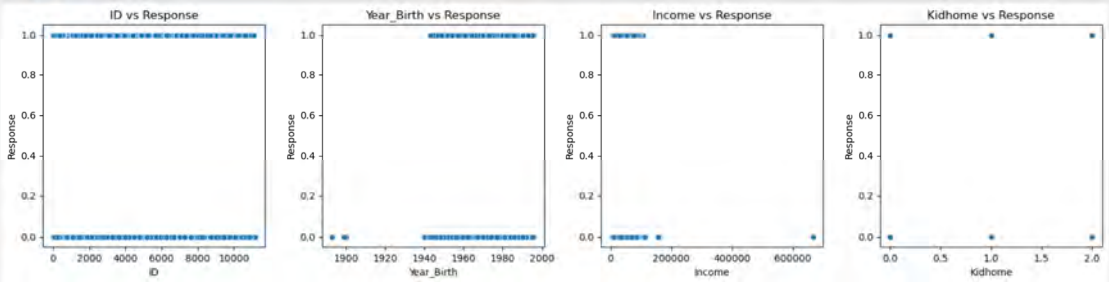


Figure 46a: Results of Performing Bivariate Analysis with Scatter Plots

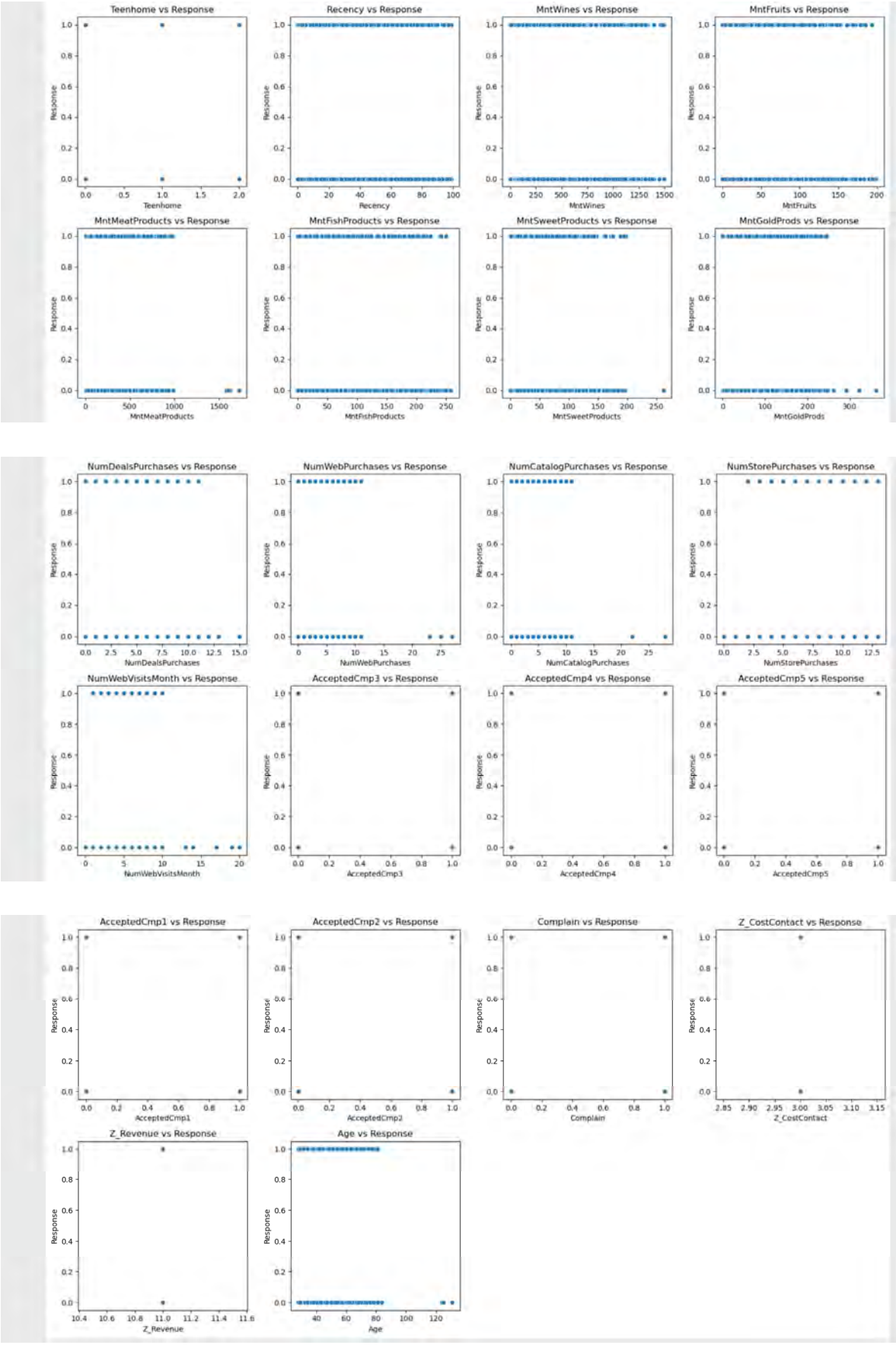
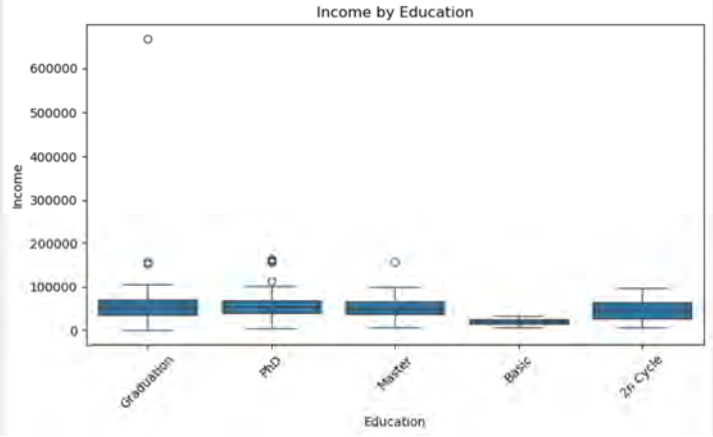
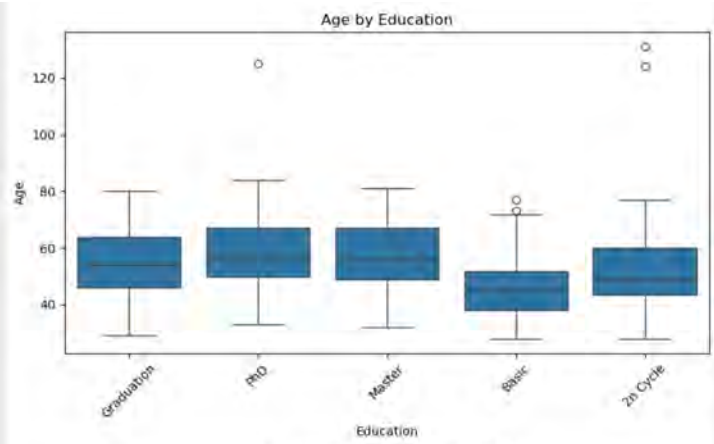


Figure 46b: Results of Performing Bivariate Analysis with Scatter Plots

```
[100]: categorical_vars = ['Education', 'Marital_Status']
numerical_vars = ['Income', 'Age', 'NumWebVisitsMonth']

for cat in categorical_vars:
    for num in numerical_vars:
        plt.figure(figsize=(8, 5))
        sns.boxplot(data=data, x=cat, y=num)
        plt.title(f'{num} by {cat}')
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.show()
```





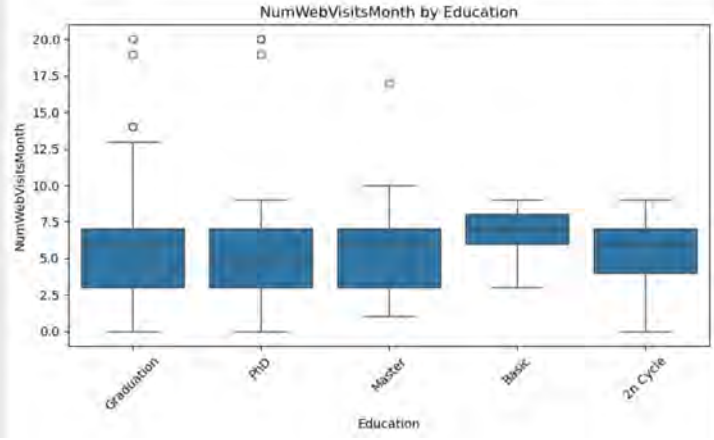


Figure 47a: Results of Performing Bivariate Analysis with Box Plots

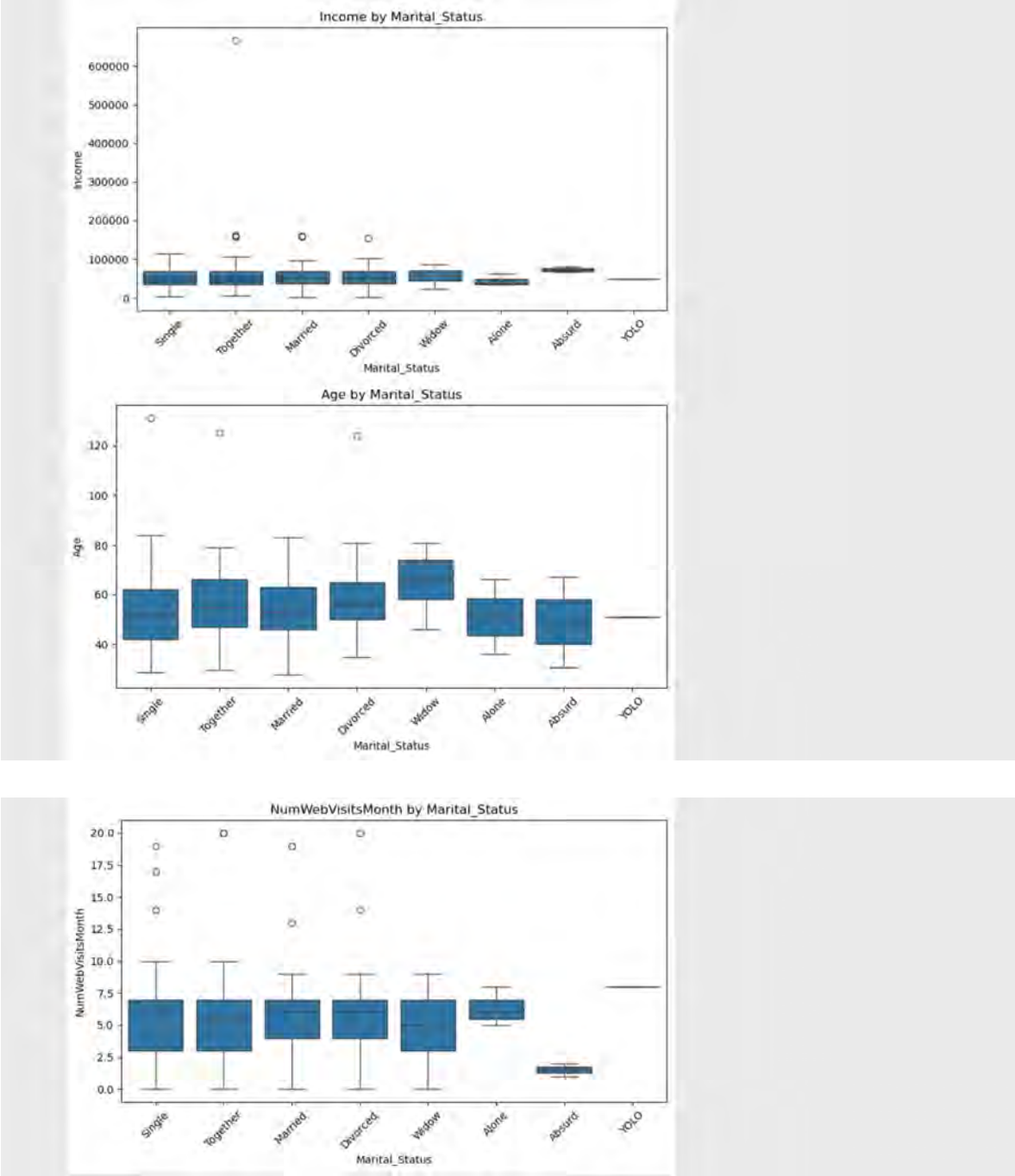


Figure 47b: Results of Performing Bivariate Analysis with Box Plots

```
[102]: fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.stripplot(data=data, x='Response', y='Income', ax=axes[0], jitter=True)
axes[0].set_title('Income vs Response')

sns.stripplot(data=data, x='Response', y='Age', ax=axes[1], jitter=True)
axes[1].set_title('Age vs Response')

plt.tight_layout()
plt.show()
```

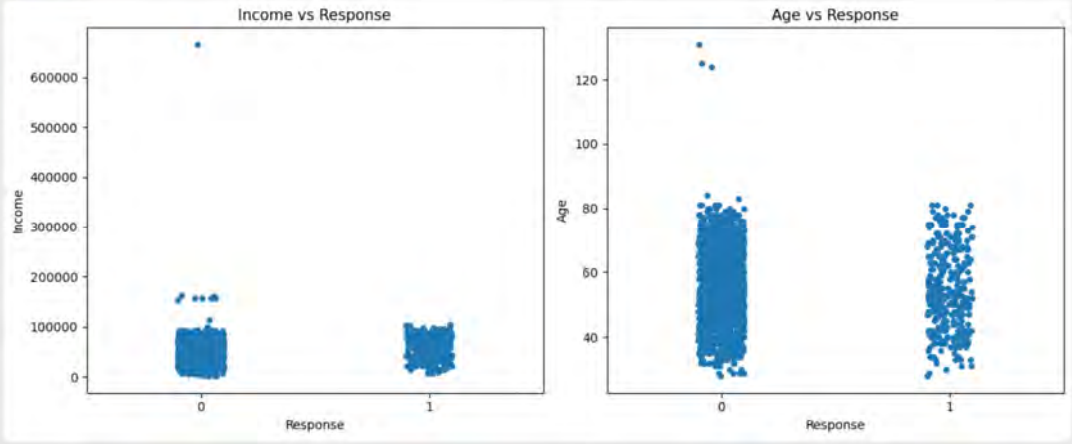


Figure 48: Results of Performing Bivariate Analysis with Strip Plot

```
[104]: fig, axes = plt.subplots(1, 2, figsize=(12, 5))

plt.figure(figsize=(8, 5))
sns.scatterplot(x='Response', y='Income', data=data)
plt.title('Income vs Response')
plt.show()

plt.figure(figsize=(8, 5))
sns.scatterplot(x='Response', y='Age', data=data)
plt.title('Age vs Response')
plt.show()

plt.tight_layout()
plt.show()
```

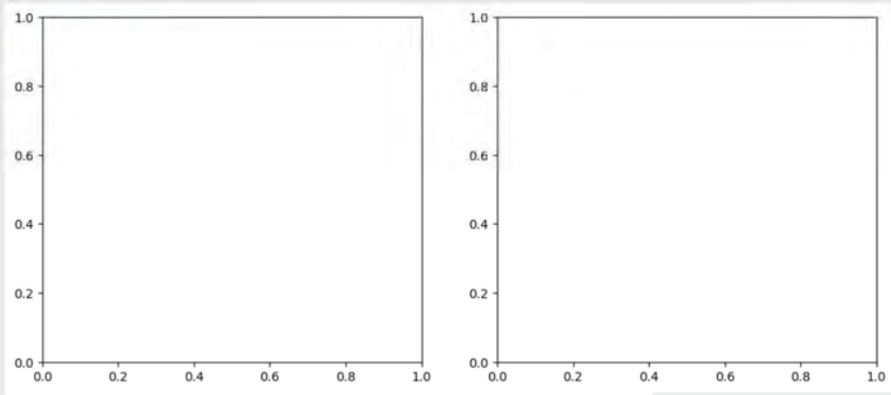


Figure 49a: Results of Performing Bivariate Analysis with Scatter Plot

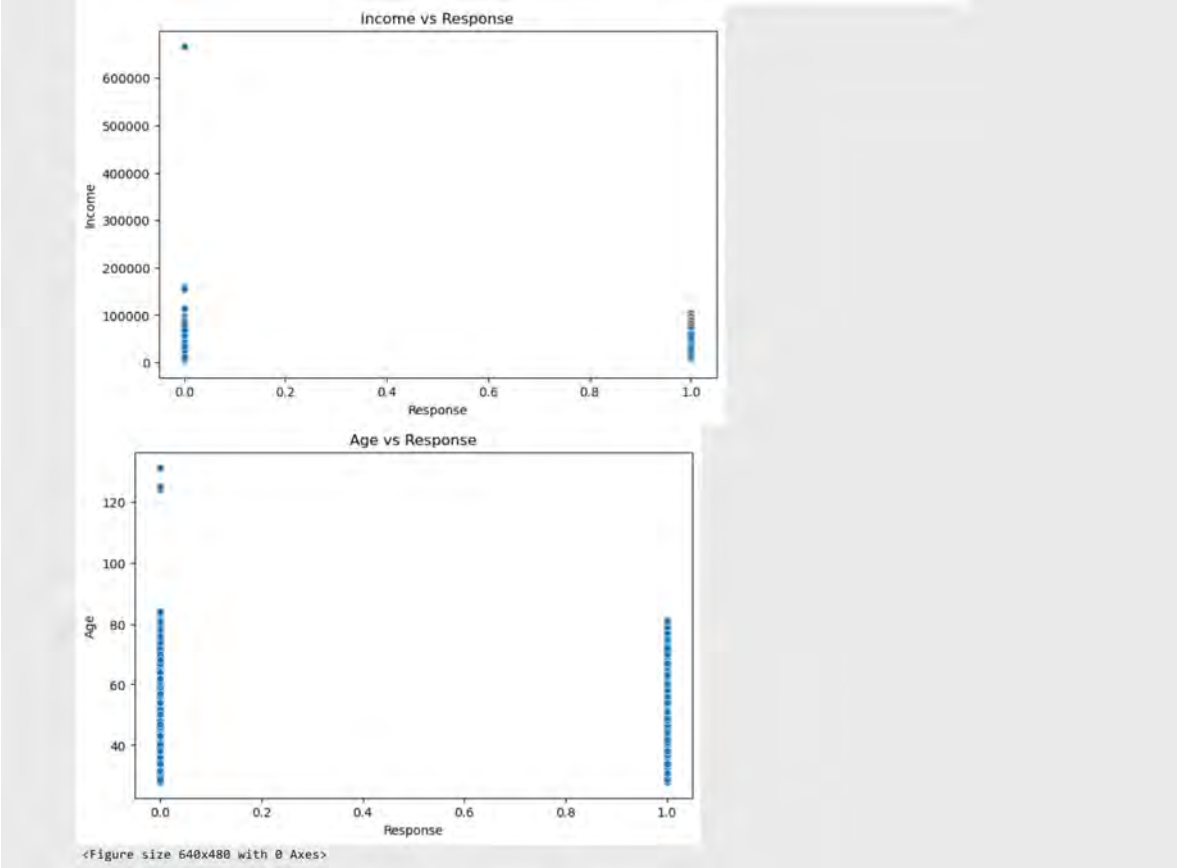


Figure 49b: Results of Performing Bivariate Analysis with Scatter Plot

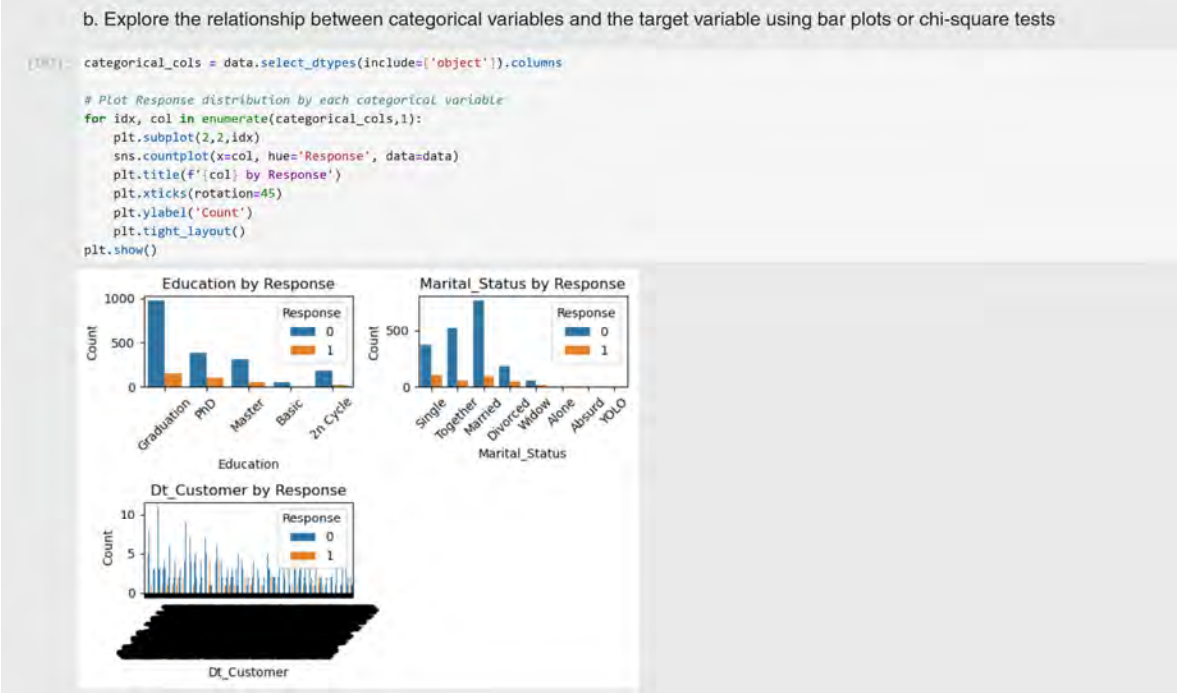
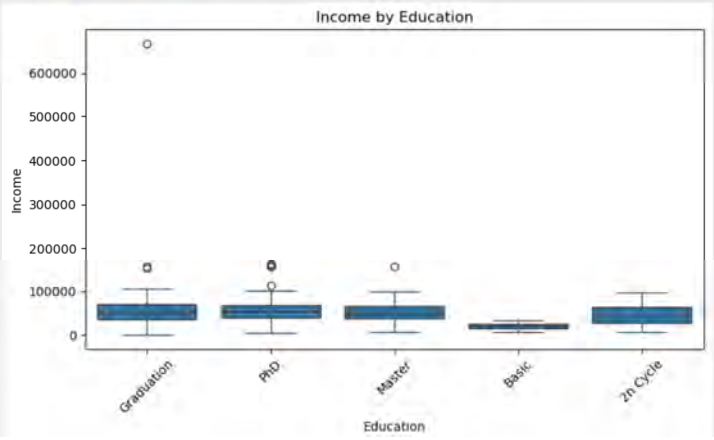


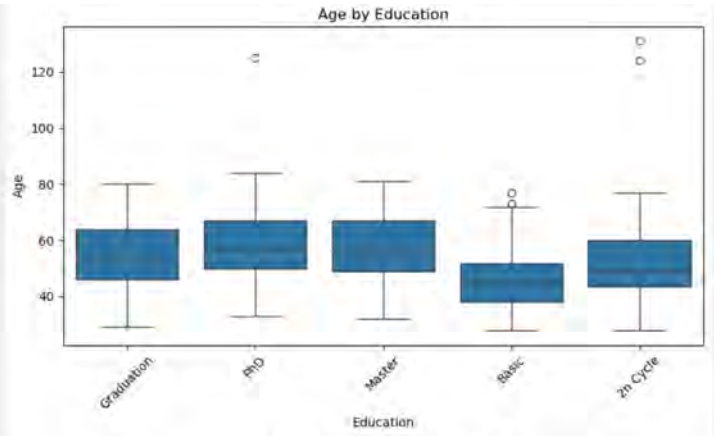
Figure 50: Results of Performing Bivariate Analysis with Count Plot

c. Explore the relationship between numerical and categorical variables using box plots or violin plots

```
categorical_vars = ['Education', 'Marital_Status']
numerical_vars = ['Income', 'Age', 'NumWebVisitsMonth']

for cat in categorical_vars:
    for num in numerical_vars:
        plt.figure(figsize=(8, 5))
        sns.boxplot(data=data, x=cat, y=num)
        plt.title(f'{num} by {cat}')
        plt.xticks(rotation=45)
        plt.tight_layout()
plt.show()
```





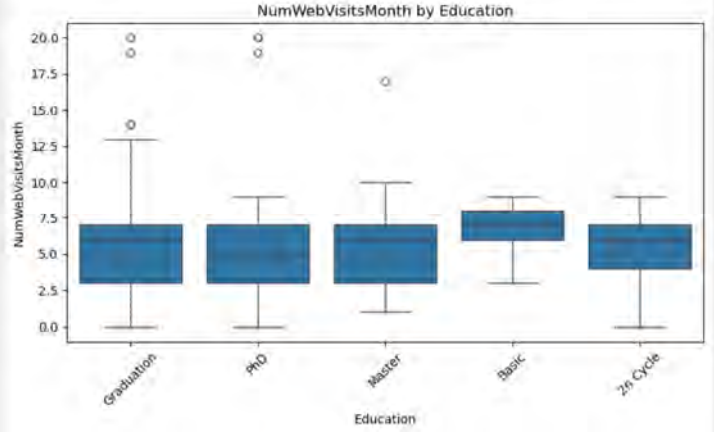


Figure 51a: Results of Performing Bivariate Analysis with Box Plot

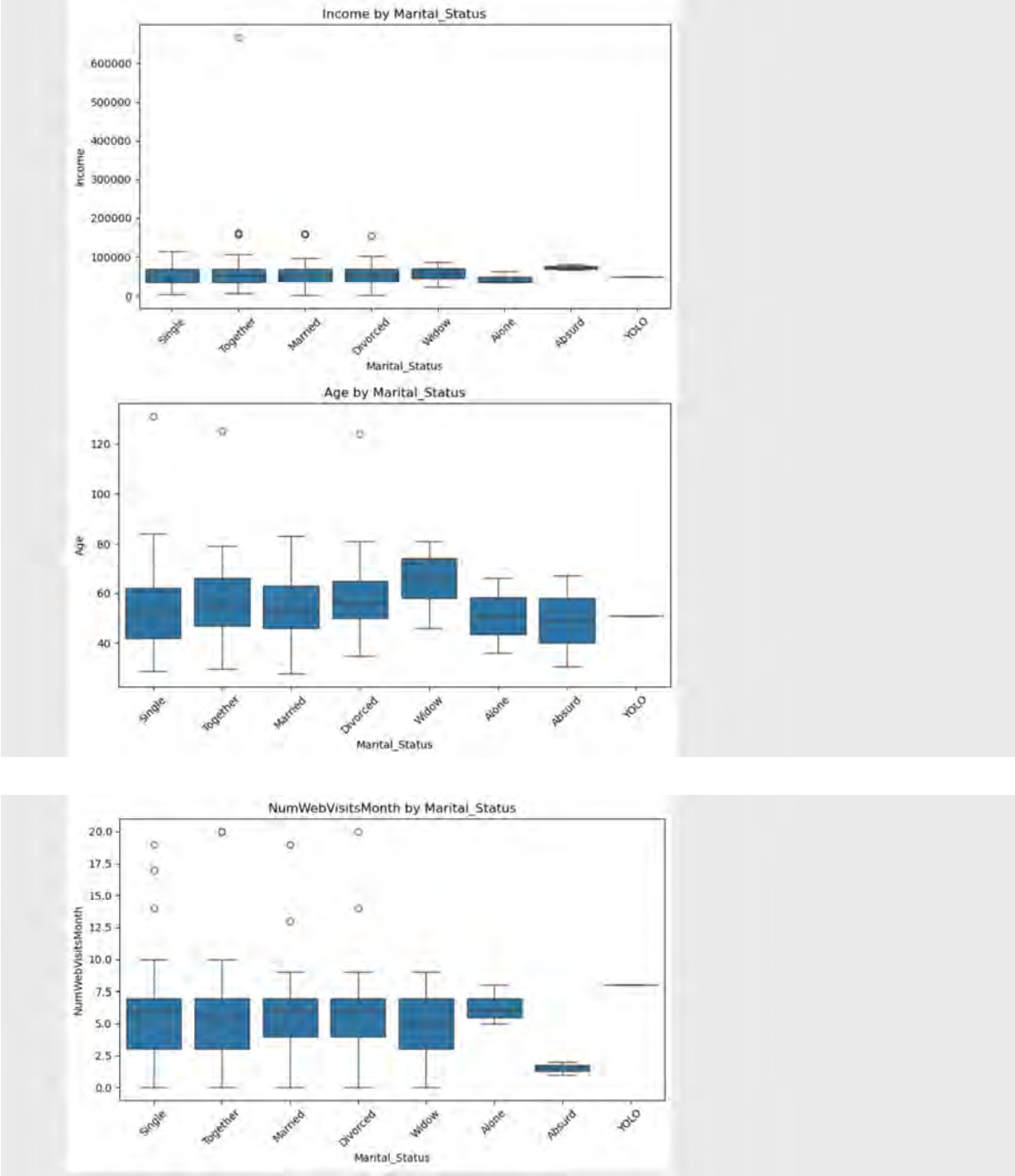


Figure 51b: Results of Performing Bivariate Analysis with Box Plot

Observations: Write an analysis report on performing exploratory data analysis (EDA) using Python in the context of building a fraud detection system for Retail Analytics

RESULTS

Exploratory Data Analysis (EDA) Report Context: Retail Analytics Fraud Detection System using Python

Objective

The objective of this Exploratory Data Analysis (EDA) is to understand the structure, patterns, and anomalies within the retail transaction data to support the development of a fraud detection system. This analysis offers critical insights into customer behavior, transaction dynamics, and feature distributions, which are essential for identifying outliers and potential fraudulent cases.

Tools and Techniques

EDA is performed using Python libraries, including:

- Pandas for data manipulation and summary statistics
- Matplotlib / Seaborn for visualizations (histograms, oxplots, violin plots, scatter plots)
- Scipy for statistical testing (not in this particular exercise, but recommended)
- Scikit-learn for correlation and feature importance assessment (not in this particular exercise, but recommended)

Data Overview

- **Total records:** N customers (e.g., 2,240 rows)
- **Features analyzed:** Customer demographics, transaction amounts (MntWines, MntFruits, etc.), web activity, and campaign responses.
- **Target variable (for fraud potential):** Not explicitly defined in this dataset, but outlier patterns in spending, frequency, and campaign response behavior were analyzed to infer potential fraud-like behavior.

Key EDA Observations

1. **Missing & Outlier Detection**

- Income had a small number of null values, which were imputed.
- Outliers were observed in high-spend product categories (e.g., MntWines > 1000) and Income (> 100K).
- Boxplots and IQR method were used to identify outlier records across monetary features systematically.

2. Feature Distributions

- Spending data (MntX columns) showed right-skewed distributions — a small number of high-spending customers.
- Age distribution showed most customers were between 35 and 65, which may help contextualize fraud by age.
- Response to campaigns skewed toward “no” Response = 0), consistent with normal customer behavior.

3. Customer Behavior Patterns

- Strong correlations were found between high income and spending in categories like wines and meat products.
- Customers with a high number of catalog purchases or frequent web visits sometimes exhibited unusual combinations of low income and high spend, which could be flagged for further review.

4. Bivariate Analysis

- Boxplots and violin plots showed significant variation in spend across marital status and education groups.
- Scatter plots of Income vs Spend and Age vs Spend revealed clusters and anomalies useful for fraud profiling.

5. Potential Fraud Indicators

- High spend + low income combinations.
- Abnormally high frequency of purchases in a short timeframe.
- Customers with unusual campaign response patterns or identical values across unrelated features.

Insights for Fraud Detection

EDA revealed that anomalous behaviors in spending, frequency, and income relationships are likely to be the most valuable indicators of fraud. Customers who do not match behavioral norms (based on demographic group, spend tier, or web/campaign interaction) should be flagged for further machine learning-based fraud analysis. Based on the data analysis produced by this Python section of this capstone project and my observations on possibly of using Python in the context of building a fraud detection system for Retail Analytics, I would first analyze the twenty-four survey respondents who had "0" for their income which was not excluded from this analysis, to analyze their demographics and buying behavior. Although they represent only 24/2240 = 0.010724 or 1.0724% of the respondents, the question to be answered is whether they accurately reflect the profile of individuals who may have committed fraud or have they intentionally omitted their income to shield their privacy.

Conclusion

The exploratory data analysis phase has laid a solid foundation for developing a fraud detection model. It revealed both expected patterns in customer behavior and outlier profiles that can be used as signals in downstream modeling. This EDA reinforces the importance of behavioral baselining and highlights specific features (e.g., TotalSpend, Income, NumWebPurchases) as key variables for fraud risk assessment.

Power BI Tasks:

4. **Interactive Dashboard Design in Power BI:**

Task 1: Visualize Customer Income Distribution:

- Visualize customer income distribution grouped by year of registration.
  - Open Power BI Desktop and load the **marketing_campaign.csv** file using the **Get Data > Xlsx** option.
  - Navigate to the **Data view** and verify data types of required columns.
  - Go to Modelling, click **New column**:
    Year Registered = YEAR('marketing_campaign'[Dt_Customer])
  - Drag **Number Diagnoses** and **Number Emergency** to the **X-axis** field.
  - In Fields pane**, right-click Income > New group > Select Bin, set size = 5000**, click OK, rename to **Income Bin**
  - Insert a **Stacked Column Chart**.
  - Drag **Income Bin** to **X-axis**.
  - Drag **Count of ID** to **Y-axis.**
  - Drag **Year Registered** to **Legend**
  - Optionally, enable **Data Labels** under the Format pane for better readability.

NOTE: Assignment correction by instructor. The 4th bullet, "**Drag Number Diagnoses and Number Emergency to the X-axis field**." does not apply to this assignment, so ignore it.

RESULTS

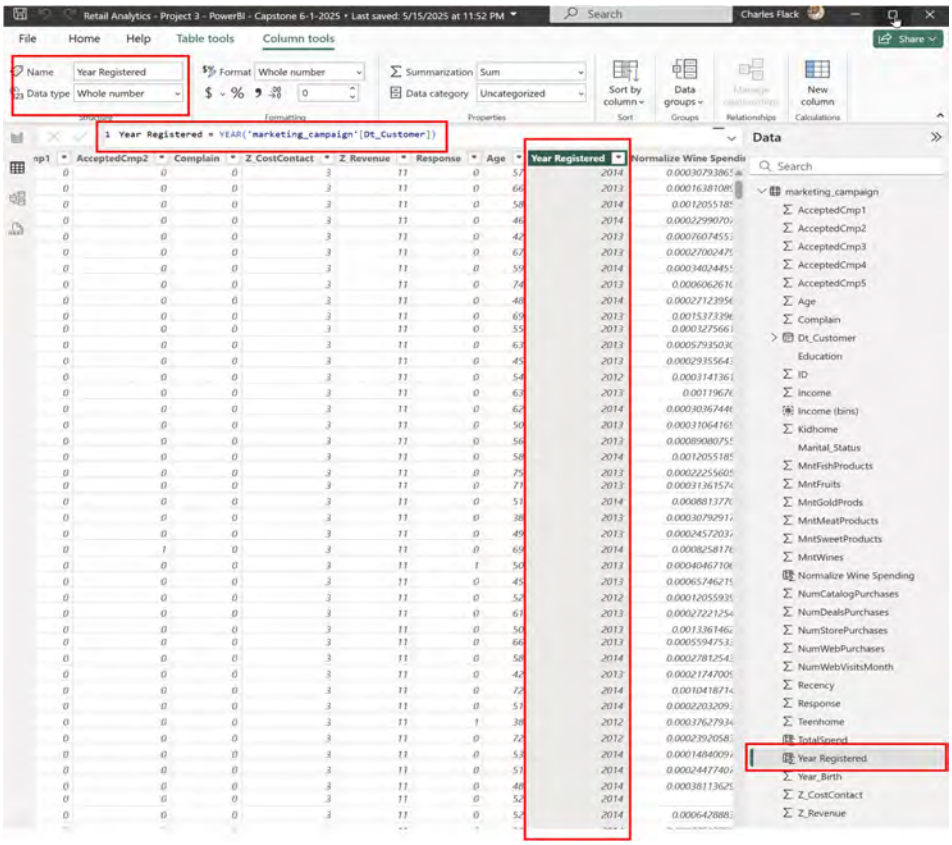


Figure 52: Created "Year Registered" Category in Dataset

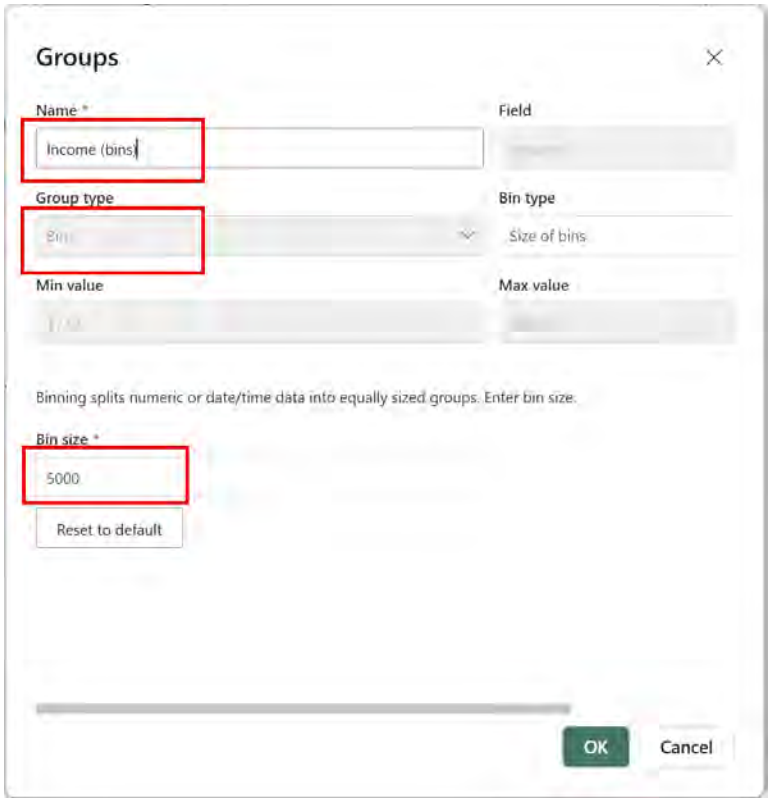


Figure 53a: Setup Steps for Visualizing Customer Income Distribution

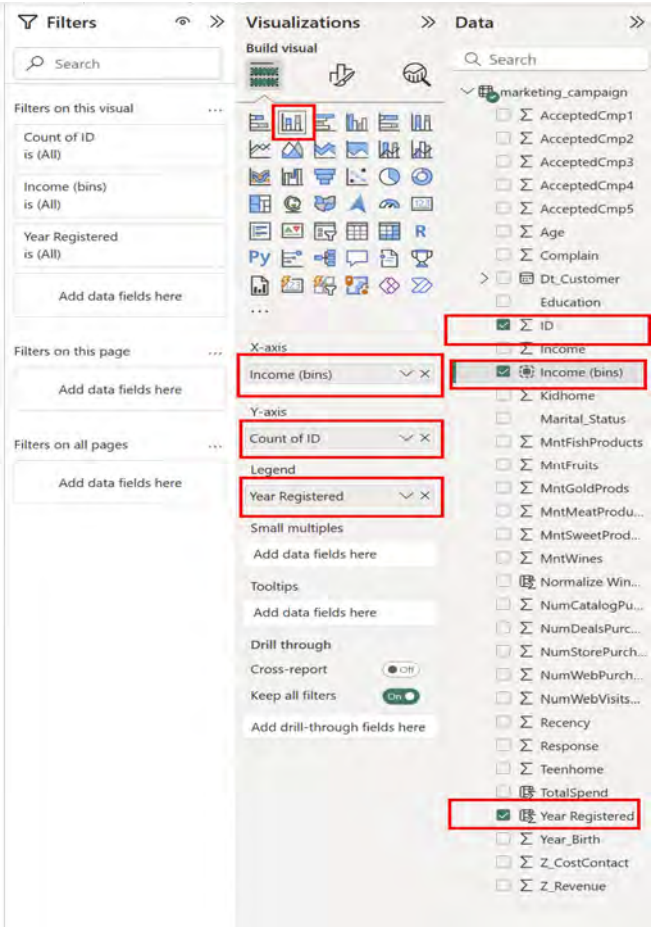


Figure 53b: Setup Steps for Visualizing Customer Income Distribution

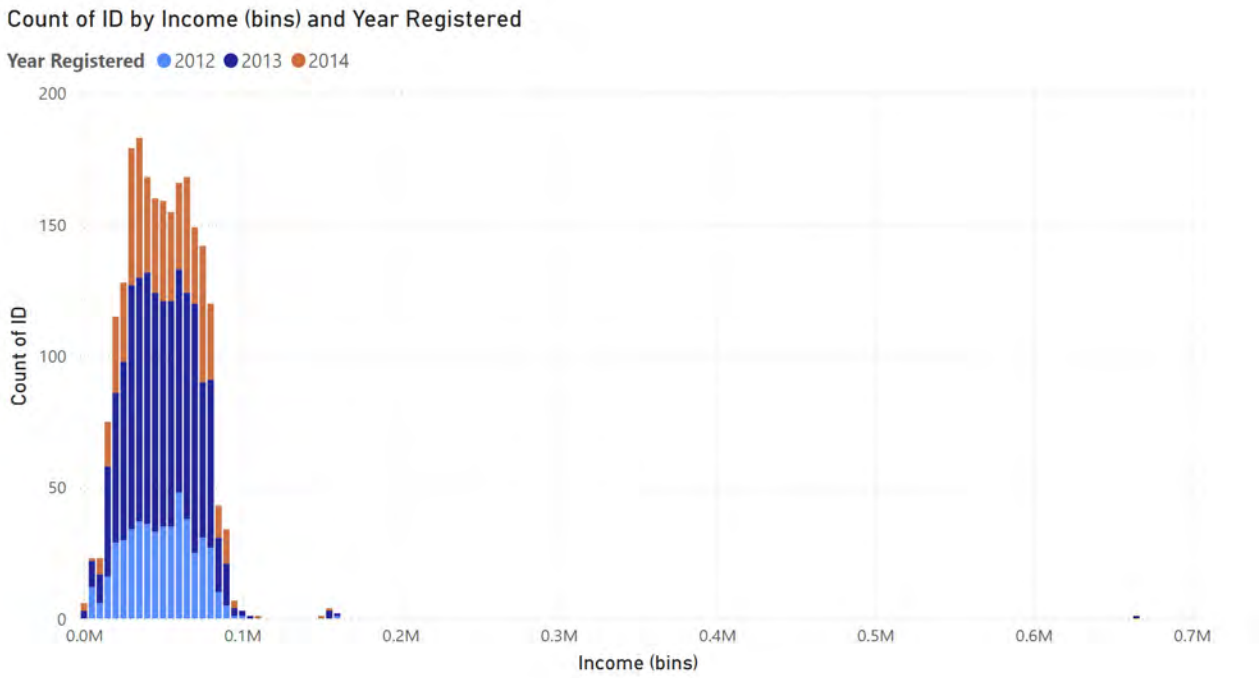


Figure 54: Results of Visualizing Customer Income Distribution

Task 2: Examine the Breakdown:

- Create Examine education levels and marital status using Clustered bar chart.
  - Create a **Clustered bar chart**.
  - Drag **Education** to **Y-axis**
  - Drag **Count of ID** to **X-axis**
  - Drag **Marital_Status** to **Legend**.
  - Optionally, enable **Data Labels** under the Format pane for better readability.

RESULTS

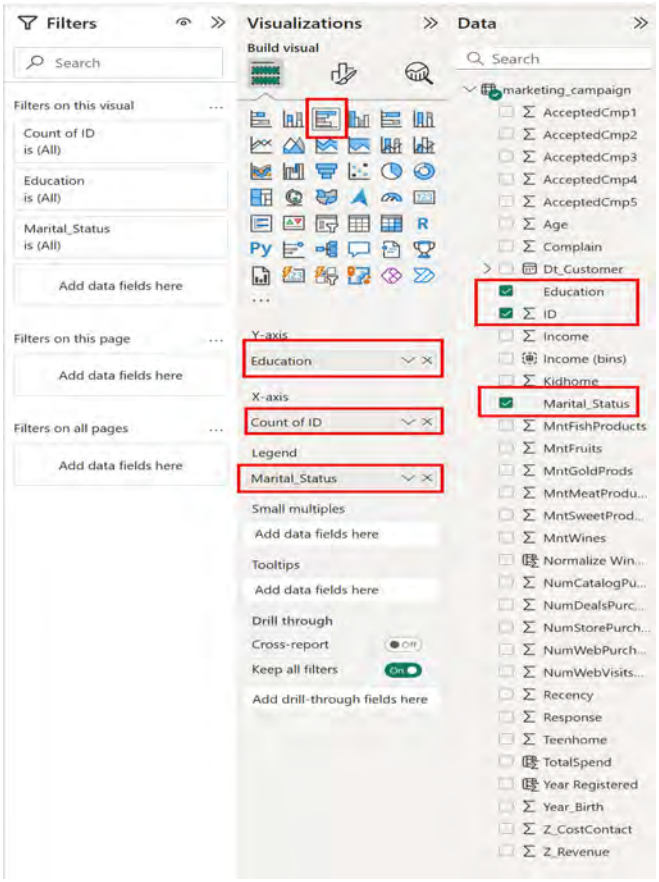


Figure 55: Setup Steps for Visualizing Education Levels and Marital Status with Clustered Bar Chart

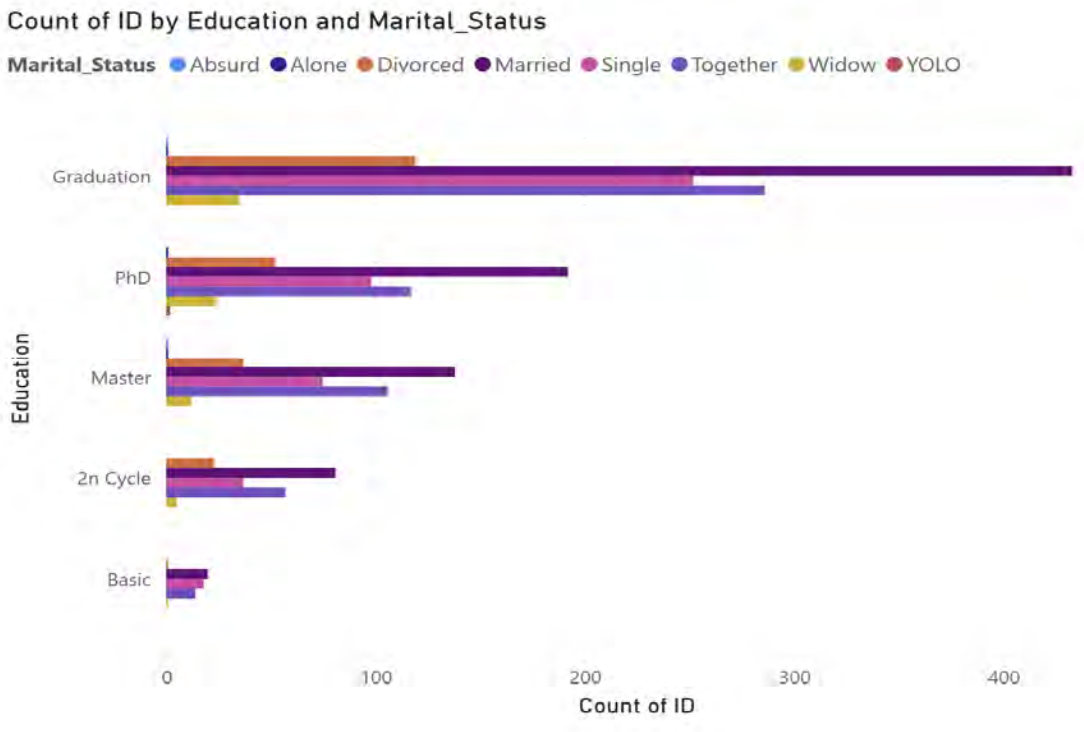


Figure 56: Results of Visualizing Education Levels and Marital Status with Clustered Bar Chart

Task 3: Examine the Relationship:

- Explore relationship between income and wine spending
  - Create a new column by clicking on Modelling
  - Use this DAX format : **Normalized Wine Spending = 'marketing_campaign'[MntWines] / 'marketing_campaign'[Income]**
  - Drag **Sum of Income** to **X-axis**
  - Drag **Sum of Normalized Wine Spending** to **Y-axis**
  - Optional, Drag Sum of **MntWines** to **Size**

RESULTS

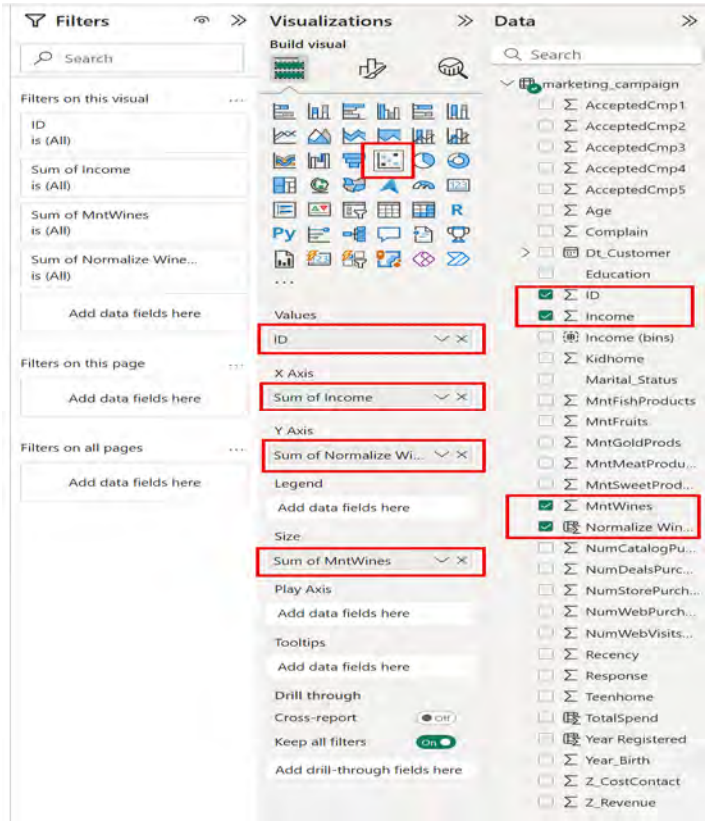


Figure 57: Setup Steps for Exploring Income and Wine Spending

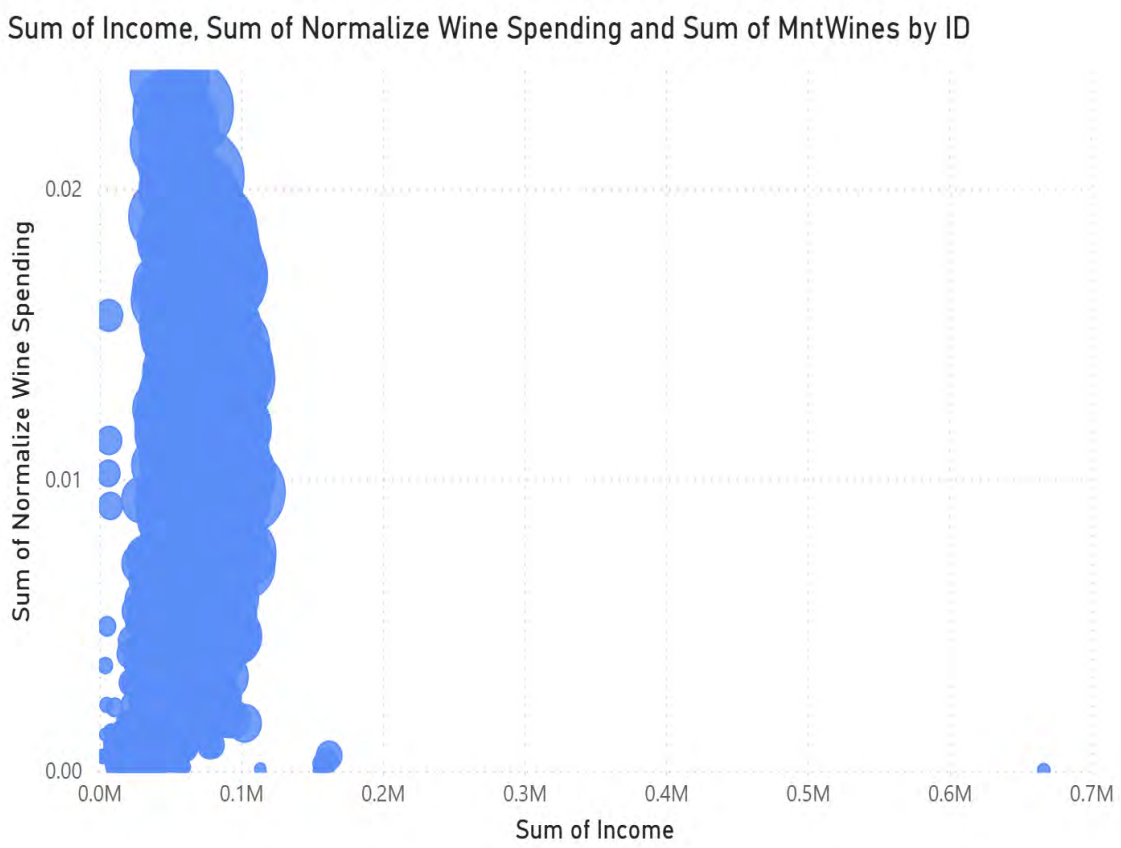


Figure 58: Results of Exploring Income and Wine Spending

Task 4: Analyze Frequency:

- Analyze frequency of purchases across categories
  - Insert a **Stacked Column Chart** on new page
  - Drag **ID** to **X-axis**
  - Drag **MntWines, MntFruits, MntMeatProducts** to **Y-axis**
  - Go to the **Filters pane** on the right.
  - Drag **ID** into the **Filters on this visual** area.
  - Change filter type to **Top N**.
  - Choose: **Show Top:** 20 (or 50, up to you)
  - **By Value:** use TotalSpend measure created earlier
  - Click **Apply Filter**

NOTE: The 8th bullet, "**By Value: use TotalSpend measure created earlier,"** was never generated earlier, so here are the steps I executed to create ir:

1. Go to "Table view"
2. Select "New Column"
3. Enter the following equation:

```
TotalSpend =
  marketing_campaign[MntWines] +
  marketing_campaign[MntFruits] +
  marketing_campaign[MntMeatProducts] +
  marketing_campaign[MntFishProducts] +
  marketing_campaign[MntSweetProducts] + marketing_campaign[MntGoldProds]
```

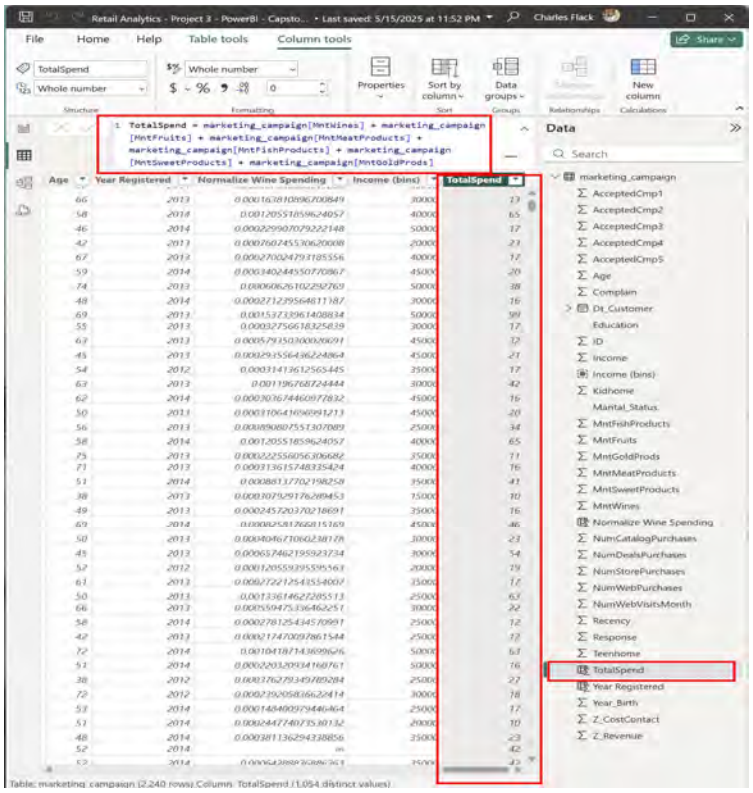


Figure 59: Created "TotalSpend" Category in Dataset

RESULTS

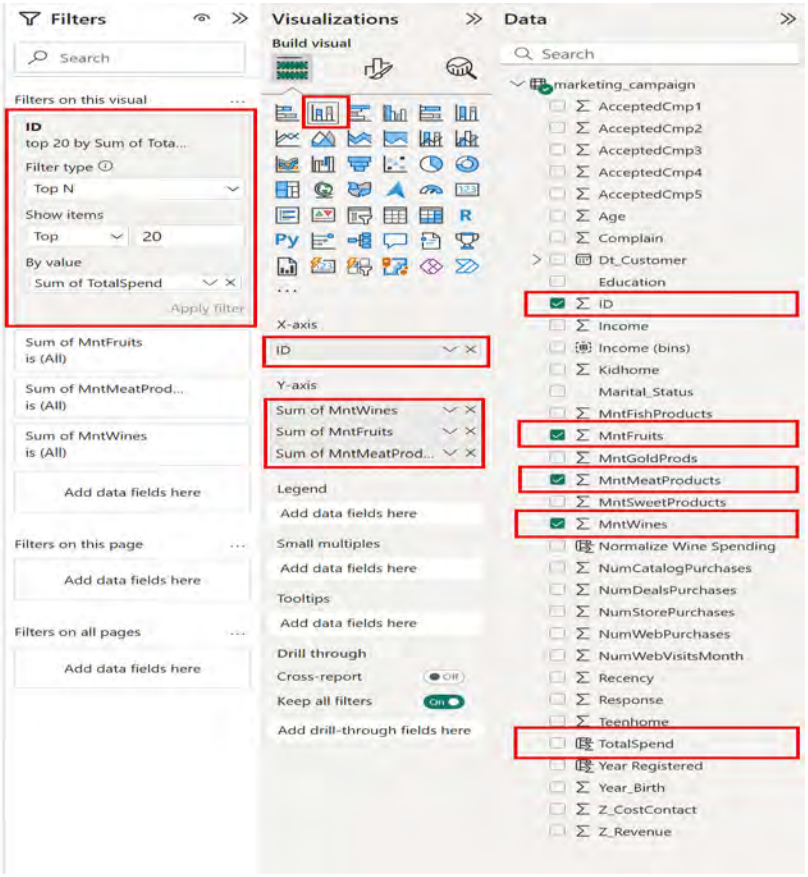


Figure 60: Setup Steps for Exploring Frequency of Purchases Across all Categories

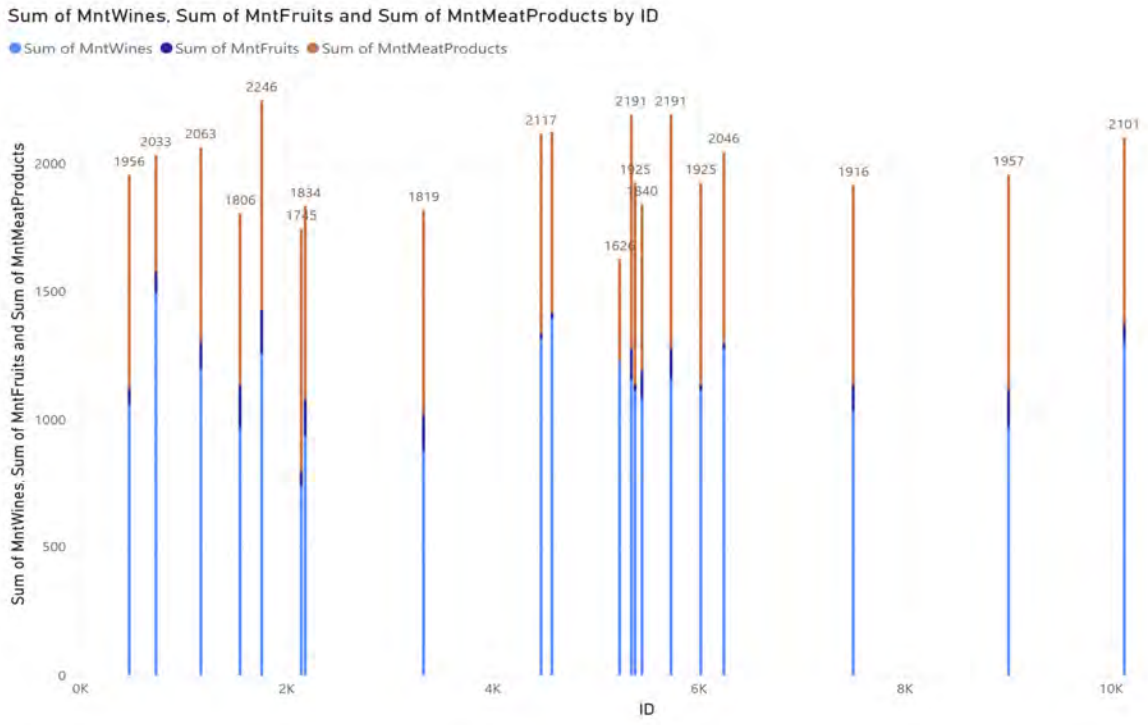


Figure 61: Results of Exploring Frequency of Purchases Across all Categories

Task 5: Create a Dashboard:

- Create a dashboard with all the visualizations
  - Click on the dashboard icon at the bottom
  - Copy the Visuals from all the pages and paste them on the new page dashboard.

RESULTS

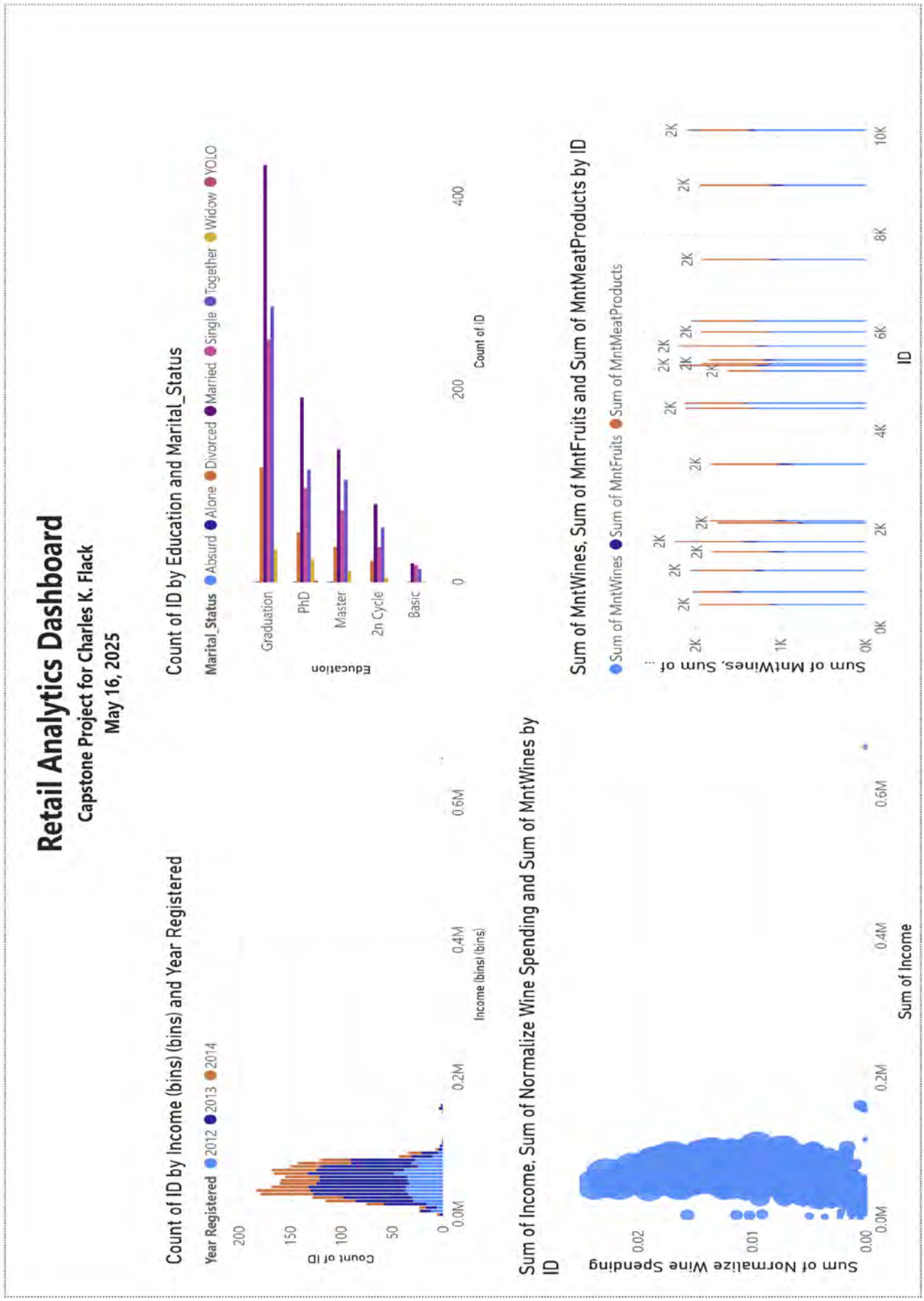


Figure 62: Results of Creating a Dashboard of all Visualizations

FINAL CONCLUSIONS:

This final capstone project was an enjoyable assignment to complete, as all the tools covered in this coursework were utilized to perform data analytics. Although I have used Microsoft Excel for many years now to perform data analytics leveraging its pivot table capabilities, I now equally enjoy the power of utilizing Microsoft Power BI, especially with its ease of creating visual dashboards.

There were two significant errors in the instructions for this assignment that were identified during execution. They have been noted in my submitted work, and I have included the workaround that I executed to enable me to complete the capstone assignment.

I successfully produced the required visual output to complete the capstone assignment for the Python section; however, the output was not as polished as I had wanted. Still, some of the graphics created by Python required additional time to fine-tune the visual outputs. I wish there were more instructional time provided on how to do this, as well as on producing output for other platforms via API interfaces or to a web server interface, to mimic the capabilities of Power BI and Tableau.

I am very disappointed that the Caltech professional certification program and capstone did not include the use of Tableau from Salesforce. Tableau is the leader in the data analytics space when compared to Power BI. It would have been beneficial to have both Power BI and Tableau as part of this Caltech professional certification program, and doing so would have positioned this program significantly ahead of other similar programs in this space. Having Tableau as a core part of this professional certification program would have exceptionally prepared me for the three jobs I have recently been recommended for at Salesforce. I will be taking the Tableau elective in the fall when it is offered, but this does not help me now with the jobs I am applying for at Salesforce this week.

Although I utilized generative AI via ChatGPT to validate command constructs used in the SQL and Python sections of this assignment, this final capstone assignment did not fully leverage the power of generative AI as could have been done in data analytics. I was able to utilize ChatGPT to validate ideas and command constructs I was exploring in my SQL and Python data analysis efforts. I also used it to validate the output generated from my MS Excel, MySQL, and Python code execution efforts, which was great to have it confirm and validate my work. I also used it to verify my ideas on how to proceed with my data analytics work. I also used it to validate and confirm conclusions I have arrived at, where it performed as my data analytics consultant. However, I believe this assignment could have gone much further in demonstrating how to leverage and utilize generative AI for data analytics effectively.

Lastly, if this course had also included “machine learning” with the generative AI component, everyone would be lining up to take this course from Caltech. I am currently seeking to enroll in and complete a professional certification program in machine learning and generative AI, aiming to enhance my marketability and value in this competitive field.

All in all, this assignment was fun!